Time-Causal Scale Space from Hemigroup Axioms — Blueprint / Shared Spec

Legend. [**T**] proved-target: the Lean formalisation must prove this from primitives together with the [A] interfaces. [**A**] analytic interface: stated in Lean as an axiom / carried hypothesis; the underlying analysis is cited, not reproven.

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Chapter 2

Axioms

The primitive object is a two-parameter family of operators indexed by ordered pairs of scales.

Definition 2.1 (causal cascade measurement family)**.** A family of linear operators

$$\Phi_{x,y} : X \to X, \qquad 0 \le x \le y < \infty,$$

is a *causal cascade measurement family* if it satisfies:

- **(A1) Boundedness.** Each $\Phi_{x,y}$ is a bounded operator on X.
- **(A2) Time-translation covariance.** $\Phi_{x,y}\,\tau_a = \tau_a\,\Phi_{x,y}$ for all $a \in \mathbb{R}$.
- **(A3) Causality.** If $f \in X$ vanishes a.e. on $(-\infty, t_0)$, then so does $\Phi_{x,y} f$.
- **(A4) Positivity.** $\Phi_{x,y} X_+ \subseteq X_+$.
- **(A5) Unit area.** $\int \Phi_{x,y} f = \int f$ for all $f \in X_+$.
- **(A6) Cascade (hemigroup).** $\Phi_{x,x} = \mathrm{Id}$, and $\Phi_{y,z}\,\Phi_{x,y} = \Phi_{x,z}$ for all $x \le y \le z$.
- **(A7) Continuity.** For every $f \in X$, the map $(x, y) \mapsto \Phi_{x,y} f$ is continuous from $\{0 \le x \le y\}$ to X.
- **(A8) Scale covariance.** There is a family $(S_\sigma)_{\sigma>0}$ of increasing bijections of $[0, \infty)$ such that
$$D_\sigma\,\Phi_{x,y} = \Phi_{S_\sigma x,\, S_\sigma y}\,D_\sigma \qquad \text{for all } 0 \le x \le y,\ \sigma > 0.$$
- **(ND) Nondegeneracy.** $\Phi_{x,y} \ne \mathrm{Id}$ whenever $x < y$.

▷ **[T]** proved-target *(Stated in Lean as `Hemigroup.CascadeFamily`, one field per clause, with (A1) carried by the type of bounded linear operators on X and (A8)'s S bundled as data. This is the one file in the development where an error would be silent — a mis-stated axiom makes the theorems vacuous rather than unprovable — so the specification is checked against a model: `Hemigroup.cascadeFamily` exhibits the kernels of 7.3 as an instance. Doing so caught two defects in an earlier statement of this definition; see the note there.*

The definition is split in two. *(A1)–(A7) and (ND) are the structure `CascadeCore`; (A8) is a separate predicate `IsScaleCovariant Fam G S`, asserting the intertwining for each σ in a set G of scale factors, and `CascadeFamily` is the core together with `IsScaleCovariant` at $G = (0, \infty)$. (A8) is the only clause whose strength varies: demanding it only for σ in a closed subgroup of $(\mathbb{R}_+, \times)$ gives a weaker axiom system, of which this one is the top stratum, and the core is common to all of them. Which sets G can occur is a theorem, not part of the statement, which is why G is left an arbitrary set. Chapter 4 is proved for the core alone — it uses (A2)–(A5) and (A7) and nothing else — so it applies to every stratum unchanged; a Lean tag naming `CascadeCore` rather than `CascadeFamily` is recording exactly that.)*

Remark 2.2 (provenance of the axioms)**.** (A1)–(A5) and (A7) are as in `@fagerstrom2005temporal`; the diagonal condition in (A6) together with (A7) is the two-parameter form of the *extended point* axiom $\lim_{\tau\to 0}\phi_\tau = \delta$, in the notation of that paper, ϕ_τ the kernel at scale τ.

The substantive change is (A6). In `@fagerstrom2005temporal` the cascade reads $\Phi_\tau\Phi_{\tau'} = \Phi_{\tau+\tau'}$, which additionally assumes that the composite measurement depends only on the total amount of scale — that measurement stages are interchangeable. (A6) drops that assumption, and every result that follows is a consequence of dropping it. Two smaller differences: `@fagerstrom2005temporal` also required the kernels to be continuous functions, which the pure delay violates, and dropping that requirement is what admits the drift term (Remark 7.5); and (ND) is a normalization not stated there.

(A8) is the two-parameter form of scaling covariance; note that an increasing bijection of $[0,\infty)$ automatically fixes 0. (ND) excludes idle stretches of the scale axis.

A *passive* variant replaces (A5) by $\int \Phi_{x,y} f \le \int f$ on X_+; all results below hold with subprobability kernels and an extra killing term, and we do not pursue it.

Chapter 3

Preliminaries and notation

Time runs over $\mathbb{R}$. We write $X = L^1(\mathbb{R}, dt)$, real, with positive cone X_+ and $\int f := \int_{\mathbb{R}} f(t)\,dt$; $M_+(\mathbb{R})$ for the finite positive Borel measures on $\mathbb{R}$, with total mass $\|\mu\| = \mu(\mathbb{R})$; translations $(\tau_a f)(t) = f(t-a)$ for $a \in \mathbb{R}$; dilations on functions $(D_\sigma f)(t) = \sigma^{-1} f(t/\sigma)$, $\sigma > 0$, which preserve mass, and on measures $D_\sigma \mu :=$ the pushforward of μ under $t \mapsto \sigma t$; and convolution $(\mu * f)(t) = \int f(t-r)\,\mu(dr)$ for $\mu \in M_+(\mathbb{R})$, $f \in X$. Then $\|\mu * f\|_1 \le \|\mu\|\,\|f\|_1$, and dilation is compatible with convolution:

$$D_\sigma(\mu * f) = (D_\sigma \mu) * (D_\sigma f).$$

For $\mu \in M_+$ supported on $[0,\infty)$ the Laplace transform is

$$\hat{\mu}(s) = \int_{[0,\infty)} e^{-st}\,\mu(dt), \qquad s \ge 0.$$

If $\mu \neq 0$ then $\hat{\mu}(s) > 0$ for all $s \ge 0$.

Definition 3.1 (completely monotone)**.** A function $f : (0,\infty) \to \mathbb{R}$ is *completely monotone* (CM) if $f \in C^\infty$ and $(-1)^n f^{(n)} \ge 0$ for all $n \ge 0$. ▷ [**T**] proved-target

Definition 3.2 (Bernstein function)**.** A function $g : (0,\infty) \to [0,\infty)$ is a *Bernstein function* if $g \in C^\infty$ and g' is completely monotone. We write $g \in \mathrm{BF}$, and $g \in \mathrm{BF}_0$ if moreover $g(0+) = 0$. ▷ [**T**] proved-target

Proposition 3.3 (the Bernstein-function toolbox)**.**

1. (Bernstein–Widder / Feller criterion.) f is the Laplace transform of a probability measure on $[0,\infty)$ if and only if f is completely monotone and $f(0+) = 1$. More generally, f is completely monotone if and only if $f(s) = \int_{[0,\infty)} e^{-st}\,F(dt)$ for a measure F on $[0,\infty)$, not necessarily finite, and F is then unique.

2. For $g : (0,\infty) \to [0,\infty)$ the following are equivalent: $g \in \mathrm{BF}$; the composition $h \circ g$ is completely monotone for every completely monotone h; and $e^{-\tau g}$ is completely monotone for every $\tau > 0$.

3. Every $g \in \mathrm{BF}$ is nonnegative, nondecreasing and concave, and has the Lévy–Khintchine representation

$$g(s) = a + bs + \int_0^\infty \left(1 - e^{-st}\right)\nu(dt), \qquad a, b \ge 0, \quad \int_0^\infty (1 \wedge t)\,\nu(dt) < \infty,$$

with $g(0+) = a$. The triple (a, b, ν) is unique.

4. BF is a convex cone, closed under pointwise limits: if $g_n \in \mathrm{BF}$ and $g_n(s) \to g(s) < \infty$ for every $s > 0$, then $g \in \mathrm{BF}$. Consequently BF is closed under mixtures $s \mapsto \int g_u(s)\, m(du)$ with $m \geq 0$, whenever the integral is finite.

▷ [**A**] analytic interface ledger A1 ledger A2 ledger A3 ledger A4 *(the classical Bernstein-function toolbox:* `@feller2009introduction`*, Ch. XIII and* `@schilling2012bernstein`*, Ch. 1 and 3. Cited, not reproven here.* ***Assignment.*** *The four parts are carried by four separate entries, one each. Part (1) is ledger A1 entire, in both the probability form and the general measure form with uniqueness. Part (2) is ledger A2 entire — all three equivalences of the cited theorem, not only the exponential one: the composition clause is what A.4 inverts a Bernstein function with. Part (3) is ledger A3 entire, including the uniqueness of the triple, which Lemma 7.1 needs and which a representation theorem without it would not supply. Part (3) is also the* bridge *between the two vocabularies this project speaks: it is what identifies 3.2 with 3.7, and therefore the single edge every Lean tag in chapters 5–7 crosses, since the development names admissibility by the representation and never by derivative signs. A node whose conclusion is stated in* BF_0 *cannot carry a Lean tag without spending this part; a node whose conclusion is stated in* LE *carries one for free. Part (4) is ledger A4 entire; it is the single external fact Theorem 5.2 rests on, and the null-array argument that produces the pointwise limit there is ours and is held [**T**]. None of the four carries anything about the hemigroup family: they are statements about* CM *and* BF *as classes of functions, and every step joining them to* $\Phi_{x,y}$ *is [**T**].)*

Which vocabulary does the work. 3.3 is used as a bridge and not as a toolkit. The object the arguments manipulate is the *representation*, part (3) — a drift and a Lévy measure — and almost every proof in Chapters 4–9 and the appendix works with it directly: the increments of 5.2 are produced as representations, not as pointwise limits of Bernstein functions; the potential kernel of A.4 is constructed as a subordinator's potential measure, not represented from complete monotonicity; the Mellin calculus of Chapter 9 mentions neither class. CM and the derivative-sign definition of BF are needed at exactly three places, and it is worth naming them: 7.12, where $B(s) = sF'(s)$ has no meaning until the representation is in hand; A.7(2), together with the draft's Thorin-subclass proposition (§13, not transcribed here), where CM is the *subject* rather than a tool; and 3.3 itself, which identifies the two vocabularies once so that no later proof has to. A reader who wants the shortest path to 7.3 or 9.8 may take (3) as the definition of admissibility and never invoke Bernstein–Widder. *(This is also the development's design: it defines admissibility by the representation and has no* CM *at all — see* `DESIGN-formalization-strategy.md`*. The paragraph records that the choice is available on paper too, and what it costs, which is nothing outside those three places.)*

Lemma 3.4 (vanishing lemma)**.** If $g \in \mathrm{BF}_0$ and $g(s_0) = 0$ for some $s_0 > 0$, then $g \equiv 0$. ▷ [**T**] proved-target *(the Lean development states this for the* representation $g(s) = b_0 s + \int (1 - e^{-st})\, \nu(dt)$ *rather than for the derivative-sign definition 3.2, following the formalisation strategy; the two notions agree by the Lévy–Khintchine theorem, which is A3 and reaches this node through 3.3(3). From the representation the proof is shorter than the one below — a sum of nonnegative terms vanishes only when each does — and needs neither concavity nor complete monotonicity.* `#print axioms` *reduces to Lean core only.)*

Proof. g is nonnegative and concave by 3.3(3), with $g(0+) = g(s_0) = 0$. A nonnegative concave function vanishing at both ends of $[0, s_0]$ vanishes on it, so $g \equiv 0$ on $(0, s_0]$ and hence $g' \equiv 0$ on $(0, s_0)$. Now g' is nonincreasing, being the derivative of a concave function, and nonnegative, being completely monotone; a nonnegative nonincreasing function that vanishes on $(0, s_0)$ vanishes identically. So $g' \equiv 0$ on $(0, \infty)$, and with $g(0+) = 0$ this gives $g \equiv 0$. □

The next two statements are not in the draft, which uses both facts without stating either. They are recorded here so that the trust they carry is visible in the dependency graph rather than buried inside proofs. Both are classical, and both are cited to the same section of `@feller2009introduction`; note that that section states the probability and the measure forms as four separate theorems, on four different pages, so the pair a given proof needs must be named individually.

Proposition 3.5 (continuity theorem for Laplace transforms)**.** Let μ_n, μ be probability measures on $[0, \infty)$. Then $\mu_n \to \mu$ weakly if and only if $\hat{\mu}_n(s) \to \hat{\mu}(s)$ for every $s > 0$. ▷ [**A**]
analytic interface ledger A5 *(Feller's continuity theorem; `@feller2009introduction`, Vol. 2, §XIII.1, Theorem 2 and Theorem 2a. A cited interface, not reproven here.* ***Assignment.*** *ledger A5 carries both clauses. The boundedness hypothesis in the second is not decoration: Feller gives a counterexample on the same page showing it cannot be dropped, and any proof here that invokes the measure form must discharge it explicitly. The probability form carries it for free, which is why Theorem 7.3 uses that one. What ledger A5 does not carry is the tightness argument that supplies weak convergence in the first place; that is [**T**].* ***Not needed in Lean*** *(resolved 2026-08-09): the development proves the form it uses rather than citing it, as `Hemigroup.tendsto_integral_of_tendsto_laplace`, whose `#print axioms` reduces to Lean core. For* causal probability *measures, convergence of the transforms together with tightness gives weak convergence; that is precisely what Theorem 7.3($\Leftarrow$) uses, and the boundedness hypothesis flagged above is carried for free by the probability form, so the trap is designed out rather than remembered. The general measure form is deliberately* not *formalised. Tightness is ours, as this clause always claimed and now checks: the Markov bound $\mu(t > T)(1 - e^{-sT}) \leq 1 - \hat{\mu}(s)$ (`measureReal_Ioi_mul_le`), uniform over the family as $\mu_{a,b}(t > T) \leq F(Bs)/(1 - e^{-sT})$ (`kernel_tail_le`), with $F(0+) = 0$ (`exists_exponent_lt`) making s choosable. The node keeps [**A**] because that is the* paper's *stance, and carries no Lean tag because what is proved is one direction of the probability case, not the statement above.)*

Proposition 3.6 (uniqueness of the Laplace transform)**.** A measure on $[0, \infty)$ is determined by its Laplace transform: if $\hat{\mu}(s) = \hat{\rho}(s)$ for all s in some interval (a, ∞), then $\mu = \rho$. ▷ [**A**]
analytic interface ledger A6 *(Feller's uniqueness theorem; `@feller2009introduction`, Vol. 2, §XIII.1, Theorem 1 and Theorem 1a. A cited interface, not reproven here.* ***Assignment.*** *ledger A6 carries the statement entire, for probability measures (Theorem 1) and for arbitrary measures (Theorem 1a). This is what licenses every step in the article of the form "the transforms agree, therefore the measures agree" — the uniqueness clause of Lemma 4.1, and Lemma 6.1, Proposition 9.2, Theorem 9.5 and Lemma 10.3(5). It says nothing about existence, and nothing about which functions are transforms; that is 3.3(1).* ***Not needed in Lean.*** *The development proves this rather than citing it: `Hemigroup.laplace_injective`, by the classical substitution $x = e^{-t}$, which carries a causal measure onto $[0, 1]$, turns $\hat{\mu}(n)$ into the nth moment of the image, and closes with the Weierstrass approximation theorem. `#print axioms` reduces to Lean core, so A6 never enters `trust-boundary.txt` and nothing proved in Lean rests on it. The node keeps [**A**] because that is the* paper's *stance, and it carries no Lean tag because what is proved is weaker than what is stated: agreement on all of $[0, \infty)$, not on a tail (a, ∞). That gap is not a formality — equality of the moments from some n_0 onwards does not by itself pin down an atom at the origin of the transported measure, and the argument that ours has none is not formalised.)*

The remaining four statements are additive too, and they are the *formalisation's* vocabulary rather than the draft's. 3.7 names the object the Lean development is written in; the three after it are the restricted cases of 3.5 and 3.6 that it proves outright, recorded as [**T**] nodes beneath the [**A**] ones they refine so that the graph shows what actually rests on ledger A5 and ledger A6. Nothing formalised does.

Definition 3.7 (Lévy exponent). A function $g : [0,\infty) \to [0,\infty]$ is a *Lévy exponent* if

$$g(s) = b_0\, s + \int_0^\infty \left(1 - e^{-st}\right) \nu(dt), \qquad b_0 \geq 0,$$

for a drift b_0 and a positive Borel measure ν on $(0,\infty)$ — with no killing term, so that $g(0) = 0$. We write $g \in \mathrm{LE}$, and call (b_0, ν) its *triple*. ▷ [**T**] proved-target *(A* name *for 3.3(3)'s right-hand side, not a new fact: by that part,* $\mathrm{LE} = \mathrm{BF}_0$ *as classes of functions, with the triple unique. Both names are kept because the two documents need different ones. The paper argues in* BF_0*, where the draft's proofs live; the Lean development argues in* LE *and never defines 3.1 at all — the reason is `blueprint/DESIGN-formalization-strategy.md`, and the payoff is that the Lévy–Khintchine theorem (ledger A3) is crossed once, here, instead of inside every proof. Stated in Lean as `Hemigroup.levyExponent`, valued in* $[0,\infty]$ *so that no integrability hypothesis is carried; the integrability condition* $\int (1 \wedge t)\, \nu(dt) < \infty$ *of 3.3(3) is equivalent, for causal* ν*, to finiteness of* g *at a single* $s > 0$*, and appears in the development as that hypothesis instead. 3.4 and 7.11 are both proved directly from this definition (`levyExponent_eq_zero_of_eq_zero`, `levyExponentD_increment`).)*

Proposition 3.8 (uniqueness of the Laplace transform, causal case). Let μ, ρ be finite measures on $\mathbb{R}$ carried by $[0,\infty)$. If $\hat{\mu}(s) = \hat{\rho}(s)$ for every $s \geq 0$, then $\mu = \rho$. ▷ [**T**] proved-target *(the case of 3.6 that this article's own arguments use, and it is* proved*: `Hemigroup.laplace_injective`, `#print axioms` reducing to Lean core. It is weaker than the cited statement in exactly one respect — agreement is assumed on all of* $[0,\infty)$*, not on a tail* (a,∞) *— and that gap is not a formality, which is why 3.6 keeps its ledger A6 tag and this node does not replace it. Every step in the formalised part of the article of the form "the transforms agree, therefore the measures agree" has both measures carried by* $[0,\infty)$ *and both transforms known on all of it, so it cites this node and not that one; the consequence is that ledger A6 does not appear in `blueprint/trust-boundary.txt`.)*

Extended 2026-08-11 to measures that are not finite*, which is what the appendix needs:* $\kappa^{(x)}$*,* $\ell^{(x)}$ *and Lebesgue measure are none of them finite, and A.5 compares them. That is 3.9 below. The finite case remains the one almost every caller uses, so it keeps this node and its tag.*

Proof. Substitute $x = e^{-t}$: the map is a measurable embedding carrying a measure on $[0,\infty)$ to one on $(0,1]$, and it turns $\hat{\mu}(n)$ into the nth moment of the image, for every $n \in \mathbb{N}$. Two finite measures on the compact $[0,1]$ with the same moments agree, by the Weierstrass approximation theorem. Transporting back along the embedding loses nothing, so $\mu = \rho$. ☐

Proposition 3.9 (uniqueness without a finiteness assumption). Let μ, ρ be measures on $\mathbb{R}$ carried by $[0,\infty)$, not assumed finite, and let $s_0 \in \mathbb{R}$ be such that $\hat{\mu}(s_0) < \infty$. If $\hat{\mu}(s) = \hat{\rho}(s)$ for every $s \geq s_0$, then $\mu = \rho$. ▷ [**T**] proved-target *(**Proved**, as `Hemigroup.laplaceL_injective_of_ne_top`, reducing to Lean core — so ledger A6 stays off the trust boundary in its general form as well as its restricted one. The hypothesis is worth reading carefully, because it is weaker than the one the chapter that needs this was expected to supply: what is assumed is not that the measures are locally finite but that* the transform converges at a single point*. Local finiteness is then a consequence, 3.11, not an assumption.*

The proof is the classical damping trick and nothing more. $e^{-s_0 t}\mu(dt)$ *is finite precisely because the transform converges at* s_0*; its transform at* s *is* $\hat{\mu}(s_0 + s)$*, so 3.8 applies to the damped pair; and the damping is undone by multiplying the density back, which is legitimate because* $e^{-s_0 t}$ *is everywhere positive and finite. Nothing here needs the tail-agreement form of ledger A6 that 3.8 declines to claim.)*

Proof. The measure $\mu_0(dt) := e^{-s_0 t}\,\mu(dt)$ is finite, its mass being $\hat{\mu}(s_0) < \infty$, and is carried by $[0,\infty)$, being absolutely continuous with respect to μ; likewise ρ_0, whose mass is $\hat{\rho}(s_0) = \hat{\mu}(s_0)$. For $s \geq 0$, $\hat{\mu_0}(s) = \hat{\mu}(s_0+s) = \hat{\rho}(s_0+s) = \hat{\rho_0}(s)$, so $\mu_0 = \rho_0$ by 3.8. Since $e^{-s_0 t}\,e^{s_0 t} = 1$ pointwise, multiplying both by the density $e^{s_0 t}$ recovers $\mu = \rho$. ☐

Corollary 3.10 (uniqueness for signed densities)**.** Let V be a measurable real function on $\mathbb{R}$ vanishing a.e. on $(-\infty, 0)$, and let $s_0 \in \mathbb{R}$ be such that $\int_0^\infty e^{-s_0 t}\,|V(t)|\,dt < \infty$. If $\int_0^\infty e^{-st}\,V(t)\,dt = 0$ for every $s \geq s_0$, then $V = 0$ a.e. ▷ **[T]** proved-target *(The Jordan-decomposition reduction to 3.9, and nothing more; recorded so that the signed applications named in the header comment rest on a stated node rather than on shorthand.)*

Proof. Write $V = V_+ - V_-$. The measures $V_+(t)\,dt$ and $V_-(t)\,dt$ are carried by $[0,\infty)$, their transforms are finite at s_0 — each is bounded by $\int e^{-s_0 t}|V|\,dt$ — and the hypothesis makes them equal for every $s \geq s_0$. By 3.9 the two measures coincide, so $V_+ = V_-$ a.e., i.e. $V = 0$ a.e. ☐

Lemma 3.11 (convergence forces local finiteness)**.** Let μ be carried by $[0,\infty)$ with $\hat{\mu}(s_0) < \infty$ for some $s_0 > 0$. Then $\mu([0,T]) < \infty$ for every T. ▷ **[T]** proved-target *(**Proved**, as `Hemigroup.measure_Icc_ne_top_of_laplaceL_ne_top`. Stated because it is what makes the hypothesis of 3.9 the natural one: "locally finite" would have been an extra assumption, and it is instead a consequence.)*

Proof. On $[0,T]$ the integrand satisfies $1 \leq e^{s_0 T} e^{-s_0 t}$, so $\mu([0,T]) \leq e^{s_0 T}\hat{\mu}(s_0) < \infty$. ☐

Proposition 3.12 (continuity theorem, causal probability case)**.** Let μ_n, μ be probability measures on $\mathbb{R}$ carried by $[0,\infty)$, and suppose

1. $\hat{\mu}_n(s) \to \hat{\mu}(s)$ for every $s \geq 0$; and
2. the family is *tight*: for every $\eta > 0$ there is a T with $\mu_n\big((T,\infty)\big) \leq \eta$ for all n, and $\mu\big((T,\infty)\big) \leq \eta$.

Then $\mu_n \to \mu$ weakly. ▷ **[T]** proved-target *(the direction and the case of 3.5 that 7.8 uses — transforms to measures, probability form — and it is* proved*: `Hemigroup.tendsto_integral_of_tendsto_laplace`, `#print axioms` reducing to Lean core, so ledger A5 does not appear in `blueprint/trust-boundary.txt` either. Tightness is a* hypothesis *here rather than a conclusion, which is the honest shape: 3.5's assignment clause always held the tightness argument **[T]**, and 3.13 is where it is discharged. Feller's boundedness hypothesis, the classical trap of the general form, is carried for free by the probability form and so cannot be forgotten. The general measure form is deliberately not formalised.)*

Proof. Push forward along $x = e^{-t}$ onto the compact $[0,1]$, as in 3.8. There, convergence of the Laplace transforms at the naturals *is* convergence of the moments, so the images converge weakly by Weierstrass; no compactness argument is needed, because the carrier is already compact.

Transporting back is where tightness is spent. A bounded continuous f on $[0,\infty)$ pulls back to $f \circ (-\log)$, which is unbounded at the origin, so it is replaced by $f \circ \big(\min(-\log, T)\big)$, bounded and continuous on $[0,1]$ and agreeing with it off (T,∞); the two integrals differ by at most $2\|f\|_\infty\,\mu_n\big((T,\infty)\big)$, which hypothesis (2) makes uniformly small. An $\varepsilon/3$ argument concludes. ☐

Lemma 3.13 (tail bound from the transform)**.** Let μ be a probability measure on $\mathbb{R}$ carried by $[0,\infty)$. Then for all $s \geq 0$ and all T,

$$\mu\big((T,\infty)\big)\,\big(1 - e^{-sT}\big) \;\leq\; 1 - \hat{\mu}(s).$$

Consequently a family of such measures whose transforms satisfy $\sup_n\big(1-\hat\mu_n(s)\big) \to 0$ as $s \to 0+$ is tight. ▷ [**T**] proved-target *(this is the tightness argument 3.5 always claimed as ours, now stated rather than left inside a proof.* `Hemigroup.measureReal_Ioi_mul_le`*,* `#print axioms` *reducing to Lean core; the divided form is* `measureReal_Ioi_le_div`*. No hypothesis on T is needed — causality supplies the sign that $T \geq 0$ would have. Its use in 7.8 is* `kernel_tail_le`*, which makes the bound uniform over the family as $\mu_{a,b}\big((T,\infty)\big) \leq F(Bs)/(1-e^{-sT})$ for $b \leq B$, and what makes that bite is $F(0+) = 0$;* `isTightMeasureSet_kernel` *repackages it in Mathlib's own vocabulary, which nothing here now consumes but any other consumer would.* The consequential sentence, for an arbitrary family rather than only the constructed kernels, *is* `Hemigroup.exists_tail_le_of_forall_laplace` *(the tail bound, in the "for every η" form* `tendsto_integral_of_tendsto_laplace` *consumes) and* `Hemigroup.isTightMeasureSet_of_forall_laplace` *(the same, in Mathlib's* `IsTightMeasureSet` *vocabulary) — same s-then-T argument, read off the hypothesis $\forall\varepsilon > 0, \exists s > 0, \forall i, 1-\hat\mu_i(s) \leq \varepsilon$ directly instead of through $F(Bs)$. Both reduce to Lean core; the kernel-family lines above are not re-derived from them, since doing so would just restate* `exists_kernel_tail_le`*'s own proof of the hypothesis rather than shorten it.)*

Proof. Pointwise: $1-\hat\mu(s) = \int\big(1-e^{-st}\big)\,\mu(dt)$, the integrand is at least $1-e^{-sT}$ on (T,∞), and it is nonnegative elsewhere because μ is carried by $[0,\infty)$ and $s \geq 0$. Dropping the complement gives the bound.

For the consequence, fix $\eta > 0$. Choose $s > 0$ with $\sup_n\big(1-\hat\mu_n(s)\big) \leq \eta/2$, then T large enough that $1-e^{-sT} \geq 1/2$; the displayed bound gives $\mu_n\big((T,\infty)\big) \leq \eta$ for every n. □

Chapter 4

Convolution representation

One device serves both lemmas. The exponential e^{-st} is unbounded on $\mathbb{R}$, so it is not a functional on X; the *clamped exponential* $w_s(t) := e^{-s\max(t,0)}$ is bounded, hence in X^*, and agrees with e^{-st} exactly where causal mass lives. For a measure μ carried by $[0,\infty)$ and $f = 1_{(0,1)}$, one has $\langle w_s, f(\cdot - r)\rangle = e^{-sr}\langle w_s, f\rangle$ for $r \geq 0$, so Fubini collapses the pairing to

$$\langle w_s, \mu * f\rangle = \langle w_s, f\rangle\,\hat{\mu}(s), \qquad \langle w_s, f\rangle \geq e^{-s} > 0: \tag{4.1}$$

a single bounded functional reads off the Laplace transform of the kernel.

Lemma 4.1 (representation)**.** Under (A1)–(A5), for each pair $0 \leq x \leq y$ there is a unique probability measure $\mu_{x,y}$ on $[0,\infty)$ with

$$\Phi_{x,y} f = \mu_{x,y} * f, \qquad f \in X,$$

and conversely every such measure defines an operator satisfying (A1)–(A5). ▷ **[T]** proved-target *(Machine-checked, `#print axioms` reducing to Lean core. The existence and uniqueness clauses are `CascadeCore.existsUnique_repr`; the converse is `mconvL1_satisfies_axioms`, whose four conjuncts are (A2)–(A5), (A1) being carried by the type. The representing measure is named `CascadeCore.repr` so that later chapters can write $\mu_{x,y}$ without re-extracting it. Note the namespace: this chapter needs only (A1)–(A5), so it is stated for the covariance-free core of 2.1 and applies verbatim to every covariance stratum.)*

Proof. Fix (x,y) with $0 \leq x \leq y$ and write $\Phi = \Phi_{x,y}$. The plan is classical: the representing measure ought to be Φ applied to a unit point mass, but δ_0 is not in L^1, so Φ is applied to an approximate identity instead, a limit measure is extracted by compactness, and the limit is then checked to represent Φ; the named steps follow that plan.

Convolution as a vector-valued integral. For $f \in L^1$ and $g \in X$,

$$f * g = \int_{\mathbb{R}} f(r)\,\tau_r g\,dr$$

as a Bochner integral *in* X: the integrand is continuous in r, because translation acts continuously on L^1, and is dominated by $|f(r)|\,\|g\|_1$. An X-valued integral commutes with every bounded operator, so (A1) and (A2) give the interchange identity

$$\Phi(f * g) = f * \Phi g, \qquad f \in L^1,\ g \in X,$$

which carries the existence argument; the remaining axioms enter where named.

The approximants. Let $\rho_\varepsilon = \varepsilon^{-1} 1_{(0,\varepsilon)}$ and $h_\varepsilon := \Phi\rho_\varepsilon$. By (A4) and (A5), $h_\varepsilon \geq 0$ with $\int h_\varepsilon = 1$; by (A3) it vanishes a.e. on $(-\infty, 0)$. So $\nu_\varepsilon := h_\varepsilon\,dt$ is a probability measure carried by $[0,\infty)$.

Tightness. By the interchange identity $\rho_1 * h_\varepsilon = \Phi(\rho_1 * \rho_\varepsilon)$, and $\rho_1 * \rho_\varepsilon \to \rho_1$ in L^1, so $\rho_1 * h_\varepsilon$ converges in X as $\varepsilon \downarrow 0$ — and a convergent sequence in L^1 has uniformly small tails. Since $\rho_1 \geq 0$ is carried by $(0,1)$ and $h_\varepsilon \geq 0$, convolving can only move mass to the right:

$$\int_T^\infty h_\varepsilon \leq \int_T^\infty \rho_1 * h_\varepsilon.$$

Along $\varepsilon_n = 1/(n+1)$ the right tails of ν_{ε_n} are therefore small uniformly in n, and (A3) kills the left tail outright, so $\{\nu_{\varepsilon_n}\}$ is tight, with compacts $[0,T]$.

The limit. By Prokhorov's theorem some subsequence $\nu_{\varepsilon_{n_k}}$ converges weakly to a probability measure μ; and $(-\infty,0)$ being open, the portmanteau theorem bounds $\mu((-\infty,0))$ by the lower limit of $\nu_{\varepsilon_{n_k}}((-\infty,0)) = 0$, so $\operatorname{supp}\mu \subseteq [0,\infty)$. Both the mass and the support clause are thus part of the extraction rather than separate arguments.

Identification. Let $\Psi \in X^*$ and $f \in X$. The map $r \mapsto \Psi(\tau_r f)$ is continuous — again because translation acts continuously on L^1 — and bounded by $\|\Psi\|\,\|f\|_1$, so it is a legitimate test function for weak convergence. Reading $f * h_\varepsilon$ with h_ε in the density slot and applying Ψ,

$$\Psi(f * h_\varepsilon) = \int_{\mathbb{R}} \Psi(\tau_r f)\,\nu_\varepsilon(dr) \xrightarrow{k\to\infty} \int_{\mathbb{R}} \Psi(\tau_r f)\,\mu(dr) = \Psi(\mu * f),$$

the last equality because $\mu * f = \int \tau_r f\,\mu(dr)$ is again an X-valued Bochner integral. On the other hand $f * h_\varepsilon = \Phi(f * \rho_\varepsilon) \to \Phi f$ in X, by the interchange identity and (A1). Hence $\Psi(\Phi f) = \Psi(\mu * f)$ for every $\Psi \in X^*$, and so $\Phi f = \mu * f$. Note that this holds for *every* $f \in X$ at once: nothing in the display is pointwise, so no restriction to C_c and no density step is needed.

Uniqueness. Take $f = 1_{(0,1)}$ and pair $\mu * f$ against the clamped exponential $w_s(t) = e^{-s\max(t,0)}$: by the factorisation 4.1, $\langle w_s, \mu * f\rangle = \langle w_s, f\rangle\,\hat\mu(s)$ with $\langle w_s, f\rangle > 0$. So Φ determines $\hat\mu$ on $[0,\infty)$, and a finite measure on $[0,\infty)$ is determined by its Laplace transform (3.6). One test function suffices; no separating family is needed.

Converse. A probability measure μ on $[0,\infty)$ makes $f \mapsto \mu * f$ bounded with norm ≤ 1 by Young's inequality (A1), translation-commuting because convolution is (A2), support-preserving on $[t_0,\infty)$ because $\operatorname{supp}\mu \subseteq [0,\infty)$ (A3), positive (A4), and mass-preserving by Tonelli (A5). □

The rest of the paper works not with the measures but with one function of them, their *exponent* $g_{x,y}(s) := -\log\hat\mu_{x,y}(s)$. The reason is the cascade: Laplace transforms turn convolution into multiplication, so $-\log$ turns the composition of two smoothings into the *sum* of their exponents, and (A6) will read as additivity in Chapter 5. That chapter's argument also needs the exponent to depend continuously on the pair of scales, so that a fine partition of $[x,y]$ gives a null array; this is what the second lemma supplies, from (A7).

Lemma 4.2 (continuity of transforms)**.** Under (A1)–(A5) and (A7), the function

$$g_{x,y}(s) := -\log\hat\mu_{x,y}(s) \in [0,\infty), \qquad s \geq 0,$$

is well defined, and $(x,y) \mapsto g_{x,y}(s)$ is continuous on $\{0 \leq x \leq y\}$ for each $s \geq 0$; moreover $g_{x,y}(0) = 0$ and $s \mapsto g_{x,y}(s)$ is continuous on $[0,\infty)$. ▷ **[T]** proved-target

(Machine-checked, `#print axioms` reducing to Lean core. The four clauses are the four conjuncts of `CascadeCore.transform_continuity`; the exponent itself is `CascadeCore.exponent`. The domain restrictions are explicit there — both continuity clauses are `ContinuousOn`, on $\{0 \leq x \leq y\}$ and on $[0,\infty)$ respectively — and $g_{x,y}(0) = 0$ is stated as the value at 0, which is what the Lean proves, since $\hat\mu(0) = 1$ holds on the nose for a probability measure.)

Proof. $\hat\mu(s) \in (0,1]$ for a probability measure on $[0,\infty)$ and $s \geq 0$: positive because the integrand is, and at most 1 because $e^{-st} \leq 1$ where the mass sits. So g is well defined and nonnegative, and $g_{x,y}(0) = -\log 1 = 0$. Continuity of $s \mapsto \hat{\mu}(s)$ on $[0, \infty)$ is dominated convergence with the constant bound 1 on the integrand. This is where causality is needed: e^{-st} is bounded on the support of μ only because that support lies in $[0, \infty)$.

For continuity in (x, y), take $f = 1_{(0,1)}$ and pair against the clamped exponential w_s; by 4.1,

$$\langle w_s, \Phi_{x,y} f\rangle = \langle w_s, \mu_{x,y} * f\rangle = \langle w_s, f\rangle\, \hat{\mu}_{x,y}(s).$$

The left side is a bounded functional applied to $\Phi_{x,y} f$, which is continuous in (x, y) on $\{0 \leq x \leq y\}$ by (A7); dividing by the constant $\langle w_s, f\rangle > 0$ gives continuity of $\hat{\mu}_{x,y}(s)$, and composing with $-\log$, legitimate because the transform never vanishes, gives it for $g_{x,y}(s)$. □

Remark 4.3 (what the representation is, and what it is not)**.** 4.1 is the Wendel step (`@wendel1952left`: the translation-invariant operators on L^1 of a group are exactly convolution by a measure), specialised to $\mathbb{R}$ and sharpened by (A3)–(A5) to a probability measure carried by $[0, \infty)$. It is proved here rather than cited; the only imported fact in this chapter is the uniqueness theorem for Laplace transforms, 3.6.

On the two routes. The classical argument regards $h_\varepsilon\, dt$ as a bounded family in $M(\mathbb{R}) = C_0(\mathbb{R})^*$ and extracts a weak-$*$ subnet by Banach–Alaoglu, recovering $\|\mu\| = 1$ from (A5) afterwards and reading the limit as a measure by Riesz–Markov. The proof given above instead makes the family *tight* — which costs one extra observation, that convolution with a density carried by $(0, 1)$ only moves mass to the right — and then Prokhorov returns a probability measure carried by $[0, \infty)$ outright. The mass and support clauses are then part of the extraction rather than separate arguments, no dual space appears, and the weak-$*$ selection apparatus — Banach–Alaoglu and Riesz–Markov — is bypassed. The second route is the one the formalisation follows, because it needs no dual space: the library has Banach–Alaoglu but not the identification of $M(\mathbb{R})$ with $C_0(\mathbb{R})^*$.

The identification step likewise has two forms. Testing against C_0 functions makes it pointwise, which forces the restriction to $f \in C_c$ and a density argument afterwards. Testing against $X^* = L^\infty$ makes it an identity in X, valid for every $f \in X$ at once, and the price is only that $f * g$ must be read as an X-valued Bochner integral — which the proof needs anyway, for the interchange identity $\Phi(f * g) = f * \Phi g$.

The multiplicative-group counterpart of this step, for the scale variable rather than time, is what Chapter 9 has to build by hand; there is no Wendel theorem to appeal to once the group is $(0, \infty)$ under multiplication and the operator is required only to be covariant.

Chapter 5

The cascade: additivity and infinite divisibility

Set

$$G(x,s) := g_{0,x}(s),$$

the exponent accumulated from scale 0 to scale x.

Lemma 5.1 (additivity)**.** Under (A1)–(A7), $\mu_{x,z} = \mu_{y,z} * \mu_{x,y}$ for $x \le y \le z$, hence

$$g_{x,z}(s) = g_{x,y}(s) + g_{y,z}(s), \qquad g_{x,x} = 0.$$

Consequently, with $G(x,s) := g_{0,x}(s)$,

$$g_{x,y}(s) = G(y,s) - G(x,s),$$

where $G(0,\cdot) = 0$, $G(\cdot,s)$ is nondecreasing and continuous for each s, and $G(x,0) = 0$. ▷ **[T]**
proved-target *(**Machine-checked**, `#print axioms` reducing to Lean core. The eight conjuncts of `CascadeCore.additivity` are the eight clauses above, in order, with G itself `CascadeCore.G`. No vocabulary question arises: every clause is an identity or an inequality between real numbers, so unlike 5.2 this one could be tagged as the draft states it. The clause $G(x,0) = 0$ is stated as the value of G at 0, which is what the last conjunct, `CascadeCore.G_atZero`, proves.*

Stated for the covariance-free core, so it holds in every stratum. That matters for the monotonicity clause in particular: `G_monotoneOn` is half of what a reading of the covariance-free stratum needs, the other half being 5.2.)

Proof. Immediate from (A6), 4.1 (under which composition of operators corresponds to convolution of measures), and 4.2. In detail: (A6) gives $\Phi_{y,z}\Phi_{x,y} = \Phi_{x,z}$, which by 4.1 reads $\mu_{y,z} * (\mu_{x,y} * f) = \mu_{x,z} * f$ for all $f \in X$, so $\mu_{x,z} = \mu_{y,z} * \mu_{x,y}$ by the uniqueness clause there. Taking Laplace transforms turns convolution into multiplication and hence $-\log$ into addition — legitimate because $\hat{\mu}(s) > 0$, which is the first clause of 4.2 — giving the additivity of g; $\Phi_{x,x} = \mathrm{Id}$ gives $\mu_{x,x} = \delta_0$ and so $g_{x,x} = 0$. Setting $x = 0$ defines G, and additivity gives $g_{x,y} = G(y,\cdot) - G(x,\cdot)$. Monotonicity in the first argument is the nonnegativity of $g_{x,y}$, continuity is 4.2, and $G(x,0) = 0$ is the corresponding clause there. □

The next step upgrades the increments from "exponents of probability measures" to Bernstein functions. In the semigroup case this is trivial, since there $\mu_\tau = \mu_{\tau/n}^{*n}$ exhibits every kernel as an n-fold convolution power and infinite divisibility is immediate. In the hemigroup case there is no such power: the increments over a partition need not be equal, or even comparable, and what replaces the argument is a null-array limit. On the half line it can be made elementary, and the proof below is the whole of it.

Theorem 5.2 (increments are Bernstein). Under (A1)–(A7), $g_{x,y} \in \mathrm{BF}_0$ for every $x \le y$. Equivalently, and this is the form the proof produces: $g_{x,y} \in \mathrm{LE}$, i.e. there are a drift $b_0(x,y) \ge 0$ and a Lévy measure $\nu_{x,y}$ on $(0,\infty)$ with

$$g_{x,y}(s) = b_0(x,y)\, s + \int_0^\infty \left(1 - e^{-st}\right) \nu_{x,y}(dt).$$

$\triangleright$ [**T**] proved-target *(the two readings are the same statement, by 3.3(3) — ledger A3 — and the second is stated because it is the only one a Lean tag can point at: the development defines admissibility by the representation and has no* CM*, hence no* BF_0*. Stating both here crosses ledger A3 once, in the statement, rather than inside every proof that consumes this theorem.*

The null-array estimate is `CascadeCore.tendsto_integral_partitionMeasure`: with Π_n the sum of the partition increments (`partitionMeasure`, a finite measure at every n), $\int(1 - e^{-st})\, d\Pi_n \to g_{x,y}(s)$ for every $s \ge 0$. Its three ingredients are 5.1, uniform continuity of $G(\cdot, s)$ on the compact $[x, y]$, and the inequality below — no limit theorem for triangular arrays. The Lean proof uses $0 \le u - (1 - e^{-u}) \le u^2$ rather than the $u^2/2$ written here: the sharper constant needs a derivative argument, u^2 follows from $e^u \ge 1 + u$ alone, and the factor multiplies a quantity that is going to zero.

The extraction of the limiting triple runs as follows. The masses $\Pi_n(\mathbb{R}) = n$ diverge — mass piles up at the origin, and that pile is *the drift — so the measures are weighted by $1 - e^{-t}$, which vanishes there at exactly the right rate. Under the change of variable $v = 1 - e^{-t}$ (`expTrans`), which carries $[0,\infty]$ onto the compact $[0, 1]$ with $\infty \mapsto 1$, the weighted measures have uniformly bounded mass and are carried by a fixed compact, so a weak limit needs no tightness argument beyond that. The two endpoints are the two degenerate parts of a triple: mass at 0 is drift, mass at 1 would be killing, and $g_{x,y}(0+) = 0$ excludes the latter.*

What has to be integrated against the weighted measure is then $k_s(v) = (1 - (1 - v)^s)/v$, `levyRatio`, which carries a removable singularity at $v = 0$ with value s — and that value is precisely why mass at the origin contributes $b_0 s$ rather than 0. Filling it in is `continuous_levyRatio`, a derivative computation: k_s is the difference quotient at 0 of $v \mapsto 1 - (1 - v)^s$, whose derivative there is s. Both it and the change-of-variable identity $k_s(1 - e^{-t})\,(1 - e^{-t}) = 1 - e^{-st}$ are machine-checked, as is the bounded form `levyRatioBdd` that weak convergence tests against. The weighted approximants themselves are `weightedPartition`, with the two facts the extraction needs: pairing k_s against ρ_n is pairing $1 - e^{-st}$ against Π_n, and the total mass of ρ_n is the $s = 1$ approximant — the latter for free, since $k_1 \equiv 1$, so the mass bound is the convergence already proved rather than a separate estimate.

Those two facts are exactly what Mathlib's Prokhorov theorem asks of a family of finite measures, with a constant *compact carrier, which also discharges its monotonicity side condition. What it returns is a cluster point rather than a limit, because Mathlib carries no metrizability instance for finite measures on $\mathbb{R}$ and so no subsequence can be extracted; that costs nothing here, since every quantity this proof reads off the limit is a continuous real function of the measure that already converges along the full sequence, and a convergent sequence has one cluster value. The limit ρ is then split at the three pieces of $[0, 1]$: the atom at 0 gives $b_0 = \rho(\{0\})$, because $k_s(0) = s$; the interior transports back through $v \mapsto -\log(1 - v)$ with density $1/v$ to give ν; and the atom at 1, the killing term, vanishes. That last step is the only genuinely analytic one, and it is short: $k_s \le 1$ on $[0, 1]$ for $s \le 1$ and $k_s \to \mathbf{1}_{\{1\}}$ pointwise as $s \downarrow 0$, so dominated convergence against the* finite *ρ reads $\rho(\{1\}) = \lim_{s\downarrow 0} g_{x,y}(s) = g_{x,y}(0) = 0$ by continuity and 4.2.*

***The proof below no longer spends ledger A4, and the change came from formalising it.** It used to close by observing that $g_{x,y}$ is a pointwise limit of Bernstein functions and invoking 3.3(4). The Lean proof cannot take that route — the development has no* CM *and hence no* BF_0 *to be closed — so it extracts the representation directly from the weak limit, and thereby spends no interface at all: its `#print axioms` output is Lean core. The extraction* *turned out to be no longer than the appeal it replaced, so the text now takes the same route. ledger A4 survives in the ledger for 3.3 itself and for 7.12, and this theorem — the key technical step of §§4–6 — is now free of it.)*

Proof. Fix $x < y$ and the uniform partition $t_i = x + \frac{i}{n}(y-x)$, $i = 0, \dots, n$. Write $\mu_i := \mu_{t_i,t_{i+1}}$ and $g_i := g_{t_i,t_{i+1}}$, so that by 5.1

$$g_{x,y}(s) = \sum_{i=0}^{n-1} g_i(s) \qquad \text{for every } n.$$

Fix $s > 0$. Since $G(\cdot, s)$ is continuous on the compact interval $[x, y]$ it is uniformly continuous there, so $m_n := \max_i g_i(s) \to 0$ as $n \to \infty$. Using $0 \le u - (1 - e^{-u}) \le u^2/2$ for $u \ge 0$,

$$\Big| g_{x,y}(s) - \sum_i \big(1 - e^{-g_i(s)}\big) \Big| \;\le\; \tfrac{1}{2} \sum_i g_i(s)^2 \;\le\; \tfrac{m_n}{2}\, g_{x,y}(s) \;\longrightarrow\; 0.$$

On the other hand, with $\Pi_n := \sum_i \mu_i \in M_+([0,\infty))$, a finite measure,

$$\sum_i \big(1 - e^{-g_i(s)}\big) = \sum_i \big(1 - \hat\mu_i(s)\big) = \int_0^\infty \big(1 - e^{-st}\big)\, \Pi_n(dt),$$

which is a Bernstein function of s vanishing at $0+$ — it is in Lévy–Khintchine form with finite Lévy measure Π_n and no drift or killing term.

It remains to pass to the limit, and the point is to extract the *representation* rather than to invoke closure of BF under pointwise limits. The total masses $\Pi_n([0,\infty)) = n$ diverge, because mass piles up at the origin — and that pile *is* the drift. So weight the measures by $1 - e^{-t}$, which vanishes at the origin at exactly the rate that compensates, and substitute $v = 1 - e^{-t}$, carrying $[0,\infty]$ onto the compact $[0,1]$ with $\infty \mapsto 1$. Weighting μ_i by $1 - e^{-t}$ and integrating gives $1 - \hat\mu_i(1)$, so the weighted measures have total mass

$$\sum_i \big(1 - \hat\mu_i(1)\big) \;=\; \sum_i \big(1 - e^{-g_i(1)}\big) \;\le\; \sum_i g_i(1) \;=\; g_{x,y}(1),$$

using $1 - e^{-u} \le u$ for $u \ge 0$ and 5.1 at $s = 1$; this bound is uniform in n. So the weighted measures live on the fixed compact $[0,1]$ with uniformly bounded mass, and a subsequence converges weakly, with no further tightness argument needed. Integrating against

$$k_s(v) \;=\; \frac{1 - (1-v)^s}{v},$$

which has a removable singularity at $v = 0$ with value s, gives

$$g_{x,y}(s) \;=\; b_0(x,y)\, s + \int_0^\infty \big(1 - e^{-st}\big)\, \nu_{x,y}(dt),$$

where the limit measure ρ on $[0,1]$ is read in three pieces. The atom at $v = 0$ gives the drift $b_0(x,y) := \rho(\{0\})$, because $k_s(0) = s$. On the interior $(0,1)$, the substitution $v \mapsto t = -\log(1-v)$ carries ρ back to a measure $\nu_{x,y}$ on $(0,\infty)$: the weight $1/v$ in k_s undoes the weighting by $1-e^{-t}$, and $1-(1-v)^s = 1-e^{-st}$, so $k_s(v)\, d\rho(v)$ on $(0,1)$ reads as $(1-e^{-st})\, d\nu_{x,y}(t)$ on $(0,\infty)$; finiteness of $\int_0^\infty (1 \wedge t)\, \nu_{x,y}(dt)$ follows from the finiteness of ρ, since $1 - e^{-t} \ge (1 - e^{-1})(1 \wedge t)$. The atom at $v = 1$ would be a killing term, and is excluded by $g_{x,y}(0) = 0$ (4.2). So $g_{x,y} \in \mathrm{BF}_0$ by the representation itself. The case $x = y$ is $g_{x,x} = 0 \in \mathrm{BF}_0$. □

Corollary 5.3 (strict monotonicity)**.** Assume in addition (ND). Then for every $s > 0$ and $x < y$,

$$G(y, s) - G(x, s) = g_{x,y}(s) > 0,$$

i.e. $G(\cdot, s)$ is strictly increasing for every $s > 0$. $\triangleright$

[T] proved-target *(no vocabulary question in the conclusion, which is a pointwise inequality, and none in the proof either: it runs 3.4, whose Lean form* `levyExponent_eq_zero_of_eq_zero` *is stated on* LE *already, so with 5.2 in representation form this followed with no further bridge and in one step — which was the reason for concluding that theorem in* LE *rather than in* BF_0*. What (ND) contributes is* `exists_exponent_ne_zero`*, "not identically zero", proved in 5.1's file; the vanishing lemma upgrades it to "nowhere zero on* $(0, \infty)$*". The Lean development also records the* G *form,* `G_strictMonoOn`*, which is what 6.2 and 7.10 will use.)*

Proof. By (ND), $\Phi_{x,y} \neq \mathrm{Id}$, so $\mu_{x,y} \neq \delta_0$ by the uniqueness clause of 4.1, and therefore $g_{x,y} \not\equiv 0$. By 5.2, $g_{x,y} \in \mathrm{BF}_0$; by 3.4 a member of BF_0 vanishing at a single point of $(0, \infty)$ vanishes identically, so $g_{x,y}(s) > 0$ for all $s > 0$. The displayed identity is 5.1. $\square$

Chapter 6

Scale covariance: the similarity form

Lemma 6.1 (Laplace form of covariance)**.** Under (A1)–(A5), the following are equivalent:

1. axiom (A8);
2. $D_\sigma\, \mu_{x,y} = \mu_{S_\sigma x,\, S_\sigma y}$ for all $0 \le x \le y$, $\sigma > 0$;
3. $g_{S_\sigma x,\, S_\sigma y}(s) = g_{x,y}(\sigma s)$ for all $0 \le x \le y$, $\sigma, s > 0$.

In particular, under (A8),

$$G(S_\sigma x,\ s) = G(x,\ \sigma s) \qquad \text{for all } x \ge 0,\ \sigma, s > 0. \tag{6.1}$$

▷ [**T**] proved-target *(1)⇒(2)∧(3) is* ***covariance_laplace****, and (2)⇒(1) is* ***isScaleCovariant_of_repr_map*** *(****Covariance.lean****, fidelity review R18); (2)⇔(3) needs no further declaration, being the transform identity $\widehat{D_\sigma\mu}(s) = \hat\mu(\sigma s)$ together with injectivity of the Laplace transform on causal probability measures and of $-\log$ on its positive values, both already in play in the proof below.*

Proof. The compatibility $D_\sigma(\mu * f) = (D_\sigma\mu) * (D_\sigma f)$ turns (A8) into the statement that $D_\sigma\mu_{x,y}$ and $\mu_{S_\sigma x, S_\sigma y}$ represent the same operator, and the uniqueness clause of 4.1 makes that an identity of measures; this is (1)⇔(2). For (2)⇔(3), $\widehat{D_\sigma\mu}(s) = \hat\mu(\sigma s)$, so taking $-\log$ gives $g_{S_\sigma x, S_\sigma y}(s) = g_{x,y}(\sigma s)$. Equation (6.1) is the case $x = 0$ of that identity together with $S_\sigma 0 = 0$, which holds because an increasing bijection of $[0, \infty)$ fixes 0; here $G(x, s) = g_{0,x}(s)$, which needs only 4.1. ☐

Lemma 6.2 (rigidity of the action)**.** Assume (A1)–(A8) and (ND). Then:

1. For each σ, S_σ is uniquely determined by (6.1).
2. $S_\sigma S_\tau = S_{\sigma\tau}$ for all $\sigma, \tau > 0$, and $S_1 = \mathrm{Id}$.
3. $\sigma \mapsto S_\sigma x$ is continuous and strictly increasing for each $x > 0$.
4. The action has no fixed point in $(0, \infty)$: if $S_\sigma x^* = x^*$ for all σ, then $x^* = 0$.

▷ [**T**] proved-target *(three of the four clauses are injectivity of $G(\cdot, s)$ — that is, 5.3 — and nothing else. (1)* is *that injectivity (****eq_of_G_eq****); (2) is two applications of (6.1) on top of it (****S_comp****,* ***S_one****); (4) adds only that $G(x, \cdot)$ is continuous at the origin with value 0, which turns "constant on $(0, \infty)$" into "identically zero" and so contradicts 5.3 above $x = 0$ (****eq_zero_of_fixed****).*

What (3) needed, and the development did not have, was strict monotonicity in the other *variable: the proof below reads it off the sign of a Bernstein function's derivative. The substitute is* ***laplace_strictAnti****, one line of measure theory — the support of $e^{-st} - e^{-s't}$ is exactly*

$\{t \neq 0\}$*, and that set carries positive mass precisely because the kernel is not* δ_0*, which is (ND). No derivative and no Bernstein class appears. With it, (3)'s continuity half does not need the inverse function the proof below invokes:* `StrictMonoOn.continuousAt_of_exists_between` *asks only for orbit points squeezed between a given bound and* $S_{\sigma_0}x$*, and continuity of* $\sigma \mapsto G(x, \sigma)$ *supplies them. That also avoids having to show the range of* $G(\cdot, s_0)$ *is a neighbourhood, which is the step the phrase "on its range" hides.)*

Proof. (1) Uniqueness. If S and S' both satisfy (6.1) then $G(S_\sigma x, s) = G(x, \sigma s) = G(S'_\sigma x, s)$ for every $s > 0$. By 5.3, $G(\cdot, s)$ is strictly increasing, hence injective, so $S_\sigma x = S'_\sigma x$.

(2) The group law. Applying (6.1) twice,

$$G(S_\sigma S_\tau x, s) = G(S_\tau x, \sigma s) = G(x, \sigma\tau s) = G(S_{\sigma\tau} x, s),$$

and injectivity of $G(\cdot, s)$ gives $S_\sigma S_\tau = S_{\sigma\tau}$. Taking $\sigma = 1$ in (6.1) gives $G(S_1 x, s) = G(x, s)$, whence $S_1 = \mathrm{Id}$.

(3) Continuity and strict monotonicity in σ*.* Fix $x > 0$ and any $s_0 > 0$. By (6.1), $G(S_\sigma x, s_0) = G(x, \sigma s_0) = g_{0,x}(\sigma s_0)$.

The map $\sigma \mapsto g_{0,x}(\sigma s_0)$ is continuous, and it is *strictly* increasing: $g_{0,x}$ is a nonzero member of BF_0 by 5.3, and a nonzero member of BF_0 is strictly increasing, because its derivative is completely monotone and not identically zero, hence strictly positive on $(0, \infty)$ — a completely monotone function vanishing at one point vanishes beyond it, and vanishing on an interval forces the representing measure, and so the function, to be zero.

The map $G(\cdot, s_0)$ is continuous (5.1) and strictly increasing (5.3), so it has a continuous strictly increasing inverse on its range. Composing, $S_\sigma x = G(\cdot, s_0)^{-1}\big(g_{0,x}(\sigma s_0)\big)$ is continuous and strictly increasing in σ.

(4) No interior fixed point. Suppose $S_\sigma x^* = x^*$ for all $\sigma > 0$. Then (6.1) gives $G(x^*, s) = G(x^*, \sigma s)$ for all $\sigma, s > 0$, so $G(x^*, \cdot)$ is constant on $(0, \infty)$; its value is $G(x^*, 0+) = 0$ by 5.1. Hence $g_{0,x^*} \equiv 0$, i.e. $\mu_{0,x^*} = \delta_0$ and $\Phi_{0,x^*} = \mathrm{Id}$. By (ND) this forces $x^* = 0$. $\square$

Proposition 6.3 (orbit coordinate; the gauge)**.** Assume (A1)–(A8) and (ND). Then $c(\sigma) := S_\sigma 1$ is a continuous strictly increasing bijection of $(0, \infty)$ onto $(0, \infty)$ with $c(1) = 1$. Consequently

$$\chi := c^{-1} : (0, \infty) \to (0, \infty), \qquad \chi(0) := 0,$$

is an increasing bijection — *the canonical gauge* — satisfying $\chi(S_\sigma x) = \sigma\, \chi(x)$, and

$$G(x, s) = F\big(\chi(x)\, s\big), \qquad F := G(1, \cdot) \in \mathrm{BF}_0,\ F \not\equiv 0. \tag{6.2}$$

Equivalently $F \in \mathrm{LE}$, with a drift $b_0 \geq 0$ and a Lévy measure ν on $(0, \infty)$. ▷
[**T**] proved-target *(the* LE *reading is the same statement, by 3.3(3), and is the one 7.9 needs: it is what feeds 7.1 and, through it, the form (7.1). F's membership comes straight from 5.2, so nothing new is asserted here — the class is carried along, not re-established.* ***Formalised****, and the interface question the formalisation note flagged (M5 of the ladder) is settled the way it warned it had to be:* `canonical_gauge` *asserts* χ *together with the conjugation* $\chi(S_\sigma x) = \sigma\, \chi(x)$ *and the similarity form* $G(x, s) = G(1, \chi(x)\, s)$*, not* χ *alone. That bundle is what 7.10 declined to guess at.*

Only surjectivity of c had to be proved; continuity and strict monotonicity are 6.2(3) and $c(1) = 1$ *is the group law, so the injectivity argument below — the iteration along* $\rho^{\pm n}$ *— is not needed in Lean at all. Surjectivity is done without limits. The orbit* $R = c((0, \infty))$ *is invariant under every* S_ρ*, which* is *the group law; were it bounded above, its supremum would be fixed by the action — nothing on the orbit can be carried above it, and nothing below it either, since* $S_{\rho^{-1}}$ *would then carry it down — and 6.2(4) forbids a fixed point above the origin, while* $1 \in R$*. Bounded below is the same argument read downwards, where a positive infimum is itself the contradiction. The intermediate value theorem then finishes.)*

Proof. *c is continuous* by 6.2(3), and $c(1) = S_1 1 = 1$ by 6.2(2).

c is injective. Suppose $c(\sigma) = c(\tau)$ with $\rho := \sigma/\tau \neq 1$. The group law 6.2(2) gives $S_\rho 1 = 1$, so by (6.1)

$$F(s) = G(1, s) = G(S_\rho 1, s) = G(1, \rho s) = F(\rho s) \qquad \text{for all } s > 0.$$

Iterating along $\rho^{\pm n}$ gives $F(s) = F(\rho^n s)$ for every $n \in \mathbb{Z}$. Replacing ρ by ρ^{-1} if necessary we may take $\rho < 1$, and then $\rho^n s \to 0$, so $F(s) = F(0+) = 0$ for every s by 5.1. Thus $F \equiv 0$, which contradicts 5.3 under (ND).

c is increasing. Being continuous and injective on the connected set $(0, \infty)$, c is strictly monotone, so it suffices to exclude the decreasing case. Suppose c were decreasing and fix $\sigma > 1$, so that $S_\sigma 1 < 1$. Since $G(\cdot, s)$ is strictly increasing (5.3),

$$G(S_\sigma 1, s) < G(1, s) = F(s) \qquad \text{for every } s > 0.$$

But (6.1) evaluates the left-hand side as $G(1, \sigma s) = F(\sigma s)$, and F is nondecreasing, being a Bernstein function, so $F(\sigma s) \geq F(s)$. The two are incompatible, so c is increasing.

c is onto $(0, \infty)$. Its limits as $\sigma \to 0+$ and $\sigma \to \infty$ exist in $[0, \infty]$ by monotonicity, and each is a fixed point of the action or a boundary point; by 6.2(4) there is no fixed point in $(0, \infty)$, so the limits are 0 and ∞. With continuity, c is a bijection onto $(0, \infty)$.

The gauge relation. By the group law, $S_\sigma x = S_\sigma S_{\chi(x)} 1 = S_{\sigma\chi(x)} 1$, so $\chi(S_\sigma x) = \sigma \chi(x)$.

The similarity form. For $x > 0$, writing $x = S_{\chi(x)} 1$ and using (6.1),

$$G(x, s) = G\big(S_{\chi(x)} 1,\, s\big) = G\big(1,\, \chi(x)\, s\big) = F\big(\chi(x)\, s\big),$$

and $F = g_{0,1} \in \mathrm{BF}_0$ by 5.2, with $F \not\equiv 0$ by (ND) through 5.3. The case $x = 0$ is $G(0, \cdot) = 0 = F(0)$. $\square$

Remark 6.4 (gauge freedom)**.** In the hemigroup setting the scale coordinate carries no intrinsic parametrization: replacing x by any increasing bijection of $[0, \infty)$ yields an equivalent family. 6.3 fixes the *canonical gauge* $\tilde{x} := \chi(x)$, in which S_σ acts by multiplication and $G(x, s) = F(\tilde{x}\, s)$. In it, scale carries the dimension of time: s is dual to time, and $\tilde{x}s$ must be dimensionless for $F(\tilde{x}s)$ to make sense. The canonical gauge is the working gauge of the rest of the paper.

The other homogeneous gauges are $c\,\tilde{x}^{\alpha}$, $c, \alpha > 0$ (the *α-gauge*). One of them recurs, for the Bessel family of Chapters 8–10: the *parabolic gauge* ξ, defined by

$$\tilde{x} = \xi^2,$$

under which that family's memory-line evolution is exactly the Bessel operator $\frac{1}{2}\,\partial_\xi^2 + \frac{\beta}{\xi}\,\partial_\xi$ — the heat-equation form; the factor $\frac{1}{2}$ arises from the substitution itself, as for Brownian motion at $a = \frac{1}{2}$. It is named at each use and never written $\tilde{x}$.

Chapter 7

Self-decomposability and the main theorem

By 6.3 the exponent of every kernel is an increment $F(bs) - F(as)$ with $0 \le a \le b$, and by 5.2 every such increment must lie in BF_0. That pins down F.

Lemma 7.1 (self-decomposable exponents)**.** For $F \in \mathrm{BF}_0$ the following are equivalent:

1. $F(b\,\cdot) - F(a\,\cdot) \in \mathrm{BF}_0$ for all $0 < a \le b$;
2. $B(s) := s\,F'(s)$ is a Bernstein function;
3. F has the representation

$$F(s) = b_0\,s + \int_0^\infty \left(1 - e^{-st}\right) \frac{k(t)}{t}\,dt, \qquad b_0 \ge 0, \tag{7.1}$$

with $k : (0,\infty) \to [0,\infty)$ nonincreasing, $\int_0^1 k < \infty$ and $\int_1^\infty k(t)/t\,dt < \infty$.

▷ [T] proved-target *(a collation, like 7.3 and for the same reason: its implications are proved by different means, at different cost, and one of them is formalised. The cycle is* (1) ⇒ (2) ⇒ (3) ⇒ (1)*, the first two steps being 7.12 and the last 7.11.* ***The asymmetry is the point.*** (3) ⇒ (1) *is elementary from the representation and is machine-checked;* (1) ⇒ (2) *and* (2) ⇒ (3) *rest on 3.3(4) and on the uniqueness clause of 3.3(3) — ledger A4 and ledger A3 — so formalising them means formalising those, which is what the trust boundary declines to do. This node is therefore not marked as proved in Lean; its expensive half is instead carried as ledger A18, which 7.9 spends and 7.8 does not. What a consumer needs is almost always one of the two halves, and citing the right one is what keeps the construction off ledger A3, ledger A4 and ledger A18 alike — a separation* ***`CIAxiomGuard.lean`*** *now checks per node.)*

Proof. (1) ⇒ (2) and (2) ⇒ (3) are 7.12; (3) ⇒ (1) is 7.11. The cycle gives the three equivalences. □

Remark 7.2 (the probabilistic reading)**.** Probabilistically, (7.1) together with the similarity structure below says that the kernels are the marginal laws of a *self-similar additive process*, a Sato subordinator: an increasing process $(T_{\tilde{x}})_{\tilde{x} \ge 0}$ with independent — but not stationary — increments, satisfying $T_{\sigma\tilde{x}} \overset{d}{=} \sigma\,T_{\tilde{x}}$.

Theorem 7.3 (Main theorem)**.** A family $(\Phi_{x,y})_{0 \le x \le y}$ satisfies (A1)–(A8) and (ND) **iff** there exist an increasing bijection χ of $[0,\infty)$ — the gauge — and a function F of the form (7.1) with $F \not\equiv 0$ such that

$$\Phi_{x,y} f = \mu_{x,y} * f, \qquad \hat{\mu}_{x,y}(s) = \exp\Big[-\big(F(\chi(y)\,s) - F(\chi(x)\,s)\big)\Big].$$

In particular, in the canonical gauge the scale-space kernels are

$$\phi_x(t) \;=\; \mathcal{L}^{-1}\Big[e^{-F(x\,s)}\Big](t), \qquad F \text{ as in (7.1)},$$

and the pair (χ, F) is unique up to the normalization $\chi(1) = 1$. ▷ **[T]** proved-target *(the variant of `@fagerstrom2005temporal`, Thm. 2, for the hemigroup axioms. Proved here, not cited: the 2005 theorem is the semigroup statement, recovered below as 7.4. **Formalisation state.** This node is a collation, and its three halves are 7.8, 7.9 and 7.10, where the formalisation state of each is recorded. All three are machine-checked, and since 2026-08-12 so is the collation itself, as `Hemigroup.SelfDecomposableExponent.main_characterization` — a statement with no content of its own beyond assembling the three. It was added because the dependency graph reported the article's main theorem as unproved while all of it was proved, which is the same gap 9.8 had.*

The halves remain where the trust boundary is read off, and the bundle cannot replace them. *(⇐) and the uniqueness clause rest on ledger A17, (⇒) on ledger A18, and neither borrows the other's — a separation `CIAxiomGuard.lean` checks per half and that a bundle necessarily destroys, since it depends on both. That is why the halves are stated separately (see the note at the head of this file), and adding the collation does not weaken the reason; it only stops the graph from understating what is done.*

One shape note. *7.8 is a* definition *in Lean — it exhibits the structure rather than asserting a property of it, which is the stronger reading. A conjunction needs a proposition, so (⇐) appears in the bundle as the existence statement that definition witnesses, with the witness being `cascadeFamily` itself.*

The round trip. *Since the fidelity review (finding R1) the (⇒) conjunct of the bundle is `Hemigroup.CascadeCore.main_analysis'`: it concludes in a gauge χ with $\chi(0) = 0$, $\chi(1) = 1$, strictly increasing and onto $[0, \infty)$, and an F of the form (7.1) with $F \not\equiv 0$ — exactly the type the (⇐) conjunct starts from — with $\mu_{x,y} = \mu^F_{\chi(x),\chi(y)}$ and $\Phi_{x,y} f = \mu^F_{\chi(x),\chi(y)} * f$. So the two directions of the **iff** meet in one type, and the uniqueness conjunct, which hypothesises $\chi(1) = 1$, is reachable from the analysis conjunct's conclusion.)*

Proof. (⇒) is 7.9, (⇐) is 7.8, and the uniqueness clause is 7.10. □

Corollary 7.4 (recovering the semigroup case)**.** Assume in addition that the family is one-parameter: $\Phi_{x,y}$ depends only on $y - x$. Then, after the normalization $g_{0,1}(1) = 1$,

$$F(s) = s^{\alpha} \quad \text{for some } 0 < \alpha \le 1,$$

i.e. the kernels are exactly the extremal stable densities of `@fagerstrom2005temporal`, Thm. 2, together with the boundary case $\alpha = 1$ — the pure delay $\mu_{0,x} = \delta_x$. ▷ **[T]** proved-target *(**Machine-checked**, `#print axioms` reducing to Lean core: the corollary spends no ledger entry, every input being chapters 4–6 or 5.2. The normalization $g_{0,1}(1) = 1$ is a* hypothesis *of the Lean statement (`hnorm : Fam.G 1 1 = 1`), not a reparametrisation performed — rescaling the scale axis achieves it and is a sentence in prose; and it does not compete with 7.3's $\chi(1) = 1$: since $g_{0,1}(s) = F(\chi(1)s)$, the two normalizations are one, and the one-parameter hypothesis is on the family* as given, *where $G(x, s) = x\, s^{\alpha}$ and the canonical gauge is $\chi(x) = x^{1/\alpha}$ (fidelity review R7). Two things the formalisation says that the proof below does not. **The multiplier of the action** is F. The proof introduces c with $S_\sigma x = c(\sigma)x$ and $c(\sigma)g(s) = g(\sigma s)$, and treats them as two unknown functions until Cauchy identifies both. Under the stated normalization they are the same function: (6.1) at $s = 1$ gives $S_\sigma x = x\, G(1, \sigma)$ outright, so $c = F$ on $(0, \infty)$ and Cauchy is applied once, not twice. **And $\alpha \le 1$ needs only subadditivity.** Membership of* BF_0 *is more than the bound uses: $F(s + t) \le F(s) + F(t)$ is $(1 - e^{-su})(1 - e^{-tu}) \ge 0$ read through the representation, and at $s = t = 1$ it gives $2^{\alpha} \le 2$. So the bound is available in* LE *and costs no bridge — the same discipline 7.11 follows.)*

Proof. Homogeneity gives $g_{x,y} = (y-x)\,g$ with $g := g_{0,1}$, and (6.1) becomes $(S_\sigma y - S_\sigma x)\,g(s) = (y-x)\,g(\sigma s)$. Hence S_σ is affine, and fixing 0 it is linear: $S_\sigma x = c(\sigma)x$ with $c(\sigma)g(s) = g(\sigma s)$.

Then c is a continuous multiplicative homomorphism of $(0,\infty)$, so $c(\sigma) = \sigma^\alpha$ by Cauchy's functional equation, as in `@fagerstrom2005temporal`, Lemma 2. Now $g(\sigma s) = \sigma^\alpha g(s)$ forces $g(s) = g(1)s^\alpha$, and the normalization makes $g(1) = 1$. Membership of BF_0 restricts α to $(0,1]$. The value $\alpha = 1$ gives $sF'(s) = s \in \mathrm{BF}$, so 7.1 admits it and it is not excluded by the present axioms. $\square$

Remark 7.5 (the boundary case $\alpha = 1$ and the drift term)**.** `@fagerstrom2005temporal` excluded $\alpha = 1$ through the pointwise-continuity requirement on the kernel functions, which the deterministic delay $\delta(t-x)$ violates. Under 2.1 the delay family — and more generally the drift term $b_0 s$ in (7.1), which shifts every kernel by $b_0 x$ — is admissible.

If that is undesired it is removed by one additional axiom: absolute continuity of the kernels for $x > 0$, or the requirement $\lim_{s\to\infty} F(s)/s = 0$, which says there is no deterministic front.

Proposition 7.6 (Choquet structure of the admissible cone)**.** The superposition map of 7.14,

$$(b_0,\rho) \;\mapsto\; F, \qquad F(s) = b_0\, s + \int_{(0,\infty)} \mathrm{Ein}(\tau s)\,\rho(d\tau),$$

is a linear *bijection* from $\big\{(b_0,\rho) : b_0 \geq 0,\ \rho \text{ a positive Borel measure on } (0,\infty),\ \int\big[\tau\wedge(1+\log_+\tau)\big]\,\rho(d\tau) < \infty\big\}$ onto the cone of 7.13, with inverse $\rho = -dk$, $k(t) = \rho\big((t,\infty)\big)$. Consequently the extreme rays of the cone are the pure delay $s \mapsto s$ together with the one-parameter family $\mathrm{Ein}(\tau\,\cdot)$, $\tau > 0$ — the Laplace exponents, with Lévy density $t^{-1}\mathbf{1}_{(0,\tau)}(t)$, of the *dilated Dickman subordinators* (`@caravenna2019dickman`, (1.1)–(1.2), p. 1) — and every admissible family is a *unique* superposition of Dickman rays and drift. ▷ **[T]** proved-target *(Narrowed 2026-08-14; the cone and the superposition are now 7.13 and 7.14, both machine-checked, and what this node keeps is exactly the two clauses that are not.* ***Injectivity*** *is uniqueness of the Lévy–Khintchine triple, ledger A3 as the proof below spends it. It is plausibly avoidable here —* $F'(s) = b_0 + \int e^{-st}k(t)\,dt$ *is proved (A.1) and the development carries Laplace injectivity for locally finite measures, so F determines k up to a null set without any statement about* BF *as a class — but that is a piece of work in its own right and it has not been done.* ***The extreme rays*** *need a Choquet argument, and the obstruction is not Mathlib's* `IsExtreme`*: it is that the decomposition* $\rho = \int \delta_\tau\,\rho(d\tau)$ *has to be read as a barycentre in a cone of measures. Neither clause is consumed by the* $\Rightarrow$ *direction, and 7.7 is where the article uses them. The integrability condition was written out here first and the draft's Remark 7.6 has since acquired it; it is forced by the bijection, being exactly finiteness of F.)*

Proof. Surjectivity onto the cone is 7.14, and linearity is 7.13. Since $\mathrm{Ein}(z) \leq z$ and $\mathrm{Ein}(z) = \log z + \gamma_E + o(1)$ as $z \to \infty$, the integral is finite for one $s > 0$ iff it is finite for all, iff $\int[\tau\wedge(1+\log_+\tau)]\,\rho(d\tau) < \infty$; so the stated domain is exactly the set of pairs the representation accepts.

The map is injective because the Lévy–Khintchine triple of F is unique (3.3(3)): the triple determines b_0 and the Lévy measure $k(t)t^{-1}dt$, hence k up to a null set, hence $\rho = -dk$.

A linear bijection carries extreme rays to extreme rays, so it suffices to identify them in the domain, which is the product of the ray $[0,\infty)$ with a cone of positive measures. Its extreme rays are $(1,0)$ and $(0,\delta_\tau)$, $\tau > 0$: any (b_0,ρ) decomposes uniquely as $b_0\,(1,0) + \int(0,\delta_\tau)\,\rho(d\tau)$, and a pair is extreme only if it admits no nontrivial such splitting, which forces $\rho = 0$ or ρ a single atom with $b_0 = 0$. Under the map, $(1,0) \mapsto s$, the pure delay; and $(0,\delta_\tau) \mapsto \mathrm{Ein}(\tau s)$, whose $k(t) = \delta_\tau((t,\infty)) = \mathbf{1}_{(0,\tau)}(t)$ gives the Lévy density $t^{-1}\mathbf{1}_{(0,\tau)}(t)$ — which is 7.14(1). $\square$

Remark 7.7 (the three corners are slices, not extremes)**.** None of the distinguished families of this paper is extreme; each is a slice cut by a different principle — the stable family by dilation symmetry (7.4, 8.1), the Gamma family as the atoms of the Thorin representation (8.2), the Bessel family by locality (8.3, 10.5). Since all three have completely monotone k, the closed convex cone the three corners generate together is exactly the Thorin subclass — strictly smaller than the whole class, the difference being the memory kernels that are monotone but not completely monotone (plateaus, kinks, hard cutoffs). The Dickman rays are the paradigm of that difference: their kernels solve the delay equation obtained from the Volterra identity of the memory chapter with the step kernel $k = \mathbf{1}_{[0,\tau)}$, and carry derivative singularities at the delays $\tau, 2\tau, \ldots$ — echo signatures unattainable within the Thorin subclass.

Two bounds hold throughout the cone off the drift ray. If $k \not\equiv 0$, every kernel $\mu_{0,x}$, $x > 0$, is a nondegenerate self-decomposable law, hence absolutely continuous (`@sato1999levy`, Thm. 27.13) and *unimodal*, by Yamazato's theorem on class L laws (`@yamazato1978unimodality`, Thm. 1, p. 523); on the drift ray $k \equiv 0$ the kernel is the point mass $\delta_{b_0 x}$. The axioms never produce multimodal kernels. And every kernel with $k \not\equiv 0$ is infinitely divisible and carried by $[0, \infty)$, so its tail is never lighter than $e^{-ct \log t}$ (`@sato1999levy`, Thm. 26.1, p. 168): no truncation of the Gaussian is an admissible kernel among these, which is the claim made in Chapter 2; the drift ray is the point mass above, whose tail is of course lighter than any such bound.

The three statements below are the halves of 7.3, stated separately because they are proved separately and, in Lean, finished at very different times. They are additive — not in the draft, which states the theorem whole — and are appended so that 7.4 and 7.5 keep the draft's numbers.

Theorem 7.8 (the construction: every admissible F gives a family)**.** Let F be of the form (7.1) with $F \not\equiv 0$. For $0 \le a \le b$ let $\mu_{a,b}$ be the probability measure on $[0, \infty)$ with

$$\hat{\mu}_{a,b}(s) = \exp\bigl[-g_{a,b}(s)\bigr], \qquad g_{a,b}(s) := F(bs) - F(as),$$

and set $\Phi_{a,b} f := \mu_{a,b} * f$. Then $(\Phi_{a,b})_{0 \le a \le b}$ is a causal cascade measurement family in the sense of 2.1, with scaling action $S_\sigma x = \sigma x$. ▷ **[T]** proved-target
*(**Machine-checked**, as `Hemigroup.SelfDecomposableExponent.cascadeFamily` — and at the level the theorem states it: not "the kernels satisfy properties we named (A1)–(A8)" but "the kernels* are *a causal cascade measurement family", the Lean structure of 2.1 instantiated. **`#print axioms`** reduces to Lean core plus ledger A17, which is the single crossing of the trust boundary and enters at one point only: the existence of $\mu_{a,b}$ given its triple. That is A17 rather than the Bernstein–Widder route the proof below takes, because the development has the Lévy triple in hand and so needs a* construction, *not a representation theorem — see `Formalization/Hemigroup/Interfaces.lean` for why the interface is phrased so that the day the compound-Poisson construction is carried out it becomes a lemma and the trust boundary returns to empty, without touching a downstream statement. **What the proof cost, against what was expected.** The passage from exponents to measures is pure transport, the transform carrying convolution to multiplication and dilation to reparametrisation, and being injective on causal measures — and that injectivity is 3.8, proved here, so it costs nothing. The lower endpoint $a = 0$ is not a separate construction: with the normalisation $k(0) = 0$ the increment density $k(u/b) - k(u/a)$ at $a = 0$ is $k(u/b)$ and the drift is $b_0 b$, which is exactly the dilated representation of $F(b\,\cdot)$ (`increment_zero_left`). So $G(x, s) = g_{0,x}(s)$, the object the similarity analysis of chapters 5–6 is built on, is available. (A1)–(A5) come from two applications of one Tonelli identity, $\int\bigl(\int g(t-r)\,\mu(dr)\bigr)dt = \|\mu\| \int g$ — once to estimate, giving (A1); once to license the Bochner swap, giving (A5). (A7) was expected to cost a second interface and did not: 3.12 is proved rather than cited, and the tightness it needs was always ours (3.13). The bridge from weak convergence to convergence in $L^1(\mathbb{R}, dt)$ is the $\varepsilon/3$ argument below, with the compact supplied by tightness, the left tail by causality, and the extension from C_c to $L^1(\mathbb{R}, dt)$*

by density against the contraction bound. ***Two defects the instantiation caught****, neither visible from reading 2.1: that the family is indexed from $x = 0$ while the construction began at $x > 0$ — repaired at the source by the normalisation above, not by narrowing the specification — and that each axiom must carry its index restriction $0 \le x \le y$ explicitly, without which (A5) would be* false *for the constructed family, $\mu_{x,y}$ being the zero measure off the range. This is the one place in the development where an error is silent, a mis-stated axiom making the theorem vacuous rather than unprovable, and an instance is the only thing that catches it.*

The general gauge. *7.3 (⇐) asks for the family in an* arbitrary *gauge χ, and this node, like the proof below, works in the canonical one. The general case is the reparametrisation $\Phi_{x,y} \mapsto \Phi_{\chi(x),\chi(y)}$, which transports (A1)–(A6) and (ND) verbatim, (A7) because an increasing bijection of $[0,\infty)$ is continuous, and (A8) with $S_\sigma \mapsto \chi^{-1} \circ S_\sigma \circ \chi$. Since the fidelity review (finding R8) that reparametrisation is machine-checked,* `Hemigroup.CascadeFamily.reparam`*, and* `Hemigroup.SelfDecomposableExponent.cascadeFamily_reparam` *is this theorem in the gauge χ: the family $\mu^F_{\chi(x),\chi(y)} * \cdot$ is a causal cascade measurement family with action $S_\sigma x = \chi^{-1}(\sigma\,\chi(x))$. Lean core for the reparametrisation itself, ledger A17 for the corollary about F's kernels, as here.)*

Proof. Work in the canonical gauge. For $0 \le a \le b$ the function $g_{a,b} = F(b\,\cdot) - F(a\,\cdot)$ lies in LE by 7.11 (for $a = 0$ the increment is $F(b\,\cdot)$ itself, which is in BF_0 by hypothesis), hence in BF_0 by 3.3(3), so $e^{-g_{a,b}}$ is completely monotone with value 1 at 0+ by 3.3(2), and by the Bernstein–Widder criterion 3.3(1) it is the Laplace transform of a unique probability measure $\mu_{a,b}$ on $[0,\infty)$. Then:

- (A1)–(A5) hold by the converse direction of 4.1.
- (A6) holds because exponents add, $g_{a,b} + g_{b,c} = g_{a,c}$, and Laplace transforms determine measures (3.8).
- (A8) holds with $S_\sigma x = \sigma x$, since $\hat{\mu}_{\sigma a,\sigma b}(s) = \hat{\mu}_{a,b}(\sigma s)$.
- (ND) holds because F is strictly increasing, by 3.4 applied to F.

For (A7): if $(a_n, b_n) \to (a,b)$ then $\hat{\mu}_{a_n,b_n} \to \hat{\mu}_{a,b}$ pointwise, by continuity of F. The family is tight by 3.13, uniformly over $b_n \le B$, because $F(0+) = 0$ makes the bound's numerator $F(Bs)$ as small as wanted; so $\mu_{a_n,b_n} \to \mu_{a,b}$ weakly by 3.12. Weak convergence of probability measures implies $\mu_{a_n,b_n} * f \to \mu_{a,b} * f$ in $L^1(\mathbb{R}, dt)$ for every $f \in X$: writing $\mu * f = \int \tau_r f\,\mu(dr)$ as an X-valued Bochner integral, the map $r \mapsto \tau_r f$ is bounded and continuous, and an $\varepsilon/3$ argument on a compact carrying mass $1 - \varepsilon$ concludes, first for compactly supported continuous f and then for all of X by density, the error not growing because each $\Phi_{a,b}$ is a contraction.

Finally, for an arbitrary gauge χ set $\Phi'_{x,y} := \Phi_{\chi(x),\chi(y)}$: (A1)–(A6) and (ND) hold verbatim, (A7) because an increasing bijection of $[0,\infty)$ is continuous, and (A8) with $S'_\sigma = \chi^{-1}(\sigma\,\chi(\cdot))$. □

Theorem 7.9 (the analysis: every family comes from an admissible F)**.** Let $(\Phi_{x,y})_{0\le x\le y}$ satisfy (A1)–(A8) and (ND). Then there are an increasing bijection χ of $[0,\infty)$ and an F of the form (7.1) with $F \not\equiv 0$ such that $\Phi_{x,y} f = \mu_{x,y} * f$ with $\hat{\mu}_{x,y}(s) = \exp\big[-\big(F(\chi(y)s) - F(\chi(x)s)\big)\big]$. ▷
[T] proved-target *(a collation: it cites eight earlier nodes and adds one step. No new analysis happens here — all of it happened in chapters 5 and 6 — which is why this is last on the ladder, and why the cost of formalising it was the cost of formalising those two chapters. Both structural questions the earlier note flagged are now settled: the conclusions of chapters 5–6 are read in* LE *(3.7), and χ is bundled with its conjugated action by 6.3.*

Everything but the last step is machine-checked*, as* `CascadeCore.similarity_form`*: the gauge, the convolution representation, the transform in the similarity form $\hat{\mu}_{x,y}(s) = \exp[-(F(\chi(y)s) - F(\chi(x)s))]$ with $F = G(1,\cdot)$, the fact that $F(b\,\cdot) - F(a\,\cdot) \in$* LE *for* every *$0 < a \le b$, and $F \not\equiv 0$. That last clause is where surjectivity of χ earns its place: an increment* *of F between arbitrary $a \le b$ is an increment of G between the scales $\chi^{-1}(a)$ and $\chi^{-1}(b)$, which is what 5.2 speaks about.* ***`#print axioms`*** *on it is Lean core, so the whole of the analysis direction up to the final appeal is interface-free.*

The last step is ledger A18*, taken as a reviewed trust-boundary entry on 2026-08-10 and not as a proof: 7.12 in the direction (1) ⇒ (3). The split is what makes this node's cost legible.* *`similarity_form`* *reduces to Lean core, so everything up to the final appeal is interface-free;* *`main_analysis`* *carries ledger A18 and nothing else; and 7.8 and 7.10 carry ledger A17 and* not *ledger A18. So the sentence this article has always written — that the analysis direction crosses the boundary where the constructive one does not — is now a fact* ***`#print axioms`*** *reports rather than a claim the prose makes.*

The conclusion lands where 7.8 starts. *The fidelity review (finding R1) found that* *`main_analysis`* *concluded in the raw data (χ, b_0, k) and a transform identity, without asserting that the exponent is finite or that the family's kernels* are *those of 7.8. It now also exports the finiteness $F(s) < \infty$ (free from ledger A18's conclusion) and the normalisation $\chi(1) = 1$, and* *`Hemigroup.CascadeCore.main_analysis'`* *packages the result as this statement reads: an F of the form (7.1) with $F \not\equiv 0$ — the same Lean structure 7.8 takes as input — with $\mu_{x,y} = \mu^F_{\chi(x),\chi(y)}$ and $\Phi_{x,y} = \mu^F_{\chi(x),\chi(y)} * \cdot$, the identification being Laplace injectivity (3.8). Because it names F's kernels, which ledger A17 constructs,* *`main_analysis'`* *carries ledger A17* and *ledger A18;* *`main_analysis`* *itself still carries ledger A18 alone, and both lines are in* ***`CIAxiomGuard.lean`****.)*

Proof. 4.1, 4.2, 5.1, 5.2, 5.3, 6.1, 6.2 and 6.3 together give the representation with $F = G(1, \cdot) \in \mathrm{BF}_0$ and $F \not\equiv 0$. Every increment $F(\chi(y)s) - F(\chi(x)s)$ lies in BF_0 by 5.2, and χ is onto $(0, \infty)$ by 6.3, so condition (1) of 7.1 holds for every pair $0 < a \le b$; 7.12 then yields (7.1). □

Proposition 7.10 (uniqueness of the gauge and the exponent)**.** Let F be of the form (7.1) with $F \not\equiv 0$, and let F' be of the same form. Suppose $\chi : [0, \infty) \to [0, \infty)$ is nondecreasing and positive on $(0, \infty)$ with $\chi(0+) = 0$ and $\chi(1) = 1$, and that $\mu'_{\chi(x),\chi(y)} = \mu_{x,y}$ for all $0 < x \le y$. Then $\chi = \mathrm{Id}$ and $F' = F$. ▷ **[T]** proved-target *(**Machine-checked**, as* *`Hemigroup.SelfDecomposableExponent.gauge_and_exponent_unique`*; ***`#print axioms`*** *is held against* ***`blueprint/trust-boundary.txt`*** *by CI, with ledger A17 reaching the statement through the construction of the kernels it quantifies over and nothing else. The argument itself adds no analysis to 7.8.* ***Why χ is a bare function here.*** *6.3 produces χ as an increasing bijection with an inverse conjugating the scaling action into multiplication, and none of that is needed: this clause only ever* quantifies over *gauges, so χ enters carrying exactly the three properties the argument consumes. That is a strictly stronger statement than the bijection form, and it leaves the interface for 7.9 — which must* build *χ — to be designed by its consumer.* ***The analytic content*** *is that F is strictly increasing when $F \not\equiv 0$ (`exponent_strictMono`), which is what makes the c with $g_{0,x} = F(c\,\cdot)$ unique. That proof runs 3.4 on the* increment*: if $g_{1,c}$ vanished anywhere it would vanish identically, giving $F(cs) = F(s)$, and iterating* downwards *sends $s_0/c^n \to 0$, where continuity at the origin forces $F \equiv 0$. The one new step is pinning the constant, the hypothesis giving only* differences *of exponents; it is run as an ε argument rather than a limit, since* ***`exists_exponent_lt`*** *is already non-sequential.)*

Proof. Equal kernels have equal exponents, the transform being injective (3.8) and exp injective, so the hypothesis reads

$$F'(\chi(y)s) - F'(\chi(x)s) = F(ys) - F(xs), \qquad 0 < x \le y,\ s \ge 0.$$

Let $x \downarrow 0$. Both subtrahends vanish, because $F(0+) = F'(0+) = 0$ and $\chi(0+) = 0$, leaving $F'(\chi(y)s) = F(ys)$. At $y = 1$, with $\chi(1) = 1$, this is $F' = F$ (without the normalization the pair is determined up to $(\chi, F) \mapsto (c\,\chi,\ F(c^{-1}\,\cdot))$, $c > 0$, which is the freedom the normalization removes).

Feeding that back gives $F(\chi(y)s) = F(ys)$ for all $y > 0$, $s \geq 0$. At $s = 1$, strict monotonicity of F — which holds because $F \not\equiv 0$, by 3.4 applied to the increments as in 5.3 — forces $\chi(y) = y$. □

The two statements below are the halves of 7.1, split for the same reason its consumers differ: one is elementary and machine-checked, the other is where ledger A3 and ledger A4 are spent. Additive, and appended, as above.

Lemma 7.11 (the dilation increment of a self-decomposable exponent)**.** Let $F \in \mathrm{LE}$ have drift $b_0 \geq 0$ and Lévy density $k(t)/t$ with $k \geq 0$ nonincreasing on $(0, \infty)$. Then for $0 < a \leq b$ and $s \geq 0$,

$$F(as) + G_{a,b}(s) = F(bs),$$

where $G_{a,b} \in \mathrm{LE}$ is the Lévy exponent with drift $b_0(b-a)$ and Lévy density $\big(k(u/b) - k(u/a)\big)/u$. In particular $F(b\,\cdot) - F(a\,\cdot) \in \mathrm{LE}$, which is condition (1) of 7.1 in representation form. ▷ [**T**] proved-target *(**Machine-checked**, as `Hemigroup.levyExponentD_increment`, `#print axioms` reducing to Lean core — no ledger entry at all, which is the whole point of separating it. **Stated additively**, $F(as) + G = F(bs)$ rather than as a difference, because that is what a cascade needs (exponents must* add *along $a \leq b \leq c$) and because the Lean exponents are valued in $[0, \infty]$, where subtraction truncates; the two forms agree wherever $F(as) < \infty$. **Why this is the cheap half.** It is a change of variables and a sign: du/u is the Haar measure of the dilation group, so a dilation moves the Lévy density and nothing else, and nonnegativity of $k(u/b) - k(u/a)$ is exactly the hypothesis that k is nonincreasing. No closure theorem, no derivative, no uniqueness of a triple — so none of 3.3. Classically one would instead read (3) ⇒ (2) ⇒ (1) off 7.12, which is shorter to write and costs ledger A3 and ledger A4; 7.8 cites this node rather than that route, and that is why its trust base is Lean core plus ledger A17 alone.)*

Proof. Write $F(cs) = b_0cs + \int_0^\infty \big(1 - e^{-cst}\big)\, k(t)\, \frac{dt}{t}$ for $c > 0$ and substitute $u = ct$. The measure dt/t is invariant, so

$$F(cs) = b_0 c\, s + \int_0^\infty \big(1 - e^{-su}\big)\, k(u/c)\, \frac{du}{u},$$

i.e. dilating the argument by c dilates the density and rescales the drift, and does nothing else. Subtracting the identity at $c = a$ from the one at $c = b$ gives

$$F(bs) - F(as) = b_0(b-a)\, s + \int_0^\infty \big(1 - e^{-su}\big)\, \big(k(u/b) - k(u/a)\big)\, \frac{du}{u},$$

which is the asserted $G_{a,b}$. It lies in LE: the drift $b_0(b-a)$ is nonnegative since $a \leq b$, and the density is nonnegative since $u/b \leq u/a$ for $u > 0$ and k is nonincreasing. Both integrals are finite because each is bounded by $F(bs) < \infty$. □

Lemma 7.12 (the derivative characterization)**.** Let $F \in \mathrm{BF}_0$. If $F(b\,\cdot) - F(a\,\cdot) \in \mathrm{BF}_0$ for all $0 < a \leq b$, then $B(s) := sF'(s)$ is a Bernstein function; and if B is a Bernstein function, then F has the representation (7.1) with k nonincreasing. The converse of each also holds. ▷ [**T**] proved-target *(**Taken as ledger A18, not proved** — a review decision recorded 2026-08-10. Both directions run through 3.3: (1)⇒(2) is closure of* BF *under pointwise limits, ledger A4, applied to a difference quotient; (2)⇒(3) needs uniqueness of the Lévy–Khintchine triple, ledger A3, to identify the density it produces with k. Neither is a statement about the hemigroup family — they are facts about* BF *as a class of functions — and neither can even be* stated *here, since both quantify over* BF *and the development has no* CM*. So what the trust boundary carries is the composite, in the direction 7.9 spends, phrased in* LE*: exactly the device ledger A17 uses. Worth recording that the* anchor *does not take the route this proof does: ledger A18 is grounded* *on* `@schilling2012bernstein`*, Prop. 5.17, whose hypothesis is complete monotonicity of the ratio* $g(\lambda)/g(c\lambda)$ *rather than membership of the increment in* BF_0*, and the passage from ours to his costs ledger A2 and not ledger A4 or ledger A3. Two independent routes to the same fact, which is a reason for confidence rather than a discrepancy; the ledger entry records both. Unlike ledger A17, expect it to be permanent — the (1)⇒(2) leg needs differentiability of Bernstein functions, and there is no route to that which does not reintroduce* CM*.* ***The derivative is unavoidable here****, which is the other reason to keep it separate:* $B(s) = sF'(s)$ *has no reading in* LE *until the representation is in hand, so unlike 7.11 this statement cannot be given an* LE *form at all. That is a fact about the mathematics, not a gap in the transcription.)*

Proof. (1) ⇒ (2). For $h > 0$ the difference quotient $s \mapsto \big(F(e^h s) - F(s)\big)/h$ lies in BF, being a positive multiple of an element of BF_0. As $h \downarrow 0$ it converges pointwise to $sF'(s)$, since F is smooth on $(0, \infty)$; the limit is therefore Bernstein by 3.3(4).

(2) ⇒ (1). $F(bs) - F(as) = \int_{\log a}^{\log b} B(e^u s)\, du$. Each integrand is Bernstein in s, so the finite mixture is Bernstein by 3.3(4); vanishing at 0+ follows by monotone convergence.

(2) ⇔ (3). Write $F(s) = b_0 s + \int (1 - e^{-st})\, \nu(dt)$, with no killing term because $F(0+) = 0$ (3.3(3)), so that

$$B(s) = sF'(s) = b_0 s + \int_0^\infty s\, t\, e^{-st}\, \nu(dt).$$

Note $B(0+) = 0$, since $0 \le sF'(s) \le F(s)$ by concavity.

Suppose $\nu(dt) = k(t)\, t^{-1} dt$ with k nonincreasing. Write $k(t) = \rho\big((t, \infty)\big)$ for the positive measure $\rho = -dk$; right-continuity of k may be assumed, and $k(\infty) = 0$ is forced by $\int^\infty k(t) t^{-1} dt < \infty$. Tonelli then gives

$$\int_0^\infty s\, e^{-st}\, k(t)\, dt = \int_0^\infty \rho(du) \int_0^u s\, e^{-st}\, dt = \int_0^\infty \big(1 - e^{-su}\big)\, \rho(du),$$

so $B(s) = b_0 s + \int (1 - e^{-su}) \rho(du) \in \mathrm{BF}_0$.

Conversely, if $B \in \mathrm{BF}$ then $B \in \mathrm{BF}_0$ by the remark above, so $B(s) = \beta s + \int (1 - e^{-su}) \rho(du)$. Running the same computation backwards and invoking uniqueness of the Lévy–Khintchine triple (3.3(3)) identifies $\beta = b_0$ and $\nu(dt) = k(t) t^{-1} dt$ with $k(t) = \rho((t, \infty))$ nonincreasing. □

The two statements below are the half of 7.6 that needs neither uniqueness of the Lévy–Khintchine triple nor a Choquet argument: the cone itself, and the superposition formula. Additive, and appended, by the same rule as above.

Lemma 7.13 (the admissible exponents form a convex cone)**.** If F_1, F_2 are admissible in the sense of 7.3 — of the form (7.1) with drift $b_0 \ge 0$ and $k \ge 0$ nonincreasing — then so are $F_1 + F_2$, with data $(b_0^1 + b_0^2,\ k_1 + k_2)$, and $c\,F_1$ for $c \ge 0$, with data $(c\,b_0^1,\ c\,k_1)$. So the admissible exponents form a convex cone in the pointwise order, and the correspondence with the pairs (b_0, k) is linear. ▷ **[T]** proved-target

Proof. Each of the six defining conditions is stable: nonnegativity and monotonicity of k are preserved by sums and by nonnegative multiples, the drift stays nonnegative, and the exponent of a sum is the sum of the exponents by additivity of the integral — so finiteness is inherited. Linearity of $(b_0, k) \mapsto F$ is the same computation read forwards. □

Lemma 7.14 (Dickman superposition)**.** Write $\mathrm{Ein}(z) := \int_0^z \big(1 - e^{-u}\big)\, u^{-1} du$.

1. *(The rays.)* For $\tau > 0$ the density $k = \mathbf{1}_{(0,\tau)}$ is admissible, with drift 0, and its exponent is $s \mapsto \mathrm{Ein}(\tau s)$.

2. *(The superposition.)* Every admissible F, with drift b_0 and density k, satisfies

$$F(s) \;=\; b_0\, s \;+\; \int_{(0,\infty)} \mathrm{Ein}(\tau s)\, \rho(d\tau) \qquad (s \ge 0),$$

where ρ is any positive measure on $(0, \infty)$ with $\rho\big((t, \infty)\big) = k(t)$ for almost every $t > 0$; such a ρ exists.

▷ [**T**] proved-target

Proof. (1) is the substitution $u = st$ in $\int_0^\tau (1 - e^{-st})\, t^{-1} dt$; admissibility is the integrability criterion of 8.7, both halves being immediate for a bounded, compactly supported k.

(2) k is nonincreasing and nonnegative and $k(\infty) = 0$, the last forced by $\int^\infty k(t)\, t^{-1} dt < \infty$; so a positive measure with tail k exists. Then Tonelli over the region $\{0 < t < \tau\}$ gives

$$\int_0^\infty \big(1 - e^{-st}\big)\, \frac{k(t)}{t}\, dt = \int_{(0,\infty)} \rho(d\tau) \int_0^\tau \big(1 - e^{-st}\big)\, \frac{dt}{t} = \int_{(0,\infty)} \mathrm{Ein}(\tau s)\, \rho(d\tau),$$

the inner integral being $\mathrm{Ein}(\tau s)$ by (1). □

Chapter 8

Examples and moments

Throughout this chapter we work in the canonical gauge, so that $\phi_x = \mathcal{L}^{-1}[e^{-F(xs)}]$, and T_x denotes a random variable with density ϕ_x.

Proposition 8.1 (the extremal stable family: the exponent)**.** For $0 < \alpha < 1$ the exponent $F(s) = s^\alpha$ is of the form (7.1), with Lévy density factor

$$k(t) = \frac{\alpha}{\Gamma(1-\alpha)}\, t^{-\alpha},$$

which is nonincreasing. ▷ [**T**] proved-target *(**Machine-checked**, as `Hemigroup.SelfDecomposableExponent.stableExponent_toRealExponent`, on the kernel `stableExponent`; `#print axioms` reduces to Lean core. The proof is the same antiderivative argument as 8.2, at the Gamma integral's exponent $1-\alpha$ in place of 1: $F'(s) = \frac{\alpha}{\Gamma(1-\alpha)} \int_0^\infty t^{-\alpha} e^{-st} dt = \alpha s^{\alpha-1}$, which is the derivative of s^α, and both sides vanish at $0+$ — the second by $F(0+) = 0$ and the first because $\alpha > 0$. So the "standard integral representation" the draft cites is not assumed here but derived, and $\Gamma(1-\alpha)$ enters as the constant Mathlib's Gamma integral produces rather than as a normalisation chosen to make the answer come out.*

This kernel is what separates the two halves of 8.7. *k is* unbounded *at the origin and integrable there only because $\alpha < 1$; it is integrable against dt/t at infinity only because $\alpha > 0$. Each side of the hypothesis $0 < \alpha < 1$ is used exactly once, on a different half, which is why the criterion could not have been stated with a single condition.)*

Proof. $s^\alpha = \frac{\alpha}{\Gamma(1-\alpha)} \int_0^\infty (1 - e^{-st})\, t^{-\alpha-1}\, dt$ is the standard integral representation, which is (7.1) with $b_0 = 0$ and $k(t)/t = \alpha t^{-\alpha-1}/\Gamma(1-\alpha)$; k is a negative power, hence nonincreasing, and both integrability conditions of 8.7 hold ($t^{-\alpha}$ integrable at 0, $t^{-\alpha-1}$ integrable at infinity), so F is finite and of the form (7.1). □

Proposition 8.2 (the Gamma family: the exponent)**.** For $\gamma > 0$ the exponent $F(s) = \gamma \log(1+s)$ is of the form (7.1) with $k(t) = \gamma\, e^{-t}$, which is nonincreasing. ▷ [**T**] proved-target *(**Machine-checked**, as `Hemigroup.SelfDecomposableExponent.gammaExponent_toRealExponent`, on the kernel `gammaExponent` of 8.7; `#print axioms` reduces to Lean core.*

The proof is not the draft's. *The draft quotes Frullani's integral, which Mathlib does not have. What it does have — once A.1 is proved — is the antiderivative reading of the same fact: $F'(s) = \gamma \int_0^\infty e^{-(1+s)t} dt = \gamma/(1+s)$ by the $a = 1$ case of the Gamma integral, and $\log(1+s)$ has the same derivative and the same limit 0 at the origin. The limit is the clause that needs an argument, and it is supplied once for the whole class as $F(0+) = 0$ (`tendsto_toRealExponent_nhdsGT_zero`), by dominated convergence against the $s = 1$ integrand. That is exactly the classical proof of Frullani's integral, so nothing is smuggled in; what changes is that the citation becomes a computation.*

One normalisation is not free: the Lean structure fixes $k(0) = 0$, so the kernel is γe^{-t} guarded by $\mathbf{1}_{t>0}$. Without the guard the field is unprovable rather than merely inelegant, and $\{0\}$ being null costs nothing downstream.)

Proof. The Frullani integral gives $\log(1+s) = \int_0^\infty (1-e^{-st})\, e^{-t}\, t^{-1}\, dt$, so F has the form (7.1) with $b_0 = 0$ and $k(t) = \gamma e^{-t}$, nonincreasing, and both integrability conditions of 8.7 hold (e^{-t} bounded near 0, integrable at infinity), so F is finite and of the form (7.1); hence F is admissible by 7.1. ☐

Proposition 8.3 (the Bessel-K family; preview of the memory line)**.** For $a > 0$ let $T_1 \stackrel{d}{=} 1/(2\gamma_a)$ be the scaled inverse-gamma law, where γ_a has the Gamma density of shape a, and let F be its Laplace exponent. Then

$$e^{-F(s)} = \frac{2^{1-a}}{\Gamma(a)} \big(\sqrt{2s}\big)^a K_a\big(\sqrt{2s}\big),$$

and T_1 is self-decomposable, so F is of the form (7.1) and the family is admissible. The moments satisfy $\mathbb{E}\, T_x^n < \infty$ if and only if $a > n$. ▷

[**A**] analytic interface ledger A8 *(self-decomposability of the generalized inverse Gaussian laws; `@halgreen1979self`. A cited interface, not reproven here.* ***Assignment.*** *ledger A8 carries exactly one clause: that T_1 is self-decomposable, which is what makes F admissible and puts the family inside 7.3. Everything else is **[T]** and proved below — the evaluation of the transform in terms of K_a is a classical integral representation, and the moments are a direct Gamma-integral computation from $T_1 = 1/(2\gamma_a)$, not routed through 8.6: the edge to that node is dropped below along with the sentence that used it (2026-08-29). Note also that Halgreen proves the stronger statement that the law is a generalized Gamma convolution; self-decomposability follows through* GGC $\subset$ SD*, which is a clause of ledger A16 — so A8 and A16 rest on the same tower and should be reviewed together.*

Two sentences moved out of the statement (2026-08-23). *The parabolic-gauge identification of these kernels with the Bessel-type media $\frac{1}{2}\partial_x^2 + \frac{\beta}{x}\partial_x$, $a = \frac{1}{2} - \beta$, is not this node's content — it is 9.13's, in Chapter 9, which already depends on this one and states the identification precisely, in its own gauge notation (6.4's parabolic gauge ξ); restating it here used the older, now-superseded symbol $\tilde{x}$ for it and said the same thing a second time. Likewise the closing pointer to the memory-line theorems is prose, not a mathematical clause of this proposition, and now lives at the point of citation instead.*

Proof. Write γ_a for a Gamma$(a,1)$ variable and $T_1 = 1/(2\gamma_a)$. Substituting $u = 1/(2t)$ in the Gamma density gives the density of T_1,

$$\phi_1(t) = \frac{t^{-a-1}\, e^{-1/(2t)}}{2^a\, \Gamma(a)}, \qquad t > 0.$$

The classical integral representation $\int_0^\infty t^{\nu-1} \exp(-A/t - Bt)\, dt = 2\,(A/B)^{\nu/2} K_\nu\big(2\sqrt{AB}\big)$, taken at $\nu = -a$, $A = \frac{1}{2}$, $B = s$, together with $K_{-a} = K_a$, gives

$$\mathbb{E}\, e^{-sT_1} = \frac{1}{2^a\Gamma(a)} \int_0^\infty t^{-a-1} e^{-st-1/(2t)}\, dt = \frac{2\,(2s)^{a/2}\, K_a\big(\sqrt{2s}\big)}{2^a\Gamma(a)} = \frac{2^{1-a}}{\Gamma(a)} \big(\sqrt{2s}\big)^a K_a\big(\sqrt{2s}\big).$$

Self-decomposability of T_1 is ledger A8; by 7.1 its exponent is of the form (7.1), and both integrability conditions of 8.7 hold for the resulting (b_0, k) by 8.8, since $F(1) = -\log \mathbb{E}\, e^{-T_1} < \infty$; so F is finite and 7.3 makes the family admissible. Finally, since $T_x \stackrel{d}{=} x\, T_1$ in the canonical gauge, for every $x > 0$ and $n \in \mathbb{N}$,

$$\mathbb{E}\, T_x^n = x^n\, \mathbb{E}\, T_1^n = \frac{x^n}{2^n}\, \mathbb{E}\big[\gamma_a^{-n}\big], \qquad \mathbb{E}\big[\gamma_a^{-n}\big] = \frac{1}{\Gamma(a)} \int_0^\infty u^{a-n-1} e^{-u}\, du = \frac{\Gamma(a-n)}{\Gamma(a)},$$

the last integral (hence the moment) finite exactly when $a - n > 0$; so $\mathbb{E}\, T_x^n < \infty$ iff $a > n$. This is a direct computation from the defining substitution $T_1 = 1/(2\gamma_a)$ and cites no Lévy-density asymptotics. $\square$

Proposition 8.4 (the mean delay and the influence curve)**.** $\mathbb{E}\, T_x = x\, F'(0+) = x\big(b_0 + \int_0^\infty k(t)\, dt\big)$, finite if and only if k is integrable at infinity. $\triangleright$
[**T**] proved-target *(monotone convergence in (7.1) and nothing else — no citation is needed and none is made.* ***The influence-curve identification*** *(`@fagerstrom2005temporal`, §7) moved out of the statement into prose at the point of citation (2026-08-23): it restates this node's linearity in different language rather than adding a further fact, and the statement of record should carry only the latter. The higher-moment criterion, which* does *need one, was split off as 8.6; before the split this node reported that entry's cost for a result that does not incur it. What is* not *claimed here is anything about the mode, which is the delay measure the stable family forces one to use — see 8.1.* ***Proved in Lean*** *(2026-08-14). The mean rate is* $[0,\infty]$*-valued, so the "finite if and only if" clause is `meanRate_ne_top_iff` with both sides allowed to fail and no finiteness hypothesis anywhere.* ***And the linearity of the influence curve does not depend on the identity****, which is the reverse of the order used above:* $\mu_{0,x}$ *is the law of* xT_1*, so every moment scales by a change of variables whether or not it is finite, and the identity is needed only at* $x = 1$*. That one instance,* $\mathbb{E}T_1 = F'(0+)$*, is reached without differentiating anything: the difference quotient* $\big(1 - e^{-st}\big)/s$ *is* nonincreasing *in* s*, so the proof is monotone convergence twice — once on the law, once inside (7.1) — joined by the squeeze* $we^{-w} \le 1 - e^{-w} \le w$ *at* $w = F(s_n) \to 0$*. The lower bound is the upper one times a factor tending to* 1*, which in* $[0,\infty]$ *multiplies through with no side condition, so the infinite case needs no separate treatment — and it is exactly the case a real-valued argument would have had to exclude.)*

Proof. Differentiating the transform at the origin, $\mathbb{E}\, T_x = -\partial_s e^{-F(xs)}\big|_{s=0+} = xF'(0+)$, and monotone convergence in (7.1) gives $F'(0+) = b_0 + \int_0^\infty k(t)\, dt$, finite exactly when k is integrable at infinity. $\square$

Remark 8.5 (heavy tails are the semigroup's doing)**.** The heavy tails of the 2005 kernels are an artifact of the semigroup axiom, not of causality together with covariance. Within 7.3 the semigroup sub-case (7.4) is the subset with k a pure power $c\,t^{-\alpha}$, $0 < \alpha < 1$, or the pure delay, $k \equiv 0$ with $b_0 > 0$ — the only data compatible with homogeneity. A pure power is never integrable at infinity, so by 8.4 no stable member has a mean. Any other nonincreasing k is admitted, and its moments are governed by 8.6 alone: $\mathbb{E}\, T_x^n < \infty$ exactly when $\int_1^\infty t^{n-1} k(t)\, dt < \infty$. For $k(t) = \gamma e^{-t}$ every moment is finite (8.12); for the Bessel family with shape a exactly the moments of order below a are finite (8.3).

Proposition 8.6 (the moment criterion)**.** $\mathbb{E}\, T_x^n < \infty$ if and only if $\int_1^\infty t^{n-1} k(t)\, dt < \infty$. $\triangleright$ [**A**] analytic interface ledger A7 *(the moment criterion for infinitely divisible laws; `@sato1999levy`, Thm. 25.3. A cited interface, not reproven here.* ***Assignment.*** *ledger A7 carries the equivalence itself and nothing else — that a moment of the law is finite exactly when the corresponding integral against the Lévy measure over* $\{t > 1\}$ *is finite. Translating that integral into the stated one is* ***[T]****: the Lévy measure of* T_x *is* $\nu(dt) = k(t)t^{-1}dt$ *up to the scaling by* x*, so* $\int_{t>1} t^n\, \nu(dt) = \int_1^\infty t^{n-1} k(t)\, dt$*. The* $n = 1$ *case is* not *carried by the ledger — it is 8.4, proved by monotone convergence — which is the reason for the split.)*

Proof. The Lévy measure of T_x is $\nu(dt) = k(t)t^{-1}dt$ up to the scaling by x, so $\int_{t>1} t^n\, \nu(dt) = \int_1^\infty t^{n-1} k(t)\, dt$; ledger A7 equates finiteness of that integral with finiteness of $\mathbb{E}\, T_x^n$. $\square$

Proposition 8.7 (admissibility criterion for a memory kernel)**.** Let $b_0 \ge 0$ and let $k : (0,\infty) \to [0,\infty)$ be nonincreasing with

$$\int_0^1 k(t)\, dt < \infty \qquad \text{and} \qquad \int_1^\infty \frac{k(t)}{t}\, dt < \infty.$$

Then $F(s) = b_0 s + \int_0^\infty (1 - e^{-st})\, k(t)\, \frac{dt}{t}$ is finite for every $s \geq 0$.

Consequently F is of the form (7.1), and if $F \not\equiv 0$ 7.3 applies to it. ▷ **[T]** proved-target *(**Machine-checked**, as `Hemigroup.levyExponentD_ne_top_of_integrableOn`; `#print axioms` reduces to Lean core. Elementary, and stated separately because it is the* only *clause with content in exhibiting a concrete admissible exponent: the other five fields of the Lean structure — nonnegativity of b_0 and of k, monotonicity of k, and the normalisation $k(0) = 0$ — are immediate for every kernel in this chapter, and this one is not.*

The closing reference to 7.3 is a consequence and not part of the statement, which is why this node's dependency list records only 3.7: convergence is all the estimates below establish, and admissibility is what 7.1 then reads off it.

The two-sided split is forced by the integrand rather than chosen for convenience. Near the origin $1 - e^{-st}$ is comparable to st, which cancels the $1/t$ and leaves k itself; at infinity it is bounded by 1, which leaves $k(t)/t$. The two halves therefore test genuinely different things, and the examples separate them: the stable family's k is unbounded *at the origin and integrable there all the same (8.1), while a Dickman ray's k has compact support and so passes the second test trivially.*

***Two instances are built in Lean from it**, as non-vacuity checks on remarks that would otherwise assert the existence of members of the class without exhibiting one: `Hemigroup.gammaExponent`, giving the leaky integrator $k = e^{-t}$ of 8.5 at $\gamma = 1$, and `Hemigroup.dickmanExponent`, giving the Dickman ray $k = \mathbf{1}_{(0,\tau)}$ that 7.7 identifies as an extreme ray of the admissible cone. Each is a witness that the cone in question is inhabited where the remark says it is; neither yet carries that remark's closed form, which is 8.2's business.)*

Proof. Split the integral at $t = 1$. On $(0, 1]$ use $1 - e^{-st} \leq st$, which is $1 + x \leq e^x$ at $x = -st$; the integrand is then at most $s\, k(t)$, integrable by the first hypothesis. On $(1, \infty)$ use $1 - e^{-st} \leq 1$; the integrand is at most $k(t)/t$, integrable by the second. The drift term is finite for every real s. □

Lemma 8.8 (the criterion is characteristic)**.** Let $b_0 \geq 0$ and let $k : (0, \infty) \to [0, \infty)$ be nonincreasing. If the $[0, \infty]$-valued exponent (7.1) at $s = 1$,

$$b_0 + \int_0^\infty (1 - e^{-t})\, k(t)\, \frac{dt}{t},$$

is finite, then $\int_0^1 k(t)\, dt < \infty$ and $\int_1^\infty k(t) t^{-1} dt < \infty$. ▷ **[T]** proved-target *(**Machine-checked**, as `Hemigroup.SelfDecomposableExponent.integrableOn_of_ne_top`; `#print axioms` reduces to Lean core. **Restated (2026-08-23) from the Lean hypothesis directly**, rather than through "F is of the form (7.1)": that phrase already packages the two integrability conditions in as part of Lemma 7.1(3), which made the previous wording tautological. What has content is that finiteness of the exponent at the single point $s = 1$ — nothing about s elsewhere, and no appeal to admissibility already holding — is enough to force both integrals; that is what changes 8.7 from a convenient sufficient condition into a characterisation, so that the two integrals are the right invariants of the kernel and not an artifact of how the estimate was arranged. It also has a practical consequence for the Lean development — every admissible F carries both integrability facts* already, *so the analysis of the appendix needs no hypothesis beyond membership of the class, and in particular F' may be differentiated under the integral sign without one.)*

The proof is the same inequality read in the other direction. Where 8.7 used $1 - e^{-u} \leq u$, this uses $1 - e^{-u} \geq u/(1 + u)$, which is $e^u \geq 1 + u$ rearranged; the two together pin $1 - e^{-u}$ between $u/(1 + u)$ and u, and every estimate on either side is one of them. A single value of s suffices, and $s = 1$ is taken.

Proof. Take $s = 1$ in (7.1), so that $\int_0^\infty (1 - e^{-t})k(t)t^{-1}dt < \infty$. For $t > 0$, $1 - e^{-t} \geq t/(1+t)$, so the integrand is at least $k(t)/(1+t)$. On $(0,1]$ that is at least $k(t)/2$, and on $(1,\infty)$ it is at least $k(t)/(2t)$; both claims follow. □

Proposition 8.9 (the Gamma family: the transforms and the mean)**.** With F as in 8.2, the kernels and the increment measures have transforms

$$e^{-F(xs)} = (1 + xs)^{-\gamma}, \qquad e^{-g_{a,b}(s)} = \left(\tfrac{1+as}{1+bs}\right)^{\gamma},$$

and the mean delay is $\mathbb{E}\, T_x = \gamma x$. ▷ [**T**] proved-target *(split from 8.2 because the costs differ and a node reporting the maximum cost of its clauses misreports its cheap ones; split again on 2026-08-15 for the same reason, the density and the higher moments leaving as 8.10 and 8.12.* ***Proved in Lean*** *(2026-08-15),* `Hemigroup/GammaKernels.lean`*, on ledger A17 alone — through the constructed family, as every statement about T_x and $\mu_{a,b}$ is. All three clauses are one computation from the exponent $\gamma \log(1+s)$; the mean is 8.4 applied to $\int_0^\infty \gamma e^{-t}dt = \gamma$, and that step is Lean core.* ***Note the contrast with 8.16****, which came* off *ledger A7 this week. There the mean diverges, and divergence propagates upward, so the $n = 1$ case settles every n. Here the mean converges, and convergence does not propagate downward — so the Gamma family's higher moments genuinely need the criterion. Two nodes of the same shape splitting along different lines, for a reason that became visible only once $n = 1$ was proved and asked to carry the rest.)*

Proof. $F(s) = \gamma \log(1 + s)$ by 8.2, so $e^{-F(xs)} = (1 + xs)^{-\gamma}$; and $g_{a,b} = F(b\cdot) - F(a\cdot)$, so the increment transform is the ratio $e^{-F(bs)}/e^{-F(as)}$, which is the stated one. For the mean, 8.4 gives $\mathbb{E}\, T_x = x\big(b_0 + \int_0^\infty k\big)$ with $b_0 = 0$ and $k(t) = \gamma e^{-t}$, whose integral is γ. □

Proposition 8.10 (the Gamma family: the kernel density)**.** With F as in 8.2 and $\gamma > 0$,

$$\phi_x(t) = \frac{t^{\gamma-1}e^{-t/x}}{\Gamma(\gamma)\, x^\gamma},$$

the Gamma density of fixed shape γ and scale x. ▷ [**T**] proved-target *(****Proved in Lean**** (2026-08-15),* `Hemigroup/GammaDensity.lean`*, on ledger A17 alone — through the constructed family, as every statement about $\mu_{0,x}$ is. Stated there as an identity of* measures, $\mu_{0,x} = \mathrm{gammaMeasure}(\gamma, 1/x)$*, rather than of densities: it is the same assertion in the vocabulary Mathlib already has, and it is the form that makes Mathlib's Gamma API available to 8.12, which is the cheaper of the two ways to retire that node's ledger entry.* ***The Phase-0 cost estimate for this node was right****, which is worth recording because the estimates recorded in* `PLAN` *usually are not. It said inverting* $(1 + xs)^{-\gamma}$ *"means computing the Gamma density's transform and appealing to 3.8 — available, but a separate piece of work". It is exactly that:* `kernel_unique` *is the appeal, and the transform had to be computed here because Mathlib's Gamma file carries the density, the total mass and the CDF and stops — no Laplace transform, no moment generating function, no characteristic function. The computation is the Gamma integral chapter 8 already used for the stable family's F'.)*

Proof. $(1+xs)^{-\gamma}$ is the Laplace transform of the stated Gamma density — the Gamma integral $\int_0^\infty t^{\gamma-1}e^{-(1/x+s)t}\, dt = \Gamma(\gamma)/(1/x + s)^\gamma$ — so 3.8 identifies ϕ_x. □

Remark 8.11 (the Gamma family as a Sato subordinator)**.** The increment measures $\mu_{a,b}$ of this family are the increments of the Gamma-type Sato subordinator. This is 7.2 read at the Gamma exponent, and it is recorded as a remark rather than a proposition for that reason: the general statement is the one with content, and no separate argument is made here. It is also why it is not formalised — the development has no notion of a stochastic process, and would need one to say anything a transform identity does not already say.

Proposition 8.12 (the Gamma family: the higher moments)**.** With F as in 8.2, every moment of T_x is finite, and $\operatorname{Var} T_x = \gamma x^2$. ▷ [**T**] proved-target *(**Proved in Lean** (2026-08-15), `Hemigroup/GammaDensity.lean`, on ledger A17 alone — and **off ledger A7**, which the route through 8.6 would have spent. **The prediction made when this node was split off held.** It said: "the alternative route is through 8.10: with the density in hand the moments are a Gamma computation and the ledger entry is not needed. Whichever is done first makes the other cheap." 8.10 was done first, and this became the cheap one — so the dge to 8.6 is gone and the proof below is rewritten to the Gamma computation. Mathlib carries no moments for its own Gamma law any more than it carries a transform, so both come from the single Gamma integral $\int_0^\infty t^{a+n-1}e^{-rt}dt = \Gamma(a+n)/r^{a+n}$, which now serves the transform, the integrability, the moments and the variance in one file. Note what remains true of the* stable *family: there the mean diverges, divergence propagates upward, and 8.16 came off ledger A7 by that route instead. Two families, two different escapes, and neither is the criterion.)*

Proof. By 8.10 the law of T_x is the Gamma law of shape γ and rate $1/x$, so $\mathbb{E}\, T_x^n = \Gamma(\gamma + n)x^n/\Gamma(\gamma)$, finite for every n. In particular $\mathbb{E}\, T_x = \gamma x$ and $\mathbb{E}\, T_x^2 = \gamma(\gamma+1)x^2$, so $\operatorname{Var} T_x = \gamma(\gamma+1)x^2 - \gamma^2 x^2 = \gamma x^2$. □

Proposition 8.13 (non-creation from the signal; the Pólya frequency members)**.** Write $S^-(g)$ for the number of sign changes of a function g on $\mathbb{R}$.

1. For integer γ the Gamma kernels are one-sided Pólya frequency densities, and convolution with them is variation-diminishing: $S^-(\mu_{0,x} * g) \le S^-(g)$ for every $x > 0$ and every integrable or bounded g, sign changes counted in the essential (almost-everywhere) sense. In particular, for the field $u(\cdot, x) := \mu_{0,x} * f$,

   $$S^-\big(u(\cdot,x) - c\big) \ \le\ S^-\big(f - c\big) \qquad \text{for every level } c \in \mathbb{R},$$

   so the field has, at every scale, no more zero-crossings than the signal; and if f is the primitive of a function $f' \in X_0$ — that is, $f \in \mathcal{D}$ — then, counting local extrema as essential sign changes of the derivative,

   $$S^-\big(\partial_t u(\cdot,x)\big) \ \le\ S^-\big(f'\big):$$

   no more local extrema either. For non-integer γ the kernels are not Pólya frequency densities.

2. An admissible family (7.3) has Pólya frequency kernels if and only if

   $$F(s) \ = \ b_0\, s \ + \ \sum_i \log\Big(1 + \frac{s}{\tau_i}\Big), \qquad \tau_i > 0, \quad \sum_i \frac{1}{\tau_i} < \infty$$

   — drift together with integer-weighted Thorin atoms, the summability condition being finiteness of the mean delay, $\mathbb{E}\, T_1 = b_0 + \sum_i \tau_i^{-1}$.

▷ [**A**] analytic interface ledger A19 ledger A20 *(total positivity, cited whole; `@karlin1968total`. Cited, not reproven. **Assignment.** ledger A19 carries the correspondence between variation diminution and total positivity for translation kernels — sufficiency for sign-regular kernels (Ch. 5, Thm 3.1(i), p. 233, stated for bounded Borel arguments; the extension to integrable ones is the standard truncation and is taken with the citation) and necessity for density kernels (Ch. 5, Thm 4.2, pp. 242–243). ledger A20 carries Schoenberg's representation: a one-sided density is Pólya frequency iff the reciprocal of its Laplace transform is entire of the form $Ce^{\delta s}\prod_i(1+\lambda_i s)$ with $C > 0$, $\delta \ge 0$, $\lambda_i \ge 0$, $\sum_i \lambda_i < \infty$ (Ch. 7, Thm 3.2(b), p. 345; the non-integrable one-sided form, Rem. 3.1, p. 346). Everything else is **[T]** in prose: reading $(1 + xs)^\gamma$ against the form, the level reduction, the derivative route $\partial_t(\mu * f) = \mu * f'$ on $\mathcal{D}$, clause (2)'s matching of linear factors to atoms, and the admissibility of the matched exponents through 7.3.)*

Proof. Two classical facts carry the proof. A density supported on $[0,\infty)$ is a Pólya frequency density if and only if the reciprocal of its Laplace transform is an entire function of the form $Ce^{\delta s}\prod_i(1+\lambda_i s)$ with $C>0$, $\delta\geq 0$, $\lambda_i\geq 0$ and $\sum_i\lambda_i<\infty$ — Schoenberg's representation theorem (ledger A20; `@karlin1968total`, Ch. 7, Thm 3.2(b), p. 345; the one-sided form, Remark 3.1, p. 346). And convolution with a kernel is variation-diminishing if and only if the kernel is totally positive (ledger A19; `@karlin1968total`, Ch. 5, Thm 3.1(i), p. 233, and Thm 4.2, pp. 242–243), which for a translation kernel is the Pólya frequency property.

(1) For the Gamma kernel, $1/\widehat{\mu}_{0,x}(s)=(1+xs)^\gamma$: for integer γ this is a polynomial with the single real negative root $-1/x$, of the displayed form with $\delta=0$ and γ equal factors, so the kernels are Pólya frequency densities and convolution with them diminishes essential sign changes of integrable and of bounded functions. The level form follows applied to $g=f-c$, since $\mu_{0,x}*(f-c)=u(\cdot,x)-c$ by unit mass, and the zero-crossing clause is the level $c=0$. For the extremum clause, $f\in\mathcal{D}$ gives $f=\mathbf{1}_{[0,\infty)}*f'$, hence $u(\cdot,x)=\mathbf{1}_{[0,\infty)}*(\mu_{0,x}*f')$ and $\partial_t u(\cdot,x)=\mu_{0,x}*f'$ a.e.; the variation-diminishing bound applied to $g=f'$ gives the display. (The count is read from the derivative and not from the levels: no single level need meet every extremum — two oscillations can occupy disjoint value ranges — so the level bounds alone do not bound the number of extrema.) For non-integer γ, $(1+xs)^\gamma$ has a branch point at $s=-1/x$ and is not entire, so by the same representation theorem the kernel is not a Pólya frequency density.

(2) The kernels are supported on $[0,\infty)$ with transform $e^{-F(xs)}$, so the family is Pólya frequency exactly when $e^{F(xs)}$ has the displayed form — for one x or, equivalently, for all: replacing s by xs rescales δ and the λ_i by x and preserves the form. Matching at $x=1$: $F(0+)=0$ forces $C=1$; the exponential factor is the drift, $\delta=b_0$; and the linear factors are atoms, $\lambda_i=1/\tau_i$, giving the display, the condition $\sum_i\lambda_i<\infty$ being $\sum_i\tau_i^{-1}<\infty$, i.e. $F'(0+)=b_0+\sum_i\tau_i^{-1}<\infty$. Every such F is admissible: it is of the form (7.1) with the nonincreasing memory kernel $k(t)=\sum_i e^{-\tau_i t}$ — a Frullani integral per atom — and $F(1)\leq b_0+\sum_i\tau_i^{-1}<\infty$, so 7.3 applies. ☐

Proposition 8.14 (scale-monotone variation diminution is impossible)**.** Let $(\Phi_{x,y})$ be an admissible family (7.3), in the canonical gauge, and suppose that *every* interior increment operator $\Phi_{x,y}$, $0<x<y$, is variation-diminishing. Then $F(s)=b_0 s$: the family is the pure delay. In particular no nondegenerate admissible family has the scale-monotone form of non-creation, in which every coarser record has no more sign changes than every finer one; the endpoint form of 8.13 is the strongest variation-diminishing property an admissible family can carry, and the integer-atom members of its clause (2) attain it. ▷ [**A**] analytic interface ledger A19 ledger A20 *(the same two interfaces as 8.13, in the same roles.* ***Assignment.*** *ledger A19 converts the hypothesis — every increment operator variation-diminishing — into the Pólya frequency property of the smoothed increment densities (the necessity leg, Thm 4.2), and ledger A20 converts that into the entire product form of $e^g(1+\varepsilon s)^2$. Everything downstream is* ***[T]*** *in prose and elementary: the Lévy-tail comparison through 3.9, the integer-atomic representing measures, the dilation-orbit replication of an atom, and the contradiction with local finiteness. No property of the smoothing beyond 8.13(1)'s Pólya frequency clause is used.)*

Proof. Step 1: the increments' Lévy tails are integer sums of exponentials. Fix $0<x<y$ and write $g(s):=F(ys)-F(xs)$, the increment exponent, a Bernstein function with drift $b_0(y-x)$ and Lévy tail $h(t):=k(t/y)-k(t/x)\geq 0$, by substitution in (7.1). For $\varepsilon>0$ let G_ε be the Gamma density of shape 2 at scale ε, with transform $(1+\varepsilon s)^{-2}$ — a one-sided Pólya frequency density, by the argument of 8.13(1). The measure $\nu_\varepsilon:=\mu_{x,y}*G_\varepsilon$ is absolutely continuous, and its convolution operator, the composition of two variation-diminishing operators, is variation-diminishing; so its density is a one-sided Pólya frequency density (ledger A19; `@karlin1968total`, Ch. 5, Thm 4.2, pp. 242–243, as in 8.13), and the Schoenberg representation (ledger A20; `@karlin1968total`, Ch. 7, Thm 3.2(b), p. 345; Rem. 3.1, p. 346) gives

$$1/\widehat{\nu}_\varepsilon(s) \;=\; e^{g(s)}\,(1+\varepsilon s)^2 \;=\; e^{\delta_\varepsilon s}\prod_i \big(1+\lambda_i^{(\varepsilon)} s\big), \qquad \lambda_i^{(\varepsilon)} \ge 0, \quad \textstyle\sum_i \lambda_i^{(\varepsilon)} < \infty,$$

the constant being 1 at $s = 0$. Both sides are exponentials of Bernstein functions; taking logarithms on $s > 0$ and matching the representations — drifts from the $s \to \infty$ limit, then Lévy tails through their transforms and 3.9 — gives, for a.e. $t > 0$,

$$h(t) + 2\,e^{-t/\varepsilon} \;=\; \sum_i e^{-t/\lambda_i^{(\varepsilon)}} \;=\; \int_{(0,\infty)} e^{-\tau t}\, M^{(\varepsilon)}(d\tau),$$

where $M^{(\varepsilon)} := \sum_i \delta_{1/\lambda_i^{(\varepsilon)}}$ is a positive integer-atomic measure, locally finite on $(0,\infty)$ because $\sum_i \lambda_i^{(\varepsilon)} < \infty$. Comparing two values $\varepsilon \neq \varepsilon'$ through 3.9 once more shows that $M := M^{(\varepsilon)} - 2\,\delta_{1/\varepsilon}$ does not depend on ε; evaluating any single point with a third value of ε whose exceptional atom lies elsewhere shows $M \ge 0$ and integer-atomic. Hence

$$k(t/y) - k(t/x) \;=\; \int_{(0,\infty)} e^{-\tau t}\, M_{x,y}(d\tau) \qquad \text{for a.e. } t > 0,$$

with $M_{x,y}$ positive, locally finite, and integer-atomic.

Step 2: scaling and additivity. The tails satisfy $h_{\sigma x,\sigma y}(t) = h_{x,y}(t/\sigma)$ and $h_{x,z} = h_{x,y} + h_{y,z}$ for $x < y < z$, so by 3.9,

$$M_{\sigma x,\sigma y} \;=\; \text{the pushforward of } M_{x,y} \text{ under } \tau \mapsto \tau/\sigma, \qquad M_{x,z} \;=\; M_{x,y} + M_{y,z}.$$

Step 3: an atom replicates into a continuum. Suppose some $M_{x,y} \neq 0$; by the scaling relation we may take $M_{1,y_0} \neq 0$, with an atom at τ_0 of mass $m_0 \ge 1$. For $\sigma \ge 1$, Step 2 gives $M_{\sigma,\sigma y_0}$ an atom at τ_0/σ of mass m_0; and for $Y \ge \sigma y_0$, additivity and positivity give $M_{1,Y} \ge M_{\sigma,\sigma y_0}$. Hence $M_{1,Y}$ carries an atom of mass at least 1 at *every* point τ_0/σ with $1 \le \sigma \le Y/y_0$ — uncountably many atoms of mass ≥ 1 in the interval $[\tau_0 y_0/Y,\ \tau_0]$ — while

$$e^{-\tau_0 t}\, M_{1,Y}\big([\tau_0 y_0/Y,\ \tau_0]\big) \;\le\; h_{1,Y}(t) \;\le\; k(t/Y) \;<\; \infty$$

for any fixed $t > 0$: a contradiction. So $M_{x,y} = 0$ for every pair, i.e. $k(t/y) = k(t/x)$ for all $0 < x < y$ and a.e. t; a nonincreasing function that is a.e. dilation-invariant is constant on $(0,\infty)$, and $\int_1^\infty k(t)\,t^{-1}dt < \infty$ forces the constant to be 0. Hence $F(s) = b_0 s$. Conversely the pure delay's increments are the shifts $\delta_{b_0(y-x)}$, which are variation-diminishing, so the degenerate case does occur and the statement is sharp. $\square$

Remark 8.15 (ladder monotonicity survives discrete covariance)**.** The impossibility is a continuous-covariance phenomenon, and it is sharp in the ratio: increments of a *single* ratio can all be variation-diminishing. For an integer-atom family (8.13(2)) and a fixed $q > 1$, the increments $\Phi_{x,qx}$, $x > 0$, are variation-diminishing precisely when the atom multiset satisfies $\mathrm{mult}(q\tau) \ge \mathrm{mult}(\tau)$ for every τ: then each pole of $e^{g_{x,qx}}$ at $-\tau/x$ cancels against a zero, the leftover product is of Schoenberg form, and the representation theorem runs backwards to sufficiency (necessity is Step 1 of the proof above, applied at ratio q). The time-causal limit kernel of `@lindeberg2016time` is the fundamental example: its Thorin measure is the geometric train $\sum_{i\ge1} \delta_{q^i/c}$, the condition holds, and every q-step increment exponent is $F(qxs) - F(xs) = \log(1 + cxs)$ — each rung one exponential (Gamma shape-1) kernel, a Pólya frequency density. Along any geometric ladder of ratio q the sign-change count of the record is therefore monotone from rung to rung: scale-monotone non-creation, impossible under full covariance, is available in the discrete-covariance stratum, and the published limit-kernel construction has it.

Proposition 8.16 (the extremal stable family: every moment is infinite)**.** With F as in 8.1, $\mathbb{E}\, T_x^n = \infty$ for every $n \geq 1$ and every $x > 0$, so the delay of the 2005 stable kernels cannot be measured by any moment. ▷ [**T**] proved-target *(split from 8.1, whose exponent is proved; the mode clause that accompanied this one is now 8.17. **Proved in Lean** (2026-08-15), `Hemigroup/StableMoments.lean`, on ledger A17 alone — through the constructed family, as every statement about T_x is, and unavoidably so, T_x being what ledger A17 constructs. **What it no longer costs is ledger A7.** Until this round the node was routed through 8.6 for general n, and that entry's own assignment note already recorded that its $n = 1$ case is not carried by the ledger, being 8.4. Once 8.4 was proved, $n = 1$ became all this node needs: infinite propagates upward, $t \leq 1 + t^n$ on the half-line giving $\mathbb{E}\, T \leq 1 + \mathbb{E}\, T^n$ on a probability measure. So the criterion is invoked here only in the direction "infinite stays infinite", which is elementary, and the proof below now says so. The failure of the mean is what makes 8.4 a statement about* some *admissible families rather than all of them, and it is why the article measures delay by the mode when it must.)*

Proof. By 8.4 the mean delay is $\mathbb{E}\, T_x = x\big(b_0 + \int_0^\infty k\big)$, finite exactly when k is integrable at infinity; here $b_0 = 0$ and $k(t) = \alpha t^{-\alpha}/\Gamma(1-\alpha)$, which is not integrable at infinity for $\alpha < 1$. So $\mathbb{E}\, T_x = \infty$. Every higher moment follows without a further criterion: $t \leq 1 + t^n$ for $t \geq 0$ and $n \geq 1$, so on the probability measure $\mu_{0,x}$ we get $\mathbb{E}\, T_x \leq 1 + \mathbb{E}\, T_x^n$, and a divergent mean forces $\mathbb{E}\, T_x^n = \infty$. □

Proposition 8.17 (the extremal stable family: the mode)**.** With F as in 8.1, T_x is unimodal, and its mode is x times the mode of T_1 in the canonical gauge; in the parametrization of the one-parameter family of 7.4, where $\chi(x) = x^{1/\alpha}$, it scales as $x^{1/\alpha}$. ▷
[**T**] proved-target *(unimodality is Yamazato's theorem on class L laws, cited via 7.7 since $k \not\equiv 0$ for the stable family; the canonical-gauge mode scaling is the trivial change of variables — $\mu_{0,x}$ is the law of xT_1, recorded in 8.4's annotation — and the semigroup-parameter scaling is the case $\chi(x) = x^{1/\alpha}$ of 7.4 — itself proved. **Not formalised, and not for want of the self-similarity**: the development has no definition of the mode, and needs none anywhere else, so introducing one to state a single node would be a definition without a consumer. That is the opposite decision from 'ein', which was defined here because two nodes needed it. If a second consumer appears — the tail-index yardstick is the candidate — both should be defined together. **Corrected (2026-08-23).** The statement previously read "the mode of T_x scales as $x^{1/\alpha}$" unqualified, which is false in the canonical gauge this chapter declares: there $T_x \stackrel{d}{=} xT_1$, so the mode scales as x, and $x^{1/\alpha}$ is only the scaling in the semigroup's own pre-gauge parameter (7.4's $\chi(x) = x^{1/\alpha}$).)*

Proof. By 7.7, T_x is unimodal, since $k \not\equiv 0$ for the stable family. In the canonical gauge $\mu_{0,x}$ is the law of xT_1, so its mode is x times the mode of T_1. The self-similarity is the case $\chi(x) = x^{1/\alpha}$ of 7.4, and the mode of $x^{1/\alpha}T_1$ is $x^{1/\alpha}$ times the mode of T_1. □

Chapter 9

The signaling form: Theorem 4′

Theorem 4 of @fagerstrom2005temporal inverts the scale-Cauchy problem into an evolution *in time* over the memory line: $\partial_t u = D^{1/\alpha}_{x,-} u$ with boundary data $u(t,0) = f(t)$ and $u(0,x) = 0$ — the signaling problem. The operator there is translation-invariant in x, which is available only in the homogeneous (semigroup) case. This chapter constructs the general inversion.

The structural observation that replaces homogeneity is this: writing $H := e^{-F}$, the Laplace-domain profiles of the field,

$$\hat{u}(s,x) = \hat{f}(s)\, H(sx),$$

are *dilates of a single profile* H, with the eigenvalue s appearing as the dilation parameter. Operators diagonal on dilation families are the subject of multiplicative harmonic analysis, so the inversion operator lives in the **Mellin calculus** — the multiplicative-group counterpart of the Wendel step of Chapter 4 on the additive group.

Throughout we work in the canonical gauge, so that $\mu_{0,x}$ is the law of $x\,T_1$ with $\mathbb{E}[e^{-sT_1}] = H(s)$; we write $\tilde{g}(z) := \int_0^\infty x^{z-1} g(x)\,dx$ for the Mellin transform, $\theta := x\,\partial_x$ for the Euler operator, and $(I^z f)(t) := \frac{1}{\Gamma(z)} \int_0^t (t-r)^{z-1} f(r)\,dr$ for the Riemann–Liouville integral of complex order z, $\operatorname{Re} z > 0$ (@samko1993fractional).

Definition 9.1 (the standing hypothesis (H))**.** An admissible exponent F satisfies the *standing hypothesis* (H) if $F(\infty) = \infty$ and

$$z_* \;:=\; \sup\bigl\{\, \zeta > 0 :\; \mathbb{E}\bigl[T_1^{-\zeta}\bigr] < \infty \,\bigr\} \;>\; 1.$$

▷ [**T**] proved-target

*(**Formalised**, as `Hemigroup.SelfDecomposableExponent.StandingHypothesis`. One thing about the formalisation had to be corrected, and the paragraph below this definition is what caught it. z_* **is valued in** $[0,\infty]$**, not in** $\mathbb{R}$**.** The supremum is genuinely infinite for the stable and Bessel-K families — the paragraph below says so, and 10.3(2)* proves *it for every local operator — while Lean's `Real.sSup` of an unbounded set is the junk value* 0. *With a real-valued z_* the hypothesis $1 < z_*$ was therefore **false** exactly for the kernels 10.5 classifies, so every theorem of chapters 9 and 10 was vacuous there, while remaining true and non-vacuous for the Gamma family, where $z_* = \gamma$ is finite. Nothing was unsound and no proof was wrong; the statements simply said nothing about the article's own examples. In $[0,\infty]$ the supremum is ∞, `zStar_eq_top_of_forall_negMoment_ne_top` says so, and (H) holds there rather than failing. **What it cost downstream.** Every strip condition now reads `ENNReal.ofReal c < F.zStar`, and `verticalStrip` takes its right edge in $[0,\infty]$ so that `verticalStrip 0 F.zStar` is unchanged at every use. The arithmetic that used to be `linarith` became three lemmas — monotonicity, a witness above the shift, and the $b-1$ shift, subtraction in $[0,\infty]$ being truncated — and nothing else changed. In particular the interval $(0,z_*)$ is still open and convex, which is what the analyticity and identity-theorem arguments need and what a bare finiteness condition*

would not supply: that is why the abscissa was kept as the primitive rather than replaced by $\mathbb{E}[T_1^{-c}] < \infty$. ***How it surfaced.*** *Not from a failing proof — everything compiled — but from asking what 10.3(2) would unlock, and finding that its conclusion contradicts the hypothesis every downstream statement carries. A hypothesis that a chapter's own theorem refutes is a defect only visible to something that tries to use both.)*

Both clauses are conditions at the zero-delay end of the kernels, and 9.23 and 9.24 supply the readings the definition used to assert in passing: the first clause says exactly that the kernels put no atom at zero delay, is equivalent to $b_0 > 0$ or $\int_0^\infty k(t)\, t^{-1} dt = \infty$, and is automatic for every admissible $F \not\equiv 0$ — so all of (H)'s bite is in the second clause — while the second implies $\int_0^\infty e^{-F(s)}\, ds = \mathbb{E}[T_1^{-1}] < \infty$, the converse failing only at the endpoint $z_* = 1$, an endpoint genuinely reached: the Gamma family at $\gamma = 1$ has $z_* = 1$ with $\mathbb{E}[T_1^{-1}] = \infty$. The second clause holds with $z_* = \infty$ whenever $b_0 > 0$, for the stable family, and for the Bessel-K family of 8.3; for the Gamma family $z_* = \gamma$, so (H) requires $\gamma > 1$ there. The rate is what 9.25 computes in general: $z_* = \lim F(s)/\log s$, which is $k(0^+)$ when $b_0 = 0$ and ∞ otherwise, so (H) reads: drift, or catalogue height above one at the origin. The construction extends below $z_* = 1$ by meromorphic continuation, which we do not pursue. The threshold 1 is the one this chapter keeps meeting — see 9.20, 9.22 and 9.8(2), where it arises three separate times for three different reasons — And it is where the cone's extreme rays sit: every extreme ray with $b_0 = 0$ has $F(s) \sim \log s$, hence $z_* = 1$ (7.6; also 9.25(2), since $k(0^+) = 1$ on every Dickman ray), so those rays lie exactly *on* the boundary of (H). The correspondence is not exact in either direction — the pure delay is an extreme ray and satisfies (H) outright, having $b_0 > 0$, and the Gamma family at $\gamma \le 1$ fails (H) without being extreme — but the coincidence of the three obligations at the extreme rays' abscissa marks the boundary of (H) as characteristic of the theory rather than of any one argument.

Lemma 9.2 (Mellin data of the profile)**.** Under (H), for $0 < \operatorname{Re} z < z_*$,

$$\tilde{H}(z) \;=\; \Gamma(z)\, \mathbb{E}\big[T_1^{-z}\big],$$

with $|\tilde{H}(c + i\tau)| \le \mathbb{E}[T_1^{-c}]\, |\Gamma(c + i\tau)|$ for $0 < c < z_*$. $\triangleright$ **[T]** proved-target ***(Machine-checked****, as* `Hemigroup.SelfDecomposableExponent.mellin_profile` *and* `norm_mellin_profile_le`*;* `#print axioms` *gives ledger A17 and nothing else, inherited through* `kernel` *because* T_1 *is* $\mu_{0,1}$*. In particular* no *chapter-11 interface enters — ledger A12 grounds 9.3, not this.*

The exchange of integrals is licensed by the strip condition and by nothing else. *Both clauses come out of one* $[0,\infty]$*-valued computation,* $\iint s^{c-1} e^{-ts}\, ds\, d\mu(t) = \Gamma(c)\, \mathbb{E}[T_1^{-c}]$*, which is simultaneously the Fubini side condition, the bound, and the finiteness underneath 9.15. Its left-hand side is the total mass of* $|\cdot|$ *for the product measure, so joint integrability holds* iff *the right-hand side is finite, which is* $c < z_*$*. That is the shape of 8.7 again: the condition that makes the analysis go through is characteristic rather than merely sufficient, and seeing that is what keeps a hypothesis from being invented.*

The first clause of (H) is load-bearing. $F(\infty) = \infty$ *is used exactly once, to give* $T_1 > 0$ *almost surely — the mass at the origin is below* $\mathbb{E}[e^{-sT_1}] = e^{-F(s)}$ *for every* s*. Without it the computation above is* false*, not merely unprovable: at an atom* $t = 0$ *the inner integral* $\int_0^\infty s^{c-1} ds$ *diverges, while* $\mathbb{E}[T_1^{-c}]$*, an integral over* $(0,\infty)$*, does not see the atom at all.*

Three design decisions were taken in writing the statement down. T_1 *is the kernel* $\mu_{0,1}$ *the construction already produces, so no new object enters and the density* ϕ_1 *is* not *needed — which matters, because* ϕ_1 *would drag in ledger A9 for a node that does not otherwise need it. The profile is the Laplace transform rather than the density, so Mathlib's* `mellin` *applies to it directly. And the negative moments are carried in* $[0,\infty]$*: divergence is the whole point of* z_**, and the appendix is the cautionary tale about departing from that convention.)*

Proof. By Tonelli,

$$\tilde{H}(z) = \int_0^\infty s^{z-1}\,\mathbb{E}[e^{-sT_1}]\,ds = \mathbb{E}\Big[\int_0^\infty s^{z-1}e^{-sT_1}ds\Big] = \mathbb{E}[T_1^{-z}]\,\Gamma(z),$$

the inner integral being the Gamma integral at rate T_1. The exchange is legitimate because the same computation run on absolute values gives $\Gamma(c)\,\mathbb{E}[T_1^{-c}]$ with $c = \operatorname{Re} z$, finite exactly when $c < z_*$. Since $|T_1^{-z}| = T_1^{-\operatorname{Re} z}$, the identity also gives the stated bound. □

Definition 9.3 (the inversion operator)**.** Fix $c \in (0, z_* - 1)$. For g such that $\tilde{g}$ is defined on the line $\operatorname{Re} z = c$ and $B(-z)\tilde{g}(z)$ is absolutely integrable there, set

$$\big(Ag\big)(x) \;:=\; \frac{1}{x}\cdot\frac{1}{2\pi i}\int_{(c)} x^{-z}\,B(-z)\,\tilde{g}(z)\,dz \;=\; \Big(\frac{1}{x}\,B(\theta)\,g\Big)(x),$$

the second expression being the functional-calculus reading, the Euler operator having Mellin symbol $-z$. ▷ [**A**] analytic interface ledger A12 *(Mellin inversion on a line of absolute convergence;* `@widder1941laplace`*, Ch. VI, §9, Thm. 9a. Cited, not reproven here. The operator itself is* **machine-checked***, as* `Hemigroup.SelfDecomposableExponent.inversionOperator`*: the contour integral is Mathlib's* `mellinInv`*, since parametrising the line by $z = c + iy$ turns dz into $i\,dy$ and cancels the i. It is defined for* every *g, the hypotheses having moved onto the theorems that compute it.* ***Assignment.*** *Narrowed 2026-08-12, when the node was formalised. ledger A12 now carries one step and not three: that absolute integrability of $B(-z)\tilde{g}(z)$ on the line produces the function $B(\theta)g$. Everything after that function exists is* ***[T]*** *and is proved — 9.18.*

The narrowing came out of writing the definition down. "$B(\theta)g$" is a functional calculus for an operator with poles, so the second display is not a rewriting of the first but an assertion that a certain function exists*; Mathlib's inversion theorem recovers a function from its own transform and says nothing about the integral of a product, which is exactly the same gap seen from the other side. So the formal statement takes the referent as given — the h with $\tilde{h} = B(-z)\tilde{g}$ on the line — and what is left over is its production, which is the cited theorem and nothing else. Whether ledger A12 retires is therefore now a question about* this article*: does every use of A exhibit its h? Each use here is A applied to a profile $H(s\,\cdot)$, where $h(x) = s\,x\,H(sx)$ is explicit and $\tilde{h}(z) = s^{-z}\tilde{H}(z+1)$ is 9.16's recursion with the denominator cleared.*

The ledger records a correction to the draft here: it cited `@samko1993fractional` *for this, which states Mellin inversion only as an unhypothesized display, equation (1.113); Widder states it with the absolute-convergence hypothesis that is exactly the one imposed above.)*

The second equality is an assertion and not a rewriting, and it is worth being explicit about what it asserts: $B(\theta)$ is a functional calculus for a symbol with poles, so what the display says is that there *exists* a function h on $(0, \infty)$ whose Mellin transform is $B(-z)\tilde{g}(z)$ on the line, and that the contour integral computes $h(x)/x$. Absolute integrability on the line is exactly the hypothesis under which the inversion integral recovers such an h. Given h, the rest is elementary and is 9.18; in particular the transform-level form is $\widetilde{Ag}(z) = B(1-z)\,\tilde{g}(z-1)$, wherever the identity defining h holds pointwise. Every application below *exhibits* its h rather than inferring it — for the profiles $g = H(s\,\cdot)$ it is $h(x) = s\,x\,H(sx)$, 9.19 — so the cited inversion theorem is available but never spent.

Definition 9.4 (the covariant Mellin class)**.** Fix $c \in (0, z_* - 1)$. An operator A' belongs to the *covariant Mellin class* at height c if there is a function $B'(-z)$ on the line $\operatorname{Re} z = c$ such that A' is of the form $x^{-1}B'(\theta)$ in the sense of 9.3: every g in its domain has a realising function h with $\tilde{h}(z) = B'(-z)\tilde{g}(z)$ at every point of the line where $\tilde{H}(z) \neq 0$, and $A'g = h/x$. The weight x^{-1} makes every member scale like ∂_t under $(t, x) \mapsto (\sigma t, \sigma x)$, which is the covariance (A8) requires of an inversion — the class is the covariant one. ▷

[**T**] proved-target *(**Not tagged.** `Hemigroup.SelfDecomposableExponent.RealisesAction` is exactly the realising-function clause above, for one g and its h: the transform identity off the zeros of $\tilde{H}$, together with `MellinConvergent h c` and `Complex.VerticalIntegrable (mellin h) c`, which are the two convergence hypotheses 9.3 imposes. What the Lean has* not *bundled is the quantifier "every g in A′'s domain" into a single class-membership predicate on operators: no theorem here needs to quantify over the class as a whole, since 9.5 and 9.8(3) each instantiate `RealisesAction` directly at the one g in play — the profile $H(s\,\cdot)$ and its exhibited eigenfunction — at every height c' in range, rather than through a bundled class predicate. So the definition above is faithful to what `RealisesAction` says pointwise, but has no single Lean declaration proving it as a statement about operators, and is left untagged.)*

Lemma 9.5 (uniqueness of the symbol)**.** If two operators of the form $x^{-1}B_1(\theta)$ and $x^{-1}B_2(\theta)$ both satisfy $A[H(s\,\cdot)] = s\,H(s\,\cdot)$ for some $s > 0$, at every height $c \in (0, z_* - 1)$ — that is, at every such c both are of that form on $H(s\,\cdot)$ at height c and both send it to $s\,H(s\,\cdot)$ — then $B_1 = B_2$ on the strip. Hence A is the unique inversion within the covariant Mellin class (9.4). ▷ [**T**] proved-target *(**Machine-checked**, both halves. Step 2 is 9.17; step 1 is `Hemigroup.SelfDecomposableExponent.sameSymbolAction_of_realisesAction`, and `eventuallyEq_inver` composes them against the article's own symbol, so the statement is not vacuous and the definite article in "*the *inversion" is earned. `#print axioms` gives ledger A17 and nothing else — in particular step 1 does* not *carry ledger A12, which is what this node was recorded as waiting for.*

"Of the form $x^{-1}B(\theta)$" is the hypothesis, and reading it that way is what unblocked the node. *It was recorded as waiting on ledger A12 because step 1 was read as needing 9.3's transform identity, hence the* production *of $B(\theta)g$ from an integrability hypothesis. But an operator of that form is one whose $B(\theta)g$ is* given*; the class quantified over in the statement is exactly the class for which the referent exists. So the citation was never needed here.*

Two things the proof does not need. *It needs no injectivity of the inverse Mellin transform — the obvious tool, and a second citation (`@widder1941laplace`, Thm. 6a, listed in the ledger against just this possibility). Two operators agreeing on $H(s\,\cdot)$ have the* same *realising function there, since $x^{-1}h_1 = x^{-1}h_2$ on $(0,\infty)$ is $h_1 = h_2$; so their transforms agree at every point and the symbols are read off directly. And it needs only* one *dilation, not all $s > 0$: the dilate contributes the factor s^{-z}, which never vanishes and cancels — which is why the statement asks for one s. What it does need is* every height*: the hypothesis on the line* $\operatorname{Re} z = c$ *yields agreement on that line, and the lines fill the strip only as c ranges over $(0, z_* - 1)$. A single height would suffice only if B_1, B_2 were assumed meromorphic on the strip, which the statement does not assume; the formal statement quantifies over the heights and assumes no meromorphy (fidelity review, finding R10).)*

Proof. Fix a height $c \in (0, z_* - 1)$. Both operators send $H(s\,\cdot)$ to $s\,H(s\,\cdot)$, so on $(0,\infty)$ their realising functions at height c agree; call the common value h. Then $B_1(-z)\,\tilde{g}(z) = \tilde{h}(z) = B_2(-z)\,\tilde{g}(z)$ at every point of the line $\operatorname{Re} z = c$ where the identity holds, and $\tilde{g}(z) = s^{-z}\tilde{H}(z)$, so cancelling the nonvanishing factor s^{-z} gives $B_1(-z)\tilde{H}(z) = B_2(-z)\tilde{H}(z)$ at every point of the line where $\tilde{H}(z) \neq 0$, and trivially at its zeros, where both sides vanish. As c ranges over $(0, z_* - 1)$ the lines fill the strip, which is the hypothesis of 9.17, and that lemma concludes. □

Lemma 9.6 (the memory line stores fractional integrals)**.** Under (H), let $f \in X_0$ be causal and $u(\cdot, x) := \Phi_{0,x} f$. Then for every $t > 0$ and $1 < \operatorname{Re} z < z_*$,

$$\widetilde{u(t,\cdot)}(z) \;=\; \int_0^\infty x^{z-1}\,u(t,x)\,dx \;=\; \tilde{H}(z)\,\big(I^z f\big)(t),$$

absolutely convergent. If moreover $f \in \mathcal{D}$, the same holds for $\partial_t u$ — the X_0-derivative of A.18, i.e. $u(\cdot, x)$ is the primitive of $\mathbb{E}[f'(\cdot - xT_1)]$ — with $I^z f$ replaced by $I^z f' = I^{z-1} f$; and in that case the identity extends to the wider strip $0 < \operatorname{Re} z < z_*$. ▷ [**T**] proved-target

Proof. In the canonical gauge $u(t,x) = \mathbb{E}\big[f(t - xT_1)\big]$. Tonelli applies, since for $c = \operatorname{Re} z > 1$, $\mathbb{E}[T_1^{-c}] \int_0^t y^{c-1} |f(t-y)|\, dy \le \mathbb{E}[T_1^{-c}]\, t^{c-1} \|f\|_1 < \infty$, and gives

$$\int_0^\infty x^{z-1}\, \mathbb{E}\big[f(t - xT_1)\big]\, dx \;=\; \mathbb{E}\big[T_1^{-z}\big] \int_0^t y^{z-1} f(t-y)\, dy \;=\; \mathbb{E}\big[T_1^{-z}\big]\, \Gamma(z)\, (I^z f)(t),$$

which is the claim by 9.2. (For bounded f the inner integral converges for every $c > 0$, which is the wider strip.) For $f \in \mathcal{D}$, f is bounded — $f(0) = 0$ and $f' \in X_0$ give $\|f\|_\infty \le \|f'\|_1$ — so the weight x^{z-1} is then integrable at the origin on its own, which is where the extension to $c > 0$ is used. Also for $f \in \mathcal{D}$: $u(\cdot, x) = \mu_{0,x} * f$ and $f = \mathbf{1}_{[0,\infty)} * f'$, so $u(\cdot, x) = \mathbf{1}_{[0,\infty)} * (\mu_{0,x} * f')$ — that is, $u(\cdot, x)$ is the primitive of $\mathbb{E}[f'(\cdot - xT_1)]$, which is what $\partial_t u = \mathbb{E}[f'(\cdot - xT_1)]$ asserts in X_0. Finally $I^z f' = I^{z-1} f$: substituting $f(r) = \int_0^r f'$ and exchanging the order of integration over $0 < \rho \le r \le t$ leaves $\int_\rho^t (t-r)^{z-2} dr = (t-\rho)^{z-1}/(z-1)$, and $\Gamma(z) = (z-1)\Gamma(z-1)$; the semigroup identity $I^{z-1} I^1 = I^z$ is not needed (9.22). □

Remark 9.7 (what the core supplies)**.** Of the three conditions defining $\mathcal{D}$ the proofs use exactly three consequences and no others: boundedness, which is what widens the strip; that f is the primitive of f', which is what the derivative clause asserts; and $f \in X_0$ itself, needed for the Laplace form of 9.8(2) and *not* implied by the other two, a primitive of an L^1 function lying in L^1 only when $\int_0^\infty f' = 0$. Density and invariance under τ_r and $\Phi_{x,y}$ — the substance of A.18 — are not used here.

The first clause is machine-checked*, as 9.20, which also carries the identification of $\mathbb{E}[f(t - xT_1)]$ with $\Phi_{0,x} f$ and so the canonical gauge at the level of measures. See that node's remark for why a representative of the field has to be named at all — the quantifiers "for each x, a.e. t" do not commute, so this is not avoided by weakening the reading.*

The derivative clause is machine-checked*, as* `Hemigroup.SelfDecomposableExponent.mellin_delay` *its two pieces being 9.22 and* `delayedField_eq_setIntegral`*; writing them down showed that* ***the node this proof cites is not the node it needs****. The derivation goes through $f \in \mathcal{D}$ and hence A.18 — density of the core, invariance under the delay semigroup and under Φ, the L^1 difference quotient — and uses none of it. What it uses is differentiation under the integral sign (which needs f to be an integral of an L^1 function, and nothing about Φ) and the identity $I^z f' = I^{z-1} f$ (which mentions neither the field nor the core). $\mathcal{D}$ enters only as a convenient source of those hypotheses.* What a proof cites is an upper bound on what a statement needs, *and this is the third place in the chapter where the two have come apart.*

And "$\partial_t u$" has to say which derivative it means, or the clause is false. *Read pointwise it asserts a derivative at every t, and that does not hold: $f \in \mathcal{D}$ is absolutely continuous, so f' exists only almost everywhere, and $\mathbb{E}[f(t - xT_1)]$ is a convolution of two L^1 functions — hence L^1, not continuous, so $\mathbb{E}[f'(t - xT_1)]$ has no pointwise values to be a derivative* at. *Continuity could be bought from absolute continuity of T_1's law (*`@sato1999levy`*, Thm. 27.13), an interface, for no gain. The article never meant the pointwise reading — ∂_t in §10 is the X_0-derivative — and what A.18's Φ-invariance argument establishes is $\mu * f = \mathbf{1}_{[0,\infty)} * (\mu * f')$: the field of f is the* primitive *of the field of f'. That is the form proved, it is an identity between two genuine functions of t, and it needs no interface. Third instance in this chapter of a statement having to name its reading, after 9.17 and 9.18.)*

The wider strip is not decoration; 9.8(2) needs it. *That node applies this lemma at $z - 1$, and states the range $1 < \operatorname{Re} z < z_*$ — which puts $z - 1$ outside the range stated here for $f \in X_0$. The chain closes because (2) hypothesises $f \in \mathcal{D}$, which is bounded; so nothing about the result changes, and what changes is what this lemma* says. *The gap was invisible on paper and surfaced immediately on chaining the two in Lean, the second clause's hypothesis having quietly been doing work the first clause's range did not advertise.)*

The Riemann–Liouville integral is defined in the development, not cited. *Mathlib has no fractional integral of any order —* `Analysis/` *carries Mellin, Fourier, convolution and* *distributions and nothing fractional — so* `Skeleton.riemannLiouville` *gives I^z its definition. That is not an interface:* `@samko1993fractional` *is cited here for the notation and the theory of I^z, and nothing in this chapter uses more of that theory than the definition itself.*

Theorem 9.8 (Theorem 4′: the signaling form)**.** Assume (H) and work in the canonical gauge. Then:

1. *(Eigenfunctions.)* For every $s > 0$ the profile $H(s\,\cdot)$ lies in the domain of 9.3 for every $c \in (0, z_* - 1)$, and
$$A\big[H(s\,\cdot)\big](x) \;=\; s\,H(sx), \qquad x > 0.$$

2. *(The field solves the signaling problem.)* Let $f \in \mathcal{D}$ and $u(\cdot, x) := \Phi_{0,x} f$. Then u is causal in t with $u(\cdot, x) \to f$ in X_0 as $x \downarrow 0$, and
   - Laplace form: for every $s > 0$, $A\big[\hat{u}(s, \cdot)\big] = s\,\hat{u}(s, \cdot)$, with $\hat{u}(s, 0+) = \hat{f}(s)$;
   - time-domain (Mellin) form: for every $t > 0$ and every z with $1 < \operatorname{Re} z < z_*$ and $\tilde{H}(z-1) \neq 0$ — all of the strip off an isolated set, 9.10 —
   $$\widetilde{\partial_t u(t, \cdot)}(z) \;=\; B(1-z)\,\widetilde{u(t, \cdot)}(z-1),$$
   both transforms absolutely convergent; that is, $\partial_t u = A\,u$ read through the Mellin transform in x.

3. *(Uniqueness.)* A is the unique operator in the covariant Mellin class (9.4) with property (1), at every height c; in particular, it is the operator of the signaling problem (2).

▷ [**T**] proved-target *(the hemigroup analogue of* `@fagerstrom2005temporal`*, Thm. 4, whose translation-invariant operator is recovered in 9.11 as an accident of homogeneity. Proved here, not cited.*

Clauses (1) and (3) are machine-checked, and so is (2)'s Mellin form*, the last as* `Hemigroup.SelfDecomposableExponent.mellin_signaling_form`*; what is left of (2) is its causality and boundary clauses, which are (A3) and (A7), and its Laplace form, which is (1) applied to $\hat{u}(s, \cdot) = \hat{f}(s)H(s\,\cdot)$. Clause (1) is 9.19, which is where its Lean tag also sits. All conjuncts are now assembled into* `Hemigroup.SelfDecomposableExponent.signaling_form`*, which is what this node carries: clause (1) is 9.19 in both its halves — the domain claim, that $H(s\,\cdot)$ lies in the domain of 9.3 at height c, is the conjunct* `RealisesSymbolAction` *(*`realisesSymbolAction_profile`*), and the eigen-equation is* `inversionOperator_profile`*; clause (3) is 9.5; and clause (2) is causality (*`delayedField_eq_zero`*), boundary attainment (*`tendsto_Phi_zero`*, i.e. (A7) with (A6), stated for every element of X_0) together with the identification $u(\cdot, x) = \Phi_{0,x} f$ (*`coeFn_Phi_zero`*), the Laplace form (*`laplaceFun_delayedField` *with* `inversionOperator_const_mul_profile`*) together with the boundary value $\hat{u}(s, 0+) = \hat{f}(s)$ (*`tendsto_laplaceFun_delayedField`*), the X_0-reading of $\partial_t u$ — $u(\cdot, x)$ is the primitive of the field of f', at every t (*`delayedField_eq_setIntegral'`*) — and the Mellin form (9.21) with the absolute convergence of both transforms (*`mellinConvergent_delayedField_pair`*). The domain claim, the boundary value, the identification, the derivative reading and the convergence were added as conjuncts by the fidelity review (findings R9, R11); before it they were proved as lemmas but not asserted by the theorem. Two side conditions are explicit in the statement and discussed at the nodes that introduce them: the Mellin form holds off the zeros of $\tilde{H}(z-1)$ (9.10), which the display now says, and "the covariant Mellin class" in clause (3) is 9.4 — membership is the hypothesis that $B(\theta)g$ exists (9.3), asked at every height c as 9.5 requires — the theorem keeps its full statement and its number, on the pattern of 9.5 and 9.17. That clause is also what empties the article's use of ledger A12: it* exhibits *the realising function rather than inferring its existence, and the Laplace form of clause (2) is clause (1) applied to*

$\hat{u}(s,\cdot) = \hat{f}(s)H(s\,\cdot)$, so no other shape of g ever reaches A. ***Clause (3)'s "hence with property (2)" is corrected to a comment.*** *The Lean conjunct proves uniqueness only against property (1): for every candidate symbol B, if $x^{-1}B(\theta)$ has the eigenfunction relation at every height, then B agrees with A's own symbol as germs on the strip; it says nothing about property (2), and (2)⇒(1) is not proved (it would need linearity plus division by $\hat{f}(s) \neq 0$, neither stated or used). Property (2) is established for A separately, in clause (2); "in particular" records that fact about the already-identified A, not a further uniqueness claim among operators satisfying (2).)*

Proof. (1) The realising function is exhibited: put $h(x) := s\,x\,H(sx)$, so that $\tilde{h}(z) = s^{-z}\tilde{H}(z{+}1)$, the weight x shifting the Mellin argument by one and the two powers of s combining as $s{\cdot}s^{-(z+1)}$. On the other side $\tilde{g}(z) := \widetilde{H(s\,\cdot)}(z) = s^{-z}\tilde{H}(z)$, so

$$B(-z)\,\tilde{g}(z) = \frac{\tilde{H}(z+1)}{\tilde{H}(z)} \cdot s^{-z}\tilde{H}(z) = s^{-z}\tilde{H}(z+1) = \tilde{h}(z)$$

at every point of the line where $\tilde{H}(z) \neq 0$, which is all but an isolated — hence null — set of it. The cancellation is the whole content, being 9.16's recursion with the denominator cleared; it should *not* be described as a product containing no division, since the division is there and clearing it is what costs the exceptional set. Both convergence hypotheses of 9.3 hold for h at height c, being 9.2 and 9.15 at height $c+1 \in (1, z_*)$, and h is continuous because H is. So h realises $B(\theta)H(s\,\cdot)$ and 9.18 gives $A[H(s\cdot)](x) = h(x)/x = s\,H(sx)$: the weight x^{-1} of 9.3 is exactly what cancels the weight x that shifted the transform, so the exhibition of h and the eigenfunction relation are one statement rather than two. This clause is 9.19.

(2) Causality and the boundary attainment are (A3) and (A7), and $u(\cdot,x) = \Phi_{0,x}f$ is the convolution representation of 7.3 read in the canonical gauge, $\mu_{0,x}$ being the law of xT_1. The Laplace form is (1) applied to $\hat{u}(s,\cdot) = \hat{f}(s)H(s\,\cdot)$, and the boundary value $\hat{u}(s,0+) = \hat{f}(s)$ is $H(sx) \to H(0) = 1$, the transform of a probability measure being continuous on $[0,\infty)$. The X_0-reading of $\partial_t u$ is the derivative clause of 9.6: $u(\cdot,x)$ is the primitive of $\mathbb{E}[f'(\cdot - xT_1)]$. The Mellin form combines 9.6 with 9.2, both transforms converging absolutely on the stated strip by that lemma:

$$B(1-z)\,\widetilde{u(t,\cdot)}(z-1) = \frac{\tilde{H}(z)}{\tilde{H}(z-1)} \cdot \tilde{H}(z-1)\,(I^{z-1}f)(t) = \tilde{H}(z)\,(I^{z-1}f)(t) = \partial_t\widetilde{u(t,\cdot)}(z).$$

(3) is 9.5. □

Remark 9.9 (reading of Lemma 11.5; embodiment)**.** 9.6 gives the motivation of `@fagerstrom2005temporal`, §5, an exact form: at each instant t, the Mellin analysis of the memory line holds precisely the analytic family of Riemann–Liouville integrals $\{(I^z f)(t)\}$ of the *past* signal, weighted by the negative delay-moments $\mathbb{E}[T_1^{-z}]$. The observer's embodiment of its past is fractional integration, with the admissible weightings classified by 7.3.

Remark 9.10 (classical interpretation; poles)**.** The time-domain equation in 9.8(2) is stated at the level of Mellin transforms, and that is the honest general formulation: B may have poles, at zeros of $\tilde{H}$, and polynomial growth along vertical lines, so $Au(t,\cdot)$ need not be given by an absolutely convergent integral for rough f. On the field the identity is nevertheless regular, the factor $\tilde{H}(z-1)$ cancelling every denominator. That last statement is about the product as a *meromorphic function*; read pointwise it needs $\tilde{H}(z-1) \neq 0$, which holds off an isolated set by 9.16, and 9.8(2) is stated accordingly. The distinction recurs through this chapter — it is the same one that makes 9.17's conclusion an equality of meromorphic functions rather than of functions — and is worth keeping in view: a cancellation performed on symbols has to be justified at each point where the identity is used.

When the symbol B is a *polynomial*, A is a local differential operator and the equation is a classical PDE. That is precisely the situation of Theorem $5'$ in Chapter 10, and of 9.13 below.

9.8(2) closes with an informal gloss — "that is, $\partial_t u = A\,u$ read through the Mellin transform in x" — and clause (3) proves uniqueness of the *operator* A within the covariant Mellin class, not uniqueness of the *solution* u: two different claims the single word "unique" can hide. The gap is not merely presentational. The eigenvalue recursion $s\,\tilde{g}(z) = B(1-z)\,\tilde{g}(z-1)$ that pins $\hat{u}(s,\cdot)$'s Mellin data has the classical period-one ambiguity: its general solution is $\tilde{g}(z) = s^{-z}\tilde{H}(z)\,p(z)$ for an *arbitrary* 1-periodic p, the same obstruction Bohr–Mollerup closes for Γ, one level down. What follows closes it here. 9.26 names what a solution of the signaling problem *is*, mirroring 9.8(2)'s own clauses rather than restating them; 9.27 kills the periodic ambiguity, under a normalisation the field satisfies for free — its own modes are dominated by the profile, uniformly over one period, the theory's counterpart of the log-convexity hypothesis Bohr–Mollerup imposes; and 9.28 assembles existence with *solution*-uniqueness under that normalisation — existence and uniqueness in the profile-dominated Mellin class, which 9.8 alone does not deliver. A classical-solution clause, under an extra decay hypothesis on the negative-moment function m that the Gamma, stable and Bessel-K families satisfy and the pure delay fails — correctly, that family belonging instead to 10.5's territory — is 9.29. And of Hadamard's three demands the arc so far supplies the first two; 9.31 completes the third from the representation — continuous dependence on the data, and on the exponent itself besides — so that the signaling problem is well posed in the stated class, with 9.32 marking the boundary of the claim.

Example 9.11 (extremal stable; recovering Theorem 4 of 2005)**.** For $F(s) = s^{\alpha}$ we have $\tilde{H}(z) = \Gamma(z/\alpha)/\alpha$ and

$$B(-z) \;=\; \frac{\Gamma\big(\frac{z+1}{\alpha}\big)}{\Gamma\big(\frac{z}{\alpha}\big)} \;\sim\; \Big(\frac{z}{\alpha}\Big)^{1/\alpha} \quad (z\to\infty),$$

a symbol of order $1/\alpha$. On the profiles $e^{-(sx)^{\alpha}}$ the right-sided Liouville derivative acts by $D^{1/\alpha}_{x,-}e^{-cx} = c^{1/\alpha}e^{-cx}$ with $c = s^{\alpha}$, i.e. it also produces the eigenvalue s; so the Mellin representative and the operator of `@fagerstrom2005temporal`, Thm. 4, agree on the span of the field profiles. The translation-invariant representative is a coincidence of homogeneity; the Mellin representative is the one that survives in general.

Example 9.12 (the Gamma family; scale-recursive form)**.** For $F(s) = \gamma\log(1+s)$ with $\gamma > 1$ we have $\tilde{H}(z) = \Gamma(z)\Gamma(\gamma-z)/\Gamma(\gamma)$ and $B(-z) = z/(\gamma-1-z)$, i.e.

$$B(\theta) = -\,\theta\,\big(\theta+\gamma-1\big)^{-1}, \qquad A = -\,\frac{1}{x}\,\theta\,\big(\theta+\gamma-1\big)^{-1},$$

a *rational* Mellin symbol — the memory-line mirror of the time-side resolvent $\gamma\,\partial_t(1+x\partial_t)^{-1}$ of A.23. Explicitly, $w := (\theta+\gamma-1)^{-1}u$ solves $x\,w_x + (\gamma-1)w = u$, with the regular-at-0 solution

$$w(t,x) = x^{-(\gamma-1)}\int_0^x y^{\gamma-2}\,u(t,y)\,dy, \qquad \partial_t u \;=\; -\tfrac{1}{x}\,\theta\,w \;=\; -\,\partial_x w \quad \text{(one scale-Volterra step)},$$

so the time evolution at scale x is computed from the field at *finer* scales by a single weighted average — scale-recursive, one-sided from the boundary, matching the causal architecture of the time side.

Example 9.13 (parabolic gauge; the Bessel family is the quadratic case)**.** In the α-gauge (similarity variable $x^{1/\alpha}s$; 6.4) the profiles are $H(x^{1/\alpha}s)$, the Euler operator acts on the profile through $\alpha^{-1}\theta$, and the inversion becomes $A = x^{-1/\alpha}\,B(\alpha\,\theta)$. Take the parabolic gauge $\alpha = \frac{1}{2}$, i.e. the coordinate ξ of 6.4, and the Bessel-K family of 8.3, $T_1 = 1/(2\gamma_a)$: then $\mathbb{E}[T_1^{-z}] = 2^z\,\Gamma(a+z)/\Gamma(a)$, so

$$B(-z) = 2z(z+a), \qquad A = \xi^{-2}\Big(\tfrac{\theta^2}{2} - a\theta\Big) = \tfrac{1}{2}\,\partial_\xi^2 + \frac{\beta}{\xi}\,\partial_\xi, \quad \beta = \tfrac{1}{2} - a.$$

The symbol is a *quadratic polynomial*, so A is local: the signaling form of the Bessel-K scale space is exactly the Bessel diffusion-with-drift medium, and 9.8(2) becomes the classical signaling problem $\partial_t u = \frac{1}{2}u_{\xi\xi} + \frac{\beta}{\xi}u_\xi$ with $u(t,0) = f(t)$ and $u(0,\xi) = 0$.

Remark 9.14 (Markov media)**.** Within the class of 9.8, the operators A that additionally satisfy the positive maximum principle (10.1) — i.e. generate a sub-Markov process on the memory half-line, so that the medium is a physical transport medium — are, after the logarithmic substitution $x = e^\xi$, exactly those whose symbol B is the Laplace exponent of a Lévy process. We record this without proof here. By the Lamperti correspondence (`@lamperti1972semi`) these media are positive self-similar Markov processes absorbed at the boundary, and the kernels are the boundary-arrival times $T_x \overset{d}{=}$ absorption time from depth x.

The spectral orientation of the jumps is *not* constrained. For spectrally positive media the process creeps downward through every level and the hemigroup (A6) holds *pathwise*, the kernels being genuine level-passage times; but media with downward jumps are also admissible, the factorization then holding in law though not pathwise, being inherited from the exponent F rather than from the paths.

The Gamma family is the standard example: unfolding 9.12, its medium is the multiplicative jump process that waits at scale x an Exp(mean x) time and jumps to σx with $\sigma \sim \mathrm{Beta}(\gamma-1, 1)$, reaching the boundary by accumulation of jumps; a first-jump computation confirms $\mathbb{E}_x[e^{-s\tau_0}] = (1+sx)^{-\gamma}$.

A necessary condition in all cases — again recorded without proof — is quadratic symbol growth, $B(\theta) = O(\theta^2)$. By 9.11 the homogeneous members with $\alpha < \frac{1}{2}$, of symbol order $1/\alpha > 2$, admit *no* Markov medium, although 9.8 always supplies the analytic signaling form: the operator exists, the process does not. The locality theorem settles the local sub-case — a polynomial symbol of degree at most two — but the general classification of which admissible F are Markov-realizable is the memory-line inverse problem, and it is left open; it is listed with the open problems in the concluding chapter.

Lemma 9.15 (the profile's transform is integrable on vertical lines)**.** Under (H), for every $c \in (0, z_*)$ the function $\tau \mapsto \tilde{H}(c+i\tau)$ is absolutely integrable on $\mathbb{R}$. ▷ [**T**] proved-target *(**Machine-checked**, as `Hemigroup.SelfDecomposableExponent.verticalIntegrable_mellin_profile`; `#print axioms` gives ledger A17 and nothing else. The Lean statement is `Complex.VerticalIntegrable`, which is* verbatim *the hypothesis Mathlib's `mellinInv_mellin_eq` asks for, so this node is the bridge between the chapter and the inversion theorem it cites.*

It was recorded twice as blocked on Mathlib, and was not. *The bound of 9.2 reduces this to the decay of $|\Gamma(c+i\tau)|$, and Mathlib has no such estimate — its Stirling file is Stirling's formula for $n!$ alone. That is true about Mathlib and false about the obligation. The classical asymptotic $|\Gamma(c+i\tau)| \sim \sqrt{2\pi}\,|\tau|^{c-1/2}e^{-\pi|\tau|/2}$ does need Stirling in the complex plane;* integrability needs only quadratic decay, *and quadratic decay is the functional equation twice. That is the proof below, and it is elementary.*

So this chapter is no longer blocked upstream. *What 9.3 still needs from Mathlib's inversion theorem is the* operator *formulation — exhibiting the function whose Mellin transform is $B(-z)\tilde{g}(z)$ — and a continuity hypothesis. Both are questions about this article, which is where ledger A12's correction said the cost would sit once the estimate was in hand.* `blueprint/AXIOMS.md` *now records four assessments of that entry and what each of the first three got wrong.)*

Proof. $|\tilde{H}(c+i\tau)| \le \mathbb{E}[T_1^{-c}]\,|\Gamma(c+i\tau)|$ by 9.2, the first factor being finite because $c < z_*$. For the second, two elementary bounds. Euler's integral gives $|\Gamma(\sigma+i\tau)| \le \Gamma(\sigma)$ for $\sigma > 0$, the imaginary part only rotating the integrand; and $\Gamma(z+2) = (z+1)z\Gamma(z)$ with $|z|, |z+1| \ge |\tau|$ — both have imaginary part τ — gives $|\Gamma(c+i\tau)|\,\tau^2 \le \Gamma(c+2)$. Adding them,

$$|\Gamma(c+i\tau)|\,(1+\tau^2) \;\le\; \Gamma(c) + \Gamma(c+2),$$

and $(1+\tau^2)^{-1}$ is integrable on the line. □

Lemma 9.16 (the inversion symbol)**.** Under (H), $\tilde{H}$ is analytic on the strip $0 < \operatorname{Re} z < z_*$ and does not vanish identically there. Define the *inversion symbol*

$$B(-z) \;:=\; \frac{\tilde{H}(z+1)}{\tilde{H}(z)} \;=\; z\,\frac{\mathbb{E}[T_1^{-z-1}]}{\mathbb{E}[T_1^{-z}]}, \qquad 0 < \operatorname{Re} z < z_* - 1.$$

Then B is meromorphic on that strip, with poles only at the isolated zeros of $\tilde{H}$. ▷ **[T]** proved-target *(**Machine-checked**, as `Hemigroup.SelfDecomposableExponent.meromorphicOn_inversionSymbol` together with `analyticAt_mellin_profile` (analyticity), `mellin_profile_ofReal_ne_zero` ($\tilde{H} \not\equiv 0$), `eventually_mellin_profile_ne_zero` (the zeros are isolated), `inversionSymbol_eq` (the closed form) and `analyticAt_inversionSymbol` (poles only at those zeros); `#print axioms` gives ledger A17 and nothing else. This is the clause every later node in the chapter consumes: 9.3 needs B and its strip, 9.5 needs the isolated zeros, 9.10 needs the poles.*

The analyticity came from the identity, not from the Mellin integral. *The direct route is Mathlib's `mellin_differentiableAt_of_isBigO_rpow`, which wants $H(s) = O(s^{-a})$ for every $a < z_*$ — true, since H is antitone and $H(s)s^a/a \le \int_0^s u^{a-1}H(u)du$, but work, and it re-derives from the Mellin integral what 9.2 already says. Reading $m(z) = \mathbb{E}[T_1^{-z}] = \mathbb{E}[e^{-z\log T_1}]$ as the complex moment-generating function of $-\log T_1$ instead makes it Mathlib's `analyticAt_complexMGF`, whose hypothesis is finiteness of $\mathbb{E}[T_1^{-c}]$ on an open interval of exponents — which is $(0, z_*)$, and is 9.2's own strip condition verbatim. So the strip of analyticity and the strip of the identity are the* same *strip for the* same *reason, rather than two conditions that coincide.*

The closed form needs no non-vanishing hypothesis. *$\Gamma(z+1) = z\Gamma(z)$ is the whole of it, and that cancellation is also why B alone grows along vertical lines where $\tilde{H}$ decays — the point ledger A12 turns on. Lean's $x/0 = 0$ makes both sides of the closed form vanish together at a zero of $\tilde{H}$, so the statement is unconditional and no reader is misled into thinking a side condition was checked.)*

Proof. $\tilde{H}$ is analytic on the strip by differentiation under the integral sign, the majorant $s^{c-1}H(s)$ being locally uniformly integrable there. It is not identically zero: by 9.2, $\tilde{H}(z) = \Gamma(z)\mathbb{E}[T_1^{-z}]$, and both factors are strictly positive at real $z \in (0, z_*)$. Hence its zeros are isolated, and the quotient $\tilde{H}(z+1)/\tilde{H}(z)$ — which is $z\,\mathbb{E}[T_1^{-z-1}]/\mathbb{E}[T_1^{-z}]$ by $\Gamma(z+1) = z\Gamma(z)$ — is meromorphic with poles only there. □

Lemma 9.17 (rigidity of the symbol)**.** Assume (H), and let B_1, B_2 satisfy

$$B_1(-z)\,\tilde{H}(z) \;=\; B_2(-z)\,\tilde{H}(z), \qquad 0 < \operatorname{Re} z < z_* - 1.$$

Then $B_1 = B_2$ as meromorphic functions on that strip: they agree on a punctured neighbourhood of every point of it. Where in addition both are continuous — in particular at any point that is a pole of neither — they agree there too. ▷ **[T]** proved-target *(**Machine-checked**, as `Hemigroup.SelfDecomposableExponent.SameSymbolAction.eventuallyEq`, with `eqOn_of_ne_zero` (agreement off the zeros of $\tilde{H}$), `eq_of_continuousAt` and `eqOn` (the pointwise readings) alongside it; `#print axioms` gives ledger A17 and nothing else. Split from 9.5, which keeps the reduction of the operator relation to the one above — that step is 9.3's transform-level identity* *and carries ledger A12. This is the half that does the mathematical work: the cancellation is what makes the eigenfunction relation* pin *the symbol rather than merely constrain it.*

"$B_1 = B_2$ on the strip" has to say which equality it means, and writing it in Lean is what forced the question. *The symbols are meromorphic, with poles at the zeros of $\tilde{H}$; at a pole a Lean function still has a value, a junk one, so pointwise equality of functions is the wrong reading and would be false as stated. The unconditional statement is agreement on a punctured neighbourhood of every point — which is exactly what equality of meromorphic functions means, and exactly what the draft's "everywhere on the strip by meromorphy" is asserting. Pointwise equality is then recovered under a continuity hypothesis that says, in so many words, that the symbols have no poles. Both forms are proved, and a consumer has to choose; that choice is information the prose was eliding.)*

Proof. Where $\tilde{H}(z) \neq 0$, cancel it. By 9.16 the zeros of $\tilde{H}$ are isolated, so that cancellation is available on a punctured neighbourhood of *every* point of the strip, the zeros included; hence B_1 and B_2 agree near every point, the point itself excepted. If both are continuous at such a point, the punctured neighbourhood determines the value there and the agreement extends to it. □

Lemma 9.18 (the operator computes the realising function)**.** Fix $c > 0$ and g, and let h be a function on $(0, \infty)$ such that

$$\tilde{h}(z) = B(-z)\,\tilde{g}(z)$$

for almost every z on the line $\operatorname{Re} z = c$, with both $\tilde{h}$ and the inversion integral absolutely convergent there. Then

$$(Ag)(x) = \frac{h(x)}{x} \quad \text{at every } x > 0 \text{ where } h \text{ is continuous}, \qquad \widetilde{Ag}(z) = \tilde{h}(z-1) \quad \text{for every } z.$$

In particular $\widetilde{Ag}(z) = B(1-z)\,\tilde{g}(z-1)$ wherever the first display holds pointwise. ▷ **[T]** proved-target *(**Machine-checked**, as `Hemigroup.SelfDecomposableExponent.inversionOperator_eq` with `mellin_inversionOperator` (the shift) and `mellin_inversionOperator_eq` (the display), the hypothesis on h being the structure `RealisesSymbolAction`; `#print axioms` gives ledger A17 and nothing else — in particular* not *ledger A12, which is the point of the split. This is everything in 9.3 that follows once $B(\theta)g$ has a referent; producing that referent is what ledger A12 is left carrying.*

The identity on the line can only be asked for almost everywhere, and this was not foreseen. *B is a quotient with poles at the zeros of $\tilde{H}$. Where $\tilde{H}$ vanishes the product $B(-z)\tilde{g}(z)$ vanishes with it — in Lean because $x/0 = 0$, in the prose because a meromorphic function has no value at a pole — while $\tilde{h}(z)$ need not. So a pointwise reading of the hypothesis would be satisfied by no g at all. The zeros are isolated by 9.16, hence meet the line in a null set, and the inversion integral does not see them: almost everywhere is both what is available and what is used. This is the second time in the chapter that "equality on the strip" has had to say which equality it means, the first being 9.17, and both times it was writing the statement that raised it.*

The shift is free. *$\widetilde{Ag}(z) = \tilde{h}(z-1)$ holds at every z, with no strip condition: once $Ag = x^{-1}h$ is known on $(0, \infty)$ the weight x^{-1} is a Mellin shift. The strip enters only in the hypothesis relating $\tilde{h}$ to the product, which is where it belongs.)*

Proof. The integrand of the contour integral agrees almost everywhere on the line with $\tilde{h}$, so the integral is the inverse Mellin transform of $\tilde{h}$ at height c; under the two convergence hypotheses that recovers $h(x)$ at every point of continuity, which is the first display. The second is then the Mellin transform of $x^{-1}h(x)$, i.e. $\tilde{h}$ with its argument shifted by one. □

Lemma 9.19 (the profiles are eigenfunctions)**.** Assume (H), let $c \in (0, z_* - 1)$ and $s > 0$. Then the dilate $H(s\,\cdot)$ lies in the domain of 9.3 — its realising function is $h(x) = s\,x\,H(sx)$, exhibited — and

$$A\big[H(s\,\cdot)\big](x) \;=\; s\,H(sx), \qquad x > 0.$$

▷ **[T]** proved-target *(**Machine-checked**, as `Hemigroup.SelfDecomposableExponent.inversionOperator_profile`, with `realisesSymbolAction_profi` carrying the domain clause; `#print axioms` gives ledger A17 and nothing else. Clause (1) of 9.8, split off when proved, on the pattern of 9.17 and 9.5: 11.6 keeps its full statement, and the Lean tag sits on the clause that carries it.*

This is what settles ledger A12, and it settles it by never using it. *That entry is left carrying one step — that absolute integrability of $B(-z)\tilde{g}(z)$ on the line* produces *the function $B(\theta)g$. Here the function is produced by exhibition instead, and the exhibition is free: $\tilde{h}(w) = s^{-w}\tilde{H}(w+1)$ against $\tilde{g}(w) = s^{-w}\tilde{H}(w)$ are one Mellin shift apart, and their ratio is B by 9.16. Since this dilate is the* only *shape in which A is ever applied — 9.8(2)'s Laplace form is this clause applied to $\hat{u}(s,\cdot) = \hat{\hat{f}}(s)H(s\,\cdot)$ — the article's use of ledger A12 is empty. The entry keeps its **[A]** tag on 9.3 because that is the* paper's *stance, as ledger A5 and ledger A6 do in Chapter 3; what is machine-checked is recorded here.*

One line of the draft's proof does not survive transcription, and it is instructive. *It reads "no pole of B intervenes, the product containing no division". That is true of the* simplified *product $s^{-z}\tilde{H}(z+1)$ and false of B read as a function: the division is there, and cancelling it needs $\tilde{H}(z) \neq 0$. So the identity holds only off the zeros, which is a null set of the line — countable, because isolated zeros form a discrete subset of the strip and $\mathbb{C}$ is hereditarily Lindelöf. The prose was doing algebra on symbols where the formalisation has to do it on functions, and that is exactly where the gap between them lives.)*

Proof. $\widetilde{H(s\cdot)}(z) = s^{-z}\tilde{H}(z)$ and $\widetilde{s\,x\,H(sx)}(z) = s^{-z}\tilde{H}(z+1)$, the second because the weight x shifts the Mellin argument by one. Their ratio is $B(-z)$ wherever $\tilde{H}(z) \neq 0$, which is almost every point of the line. The transform of h converges there by 9.2 and its inversion integral by 9.15, both at height $c+1 \in (1, z_*)$, so 9.18 applies and gives $A[H(s\cdot)](x) = h(x)/x = s\,H(sx)$. ☐

Lemma 9.20 (the delayed average is the field, and its transform)**.** Assume (H) and work in the canonical gauge; let $f \in L^1$ be causal and $t > 0$. Then for every $x > 0$,

$$\big(\Phi_{0,x} f\big)(\cdot) \;=\; \mathbb{E}\big[f(\,\cdot - xT_1)\big] \quad \text{a.e.},$$

and for $1 < \operatorname{Re} z < z_*$ the right-hand side, read as a function of x, has

$$\int_0^\infty x^{z-1}\,\mathbb{E}\big[f(t - xT_1)\big]\,dx \;=\; \tilde{H}(z)\,\big(I^z f\big)(t),$$

both sides absolutely convergent. ▷ **[T]** proved-target *(**Machine-checked**, as `Hemigroup.SelfDecomposableExponent.mellin_delayedField`, with `coeFn_Phi_zero` (the first display), `kernel_zero_eq_map_law`T_1 (the canonical gauge at the level of measures), `integrable_delayed` (the Fubini side condition) and `mul_riemannLiouville` (the reflection); `#print axioms` gives ledger A17 and nothing else. This is the first clause of 9.6; what is left there is its derivative clause.*

The quantifiers do not commute, and that is the honest reason a representative has to be chosen. *$\Phi_{0,x}$ acts on L^1 classes, so the first display holds for each x only up to a null set of times — and the null set depends on x. "For each x, for a.e. t" does not give "for a.e. t, for every x", so the second display, which is an integral over x at a fixed t, cannot be taken of $(\Phi_{0,x}f)(t)$ as it stands. Some representative must be named whichever way the identification is read. What the choice made here buys is that it is* local*: one function, $\mathbb{E}[f(t - xT_1)]$, defined* *outright, with the first display as its bridge — rather than a pointwise model of Φ built under all of Chapters 9–10 and the scale-Cauchy appendix.*

I^z is defined in the development, not cited. *Mathlib carries no fractional integral of any order, so the Riemann–Liouville integral is given its definition here. That adds no interface:* `@samko1993fractional` *is cited in this chapter for the notation and the theory of I^z, and nothing here uses more of that theory than the definition itself.*

The exchange is licensed by the two ends of the strip, one apiece. *Inside, the dilate of the past integrand is integrable because $\operatorname{Re} z > 1$: on $(0,t]$ the weight y^{c-1} is bounded by t^{c-1}, so f needs nothing beyond L^1, and beyond t causality kills the integrand outright. Outside, the dilation $x \mapsto \tau x$ leaves exactly the factor τ^{-z}, whose expectation is finite because $\operatorname{Re} z < z_*$. That is the same shape as 9.2's hinge with the lower end moved from 0 to 1 — and the move is what* makes *(H) ask for $z_* > 1$ rather than $z_* > 0$. Below 1 this lemma has no strip to live in and 9.8(2) would have nothing to say, which is also why 7.6's observation that every extreme ray has $z_* = 1$ bites here.)*

Proof. Everything factors through the past integral $\int_0^\infty y^{z-1} f(t-y)\,dy$. Fubini exchanges the two integrals; the inner one is then a dilate, contributing τ^{-z} times the past integral, and the outer integral of τ^{-z} is $\mathbb{E}[T_1^{-z}]$. The past integral is $\Gamma(z)(I^z f)(t)$ by the reflection $y \mapsto t-y$, whose domains match because causality has already confined the integral to $(0,t]$; and $\mathbb{E}[T_1^{-z}]\Gamma(z) = \tilde{H}(z)$ by 9.2. The two Γ's cancel, which is why the statement has none. ☐

Lemma 9.21 (the Mellin form of the signaling equation, transform side)**.** Assume (H), canonical gauge; let $f \in \mathcal{D}$ be causal, $t > 0$ and $1 < \operatorname{Re} z < z_*$, and suppose $\tilde{H}(z-1) \neq 0$. Then

$$B(1-z)\,\widetilde{u(t,\cdot)}(z-1) \;=\; \tilde{H}(z)\,\big(I^{z-1} f\big)(t).$$

▷ [**T**] proved-target *(**Machine-checked**, as* `Hemigroup.SelfDecomposableExponent.inversionSymbol_mul_mellin_delayedField`*;* `#print axioms` *gives ledger A17 and nothing else. This is the displayed computation in the proof of 9.8(2), entire; what stands between it and that node's conclusion is the identification of the right-hand side with $\widetilde{\partial_t u(t,\cdot)}(z)$, which is the derivative clause of 9.6 and needs A.18.*

Two hypotheses that the prose does not carry, for the same two reasons the chapter has met before. $\operatorname{Re} z > 1$ *is used at $z-1$, where the strip must start at 0 rather than 1 — so the boundedness of $f \in \mathcal{D}$ is load-bearing here and not incidental; see 9.6. And $\tilde{H}(z-1) \neq 0$ is needed because B is a quotient: 9.10 says the identity "is nevertheless regular, the factor $\tilde{H}(z-1)$ cancelling every denominator", which is true of the product as a* meromorphic function *and not of B evaluated at a point, where the division is present and has to be cancelled. That is the third time in this chapter that a symbolic cancellation has had to become a pointwise one, after 9.8(1) and 9.17.)*

Proof. $B(1-z) = \tilde{H}(z)/\tilde{H}(z-1)$ by 9.16, and $\widetilde{u(t,\cdot)}(z-1) = \tilde{H}(z-1)(I^{z-1}f)(t)$ by 9.20 at $z-1$, whose strip condition $0 < \operatorname{Re}(z-1) < z_*$ is the hypothesis. Cancel $\tilde{H}(z-1)$. ☐

Lemma 9.22 (the fractional integral absorbs a derivative)**.** Let $\operatorname{Re} z > 1$, let g be integrable on $(0,t]$ and set $f(r) := \int_0^r g$ — that is, f absolutely continuous with $f(0) = 0$ and $f' = g$. Then

$$\big(I^z g\big)(t) \;=\; \big(I^{z-1} f\big)(t).$$

▷ [**T**] proved-target *(**Machine-checked**, as* `Hemigroup.riemannLiouville_integral`*, with* `integral_cpow_sub_left` *for the inner integral;* `#print axioms` *reduces to Lean core alone — this statement is about the Riemann–Liouville family and mentions no hemigroup object, so not even ledger A17 enters.*

The semigroup property is not needed, and not needing it is cheaper. *The draft derives this as $I^z f' = I^{z-1} I^1 f' = I^{z-1} f$, i.e. from $I^{z-1} I^1 = I^z$ — a Beta-integral identity that* *would have to be proved in its own right, Mathlib having no fractional integral at all. Substituting $f(r) = \int_0^r g$ and exchanging the order of integration over the triangle $0 < \rho \le r \le t$ instead leaves the inner integral $\int_\rho^t (t-r)^{z-2} dr = (t-\rho)^{z-1}/(z-1)$, after which $\Gamma(z) = (z-1)\Gamma(z-1)$ finishes it. The condition $\operatorname{Re} z > 1$ is exactly what makes that inner integral converge — the same endpoint that 9.20 and 9.21 turn on, arrived at for a third independent reason.)*

Proof. Write $f(r) = \int_0^r g$ and exchange the order of integration in $\Gamma(z-1)(I^{z-1} f)(t) = \int_0^t (t-r)^{z-2} \int_0^r g(\rho)\, d\rho\, dr$ over the triangle $0 < \rho \le r \le t$, which is legitimate because the same computation on absolute values is bounded by $t^{z-1}\|g\|_1/(\operatorname{Re} z - 1)$. The inner integral is $\int_\rho^t (t-r)^{z-2} dr = (t-\rho)^{z-1}/(z-1)$, so the double integral is $(z-1)^{-1}\Gamma(z)(I^z g)(t)$; and $\Gamma(z) = (z-1)\Gamma(z-1)$. ☐

Lemma 9.23 (kernel readings of the standing hypothesis)**.** In the canonical gauge, with T_1 as in 9.1:

1. $F(\infty) = \infty$ if and only if the kernels have no atom at zero delay: $\mu_{0,x}(\{0\}) = \Pr[T_1 = 0] = 0$ for every $x > 0$.

2. Suppose $F(\infty) = \infty$. Then $\int_0^\infty e^{-F(s)}\, ds = \mathbb{E}[T_1^{-1}]$ in $[0,\infty]$; consequently $z_* > 1$ implies $\int_0^\infty e^{-F(s)}\, ds < \infty$, and $\int_0^\infty e^{-F(s)}\, ds < \infty$ implies $z_* \ge 1$.

▷ **[T]** proved-target *(**Machine-checked**, as `Hemigroup.SelfDecomposableExponent.standing_kernel_readings`, a conjunction of four parts, each with its own declaration: `tendsto_toRealExponent_atTop_iff` (clause 1), `lintegral_profile_eq` (clause 2's identity), and `negMoment_one_ne_top_of_one_lt_zStar` / `one_le_zStar_of_negMoment_one_ne_` (clause 2's two consequences). `#print axioms` gives ledger A17 throughout, on every part including the two consequences: they are the general downward-closure and abscissa lemmas 9.2 already uses, restated at $\zeta = 1$ for* this *F, and a statement about* this *F's z_* or negative moments quantifies over the family ledger A17 constructs however elementary its own argument is — the same point 9.1 records about z_* itself. One general fact underneath clause 1's converse, `Hemigroup.tendsto_laplaceL_atTop` (any finite causal measure's transform tends to its mass at the origin as $s \to \infty$), reduces to Lean core alone, carrying no hemigroup object. **The two consequences are proved without the hypothesis "$F(\infty) = \infty$" the statement poses them under** — they are unconditional facts about z_* and `negMoment`, so what is proved is strictly more than clause 2 asks for, not less.*

Proof. (1) $s \mapsto H(s) := \mathbb{E}[e^{-sT_1}]$ decreases to $\Pr[T_1 = 0]$ as $s \to \infty$: the integrand e^{-sT_1} is dominated by the constant 1 and converges pointwise to the indicator of $\{T_1 = 0\}$, so dominated convergence gives $H(s) \to \Pr[T_1 = 0]$. Since $H(s) = e^{-F(s)}$, $F(\infty) = \infty$ iff $H(s) \to 0$ iff $\Pr[T_1 = 0] = 0$. For general $x > 0$, $\mu_{0,x}$ is the law of xT_1 (the canonical gauge), and multiplication by $x \ne 0$ carries $\{0\}$ to itself and nowhere else, so $\mu_{0,x}(\{0\}) = \Pr[xT_1 = 0] = \Pr[T_1 = 0]$ for every $x > 0$; the case $x = 1$ already gives the stated equivalence, and every other x agrees with it.

(2) By Tonelli, $\int_0^\infty e^{-F(s)}\, ds = \int_0^\infty \mathbb{E}[e^{-sT_1}]\, ds = \mathbb{E}\big[\int_0^\infty e^{-sT_1}\, ds\big] = \mathbb{E}[T_1^{-1}]$ in $[0,\infty]$, the inner integral $\int_0^\infty e^{-sT_1} ds = T_1^{-1}$ being licensed pointwise because $T_1 > 0$ almost surely by (1). For the two consequences, the finiteness set $\{\zeta > 0 : \mathbb{E}[T_1^{-\zeta}] < \infty\}$ is downward closed — $t^{-\zeta'} \le t^{-\zeta} + 1$ for $t > 0$ and $\zeta' \le \zeta$, so a finite ζ-th moment bounds every smaller one up to an additive unit of mass — so $z_* > 1$ puts a real $\zeta > 1$ in the set, and hence 1 in it too, giving $\mathbb{E}[T_1^{-1}] < \infty$; conversely if $\mathbb{E}[T_1^{-1}] < \infty$ then 1 is itself a member of the set whose supremum is z_*, so $z_* \ge 1$. ☐

Lemma 9.24 (Lévy-data reading; the first clause is automatic)**.** Let F be admissible with representation (7.1). Then:

1. $F(\infty) = \infty$ if and only if $b_0 > 0$ or $\int_0^\infty k(t)\, t^{-1} dt = \infty$ — i.e. the exponent carries drift or infinite Lévy mass.

2. If $F \not\equiv 0$ then $F(\infty) = \infty$; within the admissible class the standing hypothesis therefore reduces to its second clause, $z_* > 1$.

▷ **[T]** proved-target *(**Not yet proved.** Target types are written, sorry-marked, as `Skeleton.tendsto_toRealExponent_atTop_iff_levy` and `Skeleton.tendsto_toRealExponent_atTop_of_ne_zero` in `Formalization/Skeleton/Chapter11.lean`; nothing here is cited by the library or listed in `CIAxiomGuard.lean`, per the article's decomposition discipline. **Why this clause and not 9.23's.** That node's clause 1 is stated in T_1's law and its proof is a direct dominated-convergence limit plus a change-of-variables that is already in the library (`kernel_zero_eq_map_lawT`$_1$). This one is stated in the* (b_0, k) *data of (7.1) — the two-vocabulary bridge `CLAUDE.md` names, that a node concluding in* BF_0 *needs the* LE *reading stated alongside it before it can carry a `\lean` tag with `\leanok`. Writing the target types down (the discipline this article keeps, statement first) is what turned "looks like the same kind of fact" into a real cost estimate: clause (1) needs a monotone-convergence argument for `levyJump` as* $s \to \infty$ *in* both *directions of an iff, together with the bridge from a* $[0, \infty]$*-valued limit to the* $\mathbb{R}$*-valued `atTop` reading `toRealExponent` already carries (the same bridge, `tendsto_toRealExponent_atTop_of_lawT`$_1$`_singleton_zero`, needed once, for one direction, in 9.23); clause (2) needs* $\int_0^{t_0} t^{-1} dt = \infty$*, a divergence fact the library does not currently carry in any form. Priced against the target types rather than the paper proof — the paper's own route, via complete monotonicity, is not available here at all, the development having no `CM` predicate — this is genuinely more expensive than 9.23, and nothing downstream needs it: every proof through chapter 10 consumes (H)'s two clauses as the Lean has always stated them, never this reading of them.)*

Proof. (1) ($\Leftarrow$) If $b_0 > 0$ then $F(s) \geq b_0 s \to \infty$. If $\int_0^\infty k(t) t^{-1} dt = \infty$, monotone convergence in (7.1) — $1 - e^{-st}$ increases pointwise to 1 on $(0, \infty)$ as $s \to \infty$ — gives $F(s) \to b_0 \cdot \infty + \int_0^\infty k(t) t^{-1} dt = \infty$. ($\Rightarrow$) Contrapositive: if $b_0 = 0$ and $\int_0^\infty k(t) t^{-1} dt =: M < \infty$, the same monotone limit gives $F(s)$ increasing to $M < \infty$, so $F(\infty) \neq \infty$.

(2) If $k \not\equiv 0$, nonincreasing k that is positive somewhere is bounded below by that value on the interval below it: pick t_0 with $k(t_0) > 0$, then $k(t) \geq k(t_0)$ for $t \in (0, t_0]$, so $\int_0^{t_0} k(t) t^{-1} dt \geq k(t_0) \int_0^{t_0} t^{-1} dt = \infty$, and (1) gives $F(\infty) = \infty$. If $k \equiv 0$ and $F \not\equiv 0$, (7.1) reduces to $F(s) = b_0 s$ with $b_0 > 0$, and (1) again gives $F(\infty) = \infty$. □

Lemma 9.25 (the log-growth reading of z_*)**.** Let $F \not\equiv 0$ be admissible, with (b_0, k) as in (7.1). Then:

1. The limit $\lim_{s \to \infty} F(s) / \log s$ exists in $[0, \infty]$ and equals z_*.

2. The limit is ∞ if $b_0 > 0$, and equals $k(0^+) := \sup_{t > 0} k(t)$ if $b_0 = 0$ (finite or infinite).

3. Consequently, (H) holds if and only if $b_0 > 0$ or $k(0^+) > 1$.

4. z_* is homogeneous of degree one on the admissible cone: $z_*(cF) = c\, z_*(F)$ for every $c > 0$.

▷ **[T]** proved-target *(**Not yet proved.** Target types are written, sorry-marked, as `Skeleton.tendsto_toRealExponent_div_log_atTop_zStar`, `Skeleton.tendsto_toRealExponent_div_log` `Skeleton.tendsto_toRealExponent_div_log_atTop_of_b`$_0$`_zero` and `Skeleton.zStar_smul` in `Formalization/Skeleton/Chapter11.lean`; nothing here is cited by the library or listed in `CIAxiomGuard.lean`. **The gap the obvious sketch has.** A subsequence/comparison argument on the Mellin integral of 9.2 shows only* $z_* = \liminf_{s \to \infty} F(s) / \log s$*: nothing controls the limsup for a general Bernstein* F*, whose* $F / \log s$ *can oscillate. **Existence of the limit is where self-decomposability enters** — but not by way of the derivative-sign (*BF_0*) machinery* *its usual proof reaches for. The textbook route identifies $B(s) := sF'(s)$ as a Bernstein function (7.1(2)), hence nondecreasing, and that implication is ledger ledger A18.* ***Writing the target type found a cheaper route: the explicit formula of A.1, already machine-checked, gives B's monotonicity directly****, with no appeal to the general Bernstein-closure fact — see the proof below, which takes it. So this node, unlike 7.1 itself, has no route through ledger A18 at all; its four target types are priced against ledger A17 and Lean core only, the way 9.2 already reduces off ledger A7. That is exactly the caution* `CLAUDE.md` *names about writing the cost estimate from the target type rather than from the paper proof — here the paper proof itself is the one that reaches for the expensive tool, and the cheaper one was sitting one chapter earlier.* ***Priced against the target types, not the paper proof.*** *Writing them down found that clauses (2)–(4) need* no *no-atom hypothesis at all — they are unconditional statements about the exponent's own growth, via the explicit formula $B(s) = b_0 s + \int_0^\infty e^{-u}k(u/s)\,du$ (substitute $u = st$) and monotone convergence in $k(u/s) \uparrow k(0^+)$ as $s \to \infty$ — only clause (1)'s identification of the limit with z_* needs it, since* `negMoment` *and* `zStar` *integrate over* `Ioi 0` *and are blind to an atom at the origin, the trap 9.24's target types were already written to avoid.* ***Shares machinery with 9.24****: both run a monotone-convergence argument on* `levyJump` *as $s \to \infty$ (there for $\int k(t)t^{-1}dt$, here for $k(u/s)$ itself), so the two nodes should be formalised together if either is attempted.* ***Nothing downstream needs this reading.*** *Every proof through chapter 10 consumes z_* as the abscissa 9.1 states it, never its log-growth rate; this node is a detector, not a dependency, and it has already caught one live error — §8's Bessel-K proof once asserted $k(t) \sim a$ as $t \to 0$, a finite limit that, with $b_0 = 0$, would force $F(s) = O(\log s)$ by clause (2), against the family's own $F(s) \sim \sqrt{2s}$; the correction is the dated comment in* `blueprint/src/parts/08-examples-moments.tex` *(2026-08-29).)*

Checked against every family the article names. *Gamma: $b_0 = 0$, $k(t) = \gamma e^{-t}$, so $k(0^+) = \gamma$, matching $z_* = \gamma$. Stable, Bessel-K and the pure delay: $z_* = \infty$, matching $b_0 > 0$ (pure delay) or $k(0^+) = \infty$ (stable and Bessel-K, whose k carries the $(2\pi t)^{-1/2}$ singularity at $t = 0^+$ corrected in 8.3's proof). Dickman rays: $k = \mathbf{1}_{(0,\tau)}$, so $k(0^+) = 1$, matching 7.6's $z_* = 1$ at every extreme ray with $b_0 = 0$. The leaky integrator, the Gamma family at $\gamma = 1$, agrees with the Gamma case.*

Proof. Since $F \not\equiv 0$ is admissible, 9.24(2) gives $F(\infty) = \infty$, and 9.23(1) turns that into $T_1 > 0$ almost surely, so $\mathbb{E}[T_1^{-\zeta}]$ is an honest integral over $(0, \infty)$ for every $\zeta > 0$.

The exponent's slope. By A.1, $F'(s) = b_0 + \int_0^\infty e^{-st}k(t)\,dt$, so $B(s) := sF'(s) = b_0 s + \int_0^\infty e^{-u}\,k(u/s)\,du$ after the substitution $u = st$. For $0 < s_1 \le s_2$: $b_0 s_1 \le b_0 s_2$, and k nonincreasing gives $k(u/s_1) \le k(u/s_2)$ pointwise, since $u/s_1 \ge u/s_2$; so B is nondecreasing — no appeal to 7.1 or to B being a Bernstein function as a class, only to the explicit formula and the definition of nonincreasing. And since $k(u/s)$ increases pointwise to $k(0^+)$ as $s \to \infty$, monotone convergence gives $B(\infty) = b_0 \cdot \infty + k(0^+)\int_0^\infty e^{-u}\,du$: ∞ if $b_0 > 0$, and $k(0^+)$ if $b_0 = 0$. That is clause (2).

(1) $G(u) := F(e^u)$ has $G'(u) = e^u F'(e^u) = B(e^u)$ nondecreasing by the above, so G is convex in $u = \log s$. A convex function's difference quotient $G(u)/u$ increases, past any zero of u, to its terminal slope, so $L := \lim_{s\to\infty} F(s)/\log s = \lim_{u\to\infty} G(u)/u = \lim_{s\to\infty} B(s)$ exists in $[0, \infty]$, with value given by clause (2). To identify L with z_*: by Tonelli, $\Gamma(\zeta)\,\mathbb{E}[T_1^{-\zeta}] = \int_0^\infty s^{\zeta-1}e^{-F(s)}\,ds$ for every $\zeta > 0$, the exchange licensed exactly as in 9.23(2), now at general ζ rather than at $\zeta = 1$. For $0 < \zeta < L$, eventually $F(s) \ge (\zeta + \varepsilon)\log s$ for some $\varepsilon > 0$, so $e^{-F(s)} \le s^{-\zeta-\varepsilon}$ eventually and the integral converges; for $\zeta > L$, eventually $F(s) \le (\zeta - \varepsilon)\log s$, so the integrand is eventually $\ge s^{\varepsilon-1}$ and the integral diverges. Hence $\{\zeta > 0 : \mathbb{E}[T_1^{-\zeta}] < \infty\}$ contains every $\zeta < L$ and excludes every $\zeta > L$, so its supremum, z_*, is L.

(3) is (1)–(2) read against (H)'s two clauses. The first is automatic here by 9.24(2), so (H) reduces to $z_* > 1$, which by (2) is exactly $b_0 > 0$ or $k(0^+) > 1$ — an equality throughout, so with no boundary subtlety.

(4) cF is admissible by 7.13, with data (cb_0, ck), so $k_{cF}(0^+) = c\,k(0^+)$ and (cF)'s drift is cb_0; applying (1)–(2) to cF in place of F gives $z_*(cF) = c\,z_*(F)$, whether $b_0 > 0$ or $b_0 = 0$. □

Definition 9.26 (a Mellin solution of the signaling problem)**.** Assume (H) and work in the canonical gauge, and let $f \in \mathcal{D}$. A *Mellin solution of the signaling problem for* f is a field $u : \mathbb{R} \times (0,\infty) \to \mathbb{R}$ such that $u(\cdot, x) \in X_0$ for every $x > 0$ — causal in t at every scale, the honest form of the boundary condition $u(0,\cdot) = 0$, since X_0 is by definition the causal elements of X — with $u(\cdot, x) \to f$ in X_0 as $x \downarrow 0$, and, writing $\hat{u}(s,x) := \int_0^\infty e^{-st} u(t,x)\,dt$ for $s > 0$:

1. *(Laplace form.)* For every $s > 0$: $x \mapsto \hat{u}(s,x)$ is continuous on $(0,\infty)$; its Mellin transform $z \mapsto \widetilde{\hat{u}(s,\cdot)}(z)$ is analytic on the whole strip $0 < \operatorname{Re} z < z_*$; at every height $c \in (0, z_* - 1)$, $\hat{u}(s,\cdot)$ lies in the domain of A (9.3); and
$$A\big[\hat{u}(s,\cdot)\big] \;=\; s\,\hat{u}(s,\cdot), \qquad \hat{u}(s,0+) \;=\; \hat{f}(s);$$
2. *(Mellin form.)* For every $t > 0$ and every z with $1 < \operatorname{Re} z < z_*$ and $\tilde{H}(z-1) \neq 0$,
$$\widetilde{\partial_t u(t,\cdot)}(z) \;=\; B(1-z)\,\widetilde{u(t,\cdot)}(z-1),$$
both transforms absolutely convergent.

▷ **[T]** proved-target *(**Not tagged:** a definition, no Lean declaration to carry. It exists so that 9.8(2)'s closing gloss — "that is, $\partial_t u = A\,u$ read through the Mellin transform in x" — has a referent, and a theorem can be proved* of *it rather than only alongside it: the way a weak solution turns a PDE's informal reading into an object a uniqueness theorem can quantify over. **Every clause is 9.8(2)'s, copied rather than paraphrased**, with three changes, each forced by turning an assertion about one particular field into a condition a general field either satisfies or does not. First, u is no longer presented as $\Phi_{0,x} f$, so causality has to be asked for directly rather than inherited from the construction — and it is asked for* at every scale x, *which is the honest reading of the draft's boundary condition $u(0,\cdot) = 0$: a solution's data lives on the whole boundary $\{t \in \mathbb{R}\}$, not at the single point $x = 0$ of it, and the field of 9.8 happens to have $u(\cdot, x) \in X_0$ for* every $x \geq 0$ *because it is built from $\Phi_{0,x}$, an axiom (A3) the general definition cannot assume. Second, domain membership — which 9.8(1) establishes for the field's own profile by exhibiting the realising function $s\,x\,H(sx)$ — is stated outright as a hypothesis, since a general Mellin solution supplies no such exhibition; and because 9.27 needs its recursion to hold as an identity of holomorphic functions with no exceptional set, the hypothesis includes that $\widetilde{\hat{u}(s,\cdot)}$ is analytic on the* whole *strip $0 < \operatorname{Re} z < z_*$, one step more than "lies in the domain of A at every height $c \in (0, z_* - 1)$" asks for on its own — the field satisfies this for free, since $\widetilde{\hat{u}(s,\cdot)}(z) = \hat{f}(s)\,s^{-z}\tilde{H}(z)$ is analytic wherever $\tilde{H}$ is (9.16), and a Mellin solution that did not would leave 9.27's hypothesis unmet by construction rather than by content, which is exactly the kind of implicit reading `CLAUDE.md` asks a formal statement to name rather than leave to context. Third, continuity of $x \mapsto \hat{u}(s,x)$ is added, which 9.8(2) never has to state because $\hat{u}(s,\cdot) = \hat{f}(s)H(s\,\cdot)$ is visibly continuous, but a general solution needs it named: it is what licenses reading the Mellin inversion integral pointwise rather than up to a null set, both in 9.27's own route and in 9.28's use of Mellin inversion to identify two solutions' modes. **Nothing else changes.** Clause (1)'s eigenfunction relation and boundary value and clause (2)'s transform identity are exactly 9.8(2)'s Laplace and Mellin forms.)*

Lemma 9.27 (rigidity of a profile-dominated mode)**.** Assume (H) and fix $s > 0$. Let $\tilde{g}$ be analytic on the strip $0 < \operatorname{Re} z < z_*$ and satisfy the pole-free eigenvalue recursion
$$s\,\tilde{g}(z)\,\tilde{H}(z-1) \;=\; \tilde{H}(z)\,\tilde{g}(z-1), \qquad 1 < \operatorname{Re} z < z_*,$$

an identity of holomorphic functions on that range, with no exceptional set. Suppose the *periodic factor*

$$p(z) \;:=\; \frac{s^z\,\tilde g(z)}{\tilde H(z)},$$

a priori meromorphic on $0 < \operatorname{Re} z < z_*$ with possible poles only at the isolated zeros of $\tilde H$ (9.16), is bounded on some period substrip $\{c \le \operatorname{Re} z \le c+1\} \subset (0, z_*)$, off those zeros. Then p is constant: there is $\kappa \in \mathbb{C}$ with

$$\tilde g(z) \;=\; \kappa\, s^{-z}\,\tilde H(z), \qquad 0 < \operatorname{Re} z < z_*.$$

$\triangleright$ [**T**] proved-target *(**Not yet proved.** A target type is written, sorry-marked, as `Skeleton.mode_rigidity` in `Formalization/Skeleton/Chapter11.lean`; nothing here is cited by the library or listed in `CIAxiomGuard.lean`, per the article's decomposition discipline. **Priced against the target type, not the paper proof**, per that discipline: the statement is entirely classical and every individual step is either elementary or already in Mathlib (`Complex.differentiableOn_update_limUnder_of_bddAbove` for removability from boundedness, `Differentiable.exists_eq_const_of_bounded` for Liouville), but the middle step — patching a function defined on translates of one finite-width strip, related by a functional equation on their overlaps, into a single entire periodic function on $\mathbb{C}$ — is scaffolding this development does not have and Mathlib's nearest relative (`Mathlib.Analysis.Complex.Periodic`'s `cuspFunction`, built for one-sided behaviour at a single end of a strip, the modular-forms use case) does not supply either. Building it is a real, self-contained piece of complex analysis — consistency of consecutive translates on their overlap, chained through intermediate strips for non-adjacent ones, is exactly where $z_* > 1$, i.e. (H), is spent — on the order of a session's work rather than a sitting's, and it was not attempted here. **What the hypothesis is, read against the mechanism.** "p bounded on one period substrip" says that $\hat u(s,\cdot)$'s Mellin data is dominated by the profile's own, uniformly over one period — the theory's counterpart of the log-convexity hypothesis Bohr–Mollerup imposes on Γ's difference equation, and the mechanism that kills the ambiguity here — Riemann removability plus Liouville, both elementary — is one level below the mechanism 10.11 (ledger ledger A15) uses one chapter later for the same shape of recursion with a linear symbol: no citation is needed at either point, and none is spent here. **The scale-space field satisfies the hypothesis trivially.** For $\tilde g = \widehat{\hat u(s,\cdot)}$ with $\hat u(s,\cdot) = \hat f(s) H(s\,\cdot)$, the periodic factor is the* constant *$\hat f(s)$ — see 9.28(1) — so the hypothesis is not a genuine restriction on the field, only on which other candidate solutions the uniqueness clause below can rule out. **Nothing weaker suffices, in general.** Boundedness of the* mode *itself, $|\tilde g(c+i\tau)| \le |\tilde g(c)|$ say, does not bound p: $1/|\tilde H(c+i\tau)|$ grows like $\mathrm{e}^{\pi|\tau|/2}$ along the line (the reciprocal of 9.15's decay), so a mode merely bounded on the line can still have an unbounded, and hence non-constant, periodic factor. The normalisation has to compare the mode to the profile, not bound it absolutely.)*

Proof. Where $\tilde H(z) \neq 0$ and $\tilde H(z-1) \neq 0$, dividing the recursion by $s\,\tilde H(z)\tilde H(z-1)$ and multiplying by s^z gives $p(z) = p(z-1)$, so the periodic factor satisfies the shift relation off the (isolated) zeros of $\tilde H$ on both sides.

Step 1: p extends to a meromorphic, 1-periodic function on $\mathbb{C}$. $\tilde H$'s zeros in $(0, z_*)$ are isolated (9.16), hence locally finite there: every compact subset of the open strip contains finitely many of them — accumulation at the strip's *boundary* is not excluded, and is not needed; so, writing $S := (0, z_*)$ as a set of vertical strips $S_n := S + n = \{z : n < \operatorname{Re} z < n + z_*\}$ for $n \in \mathbb{Z}$, and $p_n(z) := p(z-n)$, meromorphic on S_n with possible poles only at the (locally finite) translated zeros of $\tilde H$: for $n \le k \le m$ and $z \in S_n \cap S_m$, z lies in every intermediate S_k (since $\operatorname{Re} z > m \ge k$ and $\operatorname{Re} z < n + z_* \le k + z_*$), so the shift relation applied at each consecutive pair $(k, k+1)$, $n \le k < m$, gives $p_n(z) = p_{n+1}(z) = \cdots = p_m(z)$: every two translates agree wherever they overlap. Since $z_* > 1$ (H), consecutive strips overlap and $\{S_n\}_{n\in\mathbb{Z}}$ covers $\mathbb{C}$, so there is a well-defined meromorphic $P : \mathbb{C} \to \mathbb{C}$ with $P|_{S_n} = p_n$ for every n; it is 1-periodic, since for $z \in S_n$, $z+1 \in S_{n+1}$ and $P(z+1) = p_{n+1}(z+1) = p(z-n) = p_n(z) = P(z)$.

Step 2: P is bounded off its singular set, on all of $\mathbb{C}$. $P|_{S_0} = p$ is bounded on the given substrip $\{c \le \operatorname{Re} z \le c+1\}$, off the zeros of $\tilde{H}$ there, by hypothesis. By periodicity, P is bounded by the *same* constant on every unit translate $\{c+n \le \operatorname{Re} z \le c+1+n\}$, $n \in \mathbb{Z}$, off P's singular set there; and these translates cover $\mathbb{C}$, consecutive ones sharing an edge. So P is bounded, by one constant, off its singular set, everywhere.

Step 3: removability and Liouville. At each point w_0 of P's singular set — a priori a pole, though this step shows it is not — P is bounded on a punctured neighbourhood of w_0 (Step 2), so by Riemann's removable singularity theorem P extends to be analytic at w_0, with value the limit from nearby — hence bounded by the same constant, since nearby values are. Doing this at every such point is licensed because the singular set is locally finite in $\mathbb{C}$: every point of $\mathbb{C}$ has a compact neighbourhood inside some open S_n — again $z_* > 1$, the open intervals $(n, n+z_*)$ covering $\mathbb{R}$ — and there the possible poles are the finitely many translated zeros of Step 1. The result is an *entire* function $\hat{P}$, equal to P off its singular set and bounded on all of $\mathbb{C}$ by the constant of Step 2. Liouville's theorem gives $\hat{P} \equiv \kappa$ for some $\kappa \in \mathbb{C}$.

Closing. On $S_0 = (0, z_*)$, $\hat{P} = P = p$ off the zeros of $\tilde{H}$, so $s^z \tilde{g}(z) = \kappa \tilde{H}(z)$ for every $z \in (0, z_*)$ off that isolated set. Both sides are analytic on the whole (connected, open) strip $(0, z_*)$ — the left because $\tilde{g}$ is, by hypothesis; the right because $\tilde{H}$ is (9.16) — and agree off a set with no accumulation point in the strip, so by the identity theorem they agree everywhere on it. Dividing by $s^z \neq 0$ gives $\tilde{g}(z) = \kappa\, s^{-z} \tilde{H}(z)$ for every $z \in (0, z_*)$. □

Corollary 9.28 (the signaling problem: existence and uniqueness in the profile-dominated Mellin class)**.** Assume (H), let $f \in \mathcal{D}$, and let A be the inversion operator of 9.3. Then:

1. *(Existence.)* The scale-space field $u(t,x) := (\Phi_{0,x} f)(t)$ is a Mellin solution of the signaling problem for f (9.26); its periodic factor $p_s(z) := s^z \widetilde{\hat{u}(s,\cdot)}(z)/\tilde{H}(z)$ (9.27) is the *constant* $\hat{f}(s)$, for every $s > 0$ — in particular profile-dominated.

2. *(Uniqueness.)* It is the *unique* Mellin solution of the signaling problem for f whose modes are profile-dominated — i.e., such that for every $s > 0$ the periodic factor of 9.27 is bounded on some period substrip: if v is another such solution, then $v(\cdot, x) = u(\cdot, x) = (\Phi_{0,x} f)(\cdot)$ in X_0, for every $x > 0$.

▷ **[T]** proved-target *(**Not tagged. Clause (1)** is 9.8(1)+(2) verbatim, with the two additions 9.26 needs: $x \mapsto \hat{u}(s,x) = \hat{f}(s) H(sx)$ is continuous because H is, and its Mellin transform $\hat{f}(s) s^{-z} \tilde{H}(z)$ is analytic on the whole strip* (9.16)*, not merely at the heights* $(0, z_* - 1)$ *domain membership asks for. **Clause (2) reuses ledger A12, once, and the reuse does not widen the entry.** The step is Mellin inversion applied directly to two* functions *sharing a transform — Widder Thm. 9a, the same citation 9.3 already carries, used here without the operator's functional-calculus reading (no symbol B, no realising function) and hence, if ever formalised, through Mathlib's `mellinInv_mellin_eq` applied directly rather than through `RealisesAction` — the same pattern 9.19 found for clause (1) of 9.8: a use of* A12*'s citation that costs nothing beyond ledger A17 once written down, because the referent it needs is exhibited rather than inferred.* `blueprint/AXIOMS.md` *records this as a new reading under the existing entry; no new ledger name is created. **Why solution-uniqueness needs the normalisation and operator-uniqueness (Theorem 4′(3)) does not.** Theorem 4′(3) compares two* operators *that both realise the eigenfunction relation on* every *profile dilate $H(s\cdot)$, $s > 0$ — a one-parameter family rich enough to pin a symbol down by itself, with no growth condition. A single solution u supplies only* one *mode per s, and 9.27 is exactly the statement that one mode is not enough without a normalisation: the periodic ambiguity is real, and profile-domination is what removes it. **The route was checked against the alternative, and rejected.** Boundedness of the* *mode alone does not suffice — see 9.27's own annotation — so no weaker hypothesis earns the same conclusion; profile-domination is the minimal normalisation this argument supports, not merely a convenient one.)*

Proof. (1) is 9.8(1) for the domain and eigenfunction clauses and 9.8(2) for causality, the boundary limit, the Laplace and Mellin forms; continuity of $x \mapsto \hat{u}(s,x) = \hat{f}(s)H(sx)$ and analyticity of its transform on the whole strip are as in the statement. Since $\wideparen{\hat{u}(s,\cdot)}(z) = \hat{f}(s)\, s^{-z}\tilde{H}(z)$, the periodic factor is $p_s(z) = s^z \cdot \hat{f}(s)\, s^{-z}\tilde{H}(z)/\tilde{H}(z) = \hat{f}(s)$ identically.

(2) Let v be a Mellin solution for f with profile-dominated modes, and fix $s > 0$. Write $\tilde{g}(z) := \wideparen{\hat{v}(s,\cdot)}(z)$, analytic on the whole strip by 9.26(1). At every height $c \in (0, z_* - 1)$, $\hat{v}(s,\cdot)$ lies in the domain of A with $A[\hat{v}(s,\cdot)] = s\,\hat{v}(s,\cdot)$; taking Mellin transforms (9.3's transform identity, as in the proof of 9.8(1)) gives $s\,\tilde{g}(z) = B(1-z)\tilde{g}(z-1)$ off the zeros of $\tilde{H}(z-1)$ on the line $\operatorname{Re} z = c+1$; multiplying out $B(1-z) = \tilde{H}(z)/\tilde{H}(z-1)$ clears the denominator to $s\,\tilde{g}(z)\tilde{H}(z-1) = \tilde{H}(z)\tilde{g}(z-1)$ off an isolated set of the line, and both sides are analytic on the strip $1 < \operatorname{Re} z < z_*$ (the left by hypothesis, the right by 9.16), so by the identity theorem the identity holds *everywhere* on that range — the pole-free recursion 9.27 asks for. By hypothesis its periodic factor is bounded on some period substrip, so 9.27 gives $\kappa_s \in \mathbb{C}$ with $\tilde{g}(z) = \kappa_s\, s^{-z}\tilde{H}(z)$ for every $z \in (0, z_*)$ — that is, $\wideparen{\hat{v}(s,\cdot)}$ and $\wideparen{\kappa_s H(s\cdot)}$ agree on the strip.

Both $x \mapsto \hat{v}(s,x)$ (continuous, 9.26(1)) and $x \mapsto \kappa_s H(sx)$ (continuous, H a Laplace transform) have this same Mellin transform, absolutely convergent on the line $\operatorname{Re} z = c+1$ for any $c \in (0, z_* - 1)$ (9.15, up to the constant κ_s). Mellin inversion on a line of absolute convergence recovers a continuous function from its transform, so

$$\hat{v}(s,x) \;=\; \kappa_s\, H(sx), \qquad x > 0.$$

Letting $x \downarrow 0$ on both (continuous) sides and using $\hat{v}(s, 0+) = \hat{f}(s)$ (9.26(1)) and $H(0) = 1$ (H the Laplace transform of the law of T_1) gives $\kappa_s = \hat{f}(s)$. So $\hat{v}(s,x) = \hat{f}(s)H(sx) = \hat{u}(s,x)$ for every $x > 0$ and every $s > 0$: $t \mapsto v(t,x)$ and $t \mapsto u(t,x)$, both in X_0 (9.26(1) for v, A.18 for $u = \Phi_{0,x}f \in \mathcal{D} \subset X_0$), have the same Laplace transform for every $s > 0$, so by 3.8 they agree: $v(\cdot,x) = u(\cdot,x)$ in X_0, for every $x > 0$. □

Corollary 9.29 (classical solutions of the signaling problem)**.** Assume (H), let $f \in \mathcal{D}$ with f' additionally bounded and continuous, and let $u(t,x) := (\Phi_{0,x}f)(t)$ be the scale-space field. Suppose that for some $c \in (0, z_* - 1)$,

$$\int_{\mathbb{R}} (1+|\tau|)\,\big|m(c+1+i\tau)\big|\,d\tau \;<\; \infty, \qquad m(z) := \mathbb{E}\big[T_1^{-z}\big]$$

— decay of the negative-moment function along one vertical line. Then for every $t > 0$, the function $x \mapsto u(t,x)$ lies in the domain of A at height c, with realising function $h(x) = x\,\partial_t u(t,x)$, and

$$\partial_t u(t,x) \;=\; \big(A\,u(t,\cdot)\big)(x) \qquad \text{at every } x > 0:$$

the field solves the signaling equation *classically*, pointwise in x and at every t, not merely at the level of Mellin transforms. ▷ **[T]** proved-target *(**Not tagged. What buys the two things.** f' bounded and continuous is what makes $\partial_t u(t,x) = \mathbb{E}[f'(t - xT_1)]$ a genuine* pointwise *derivative at every t, not merely the X_0-derivative 9.6 already supplies — 9.7's remark that continuity of the field "could be bought from absolute continuity of T_1's law … for no gain" is answered here, at no extra interface, by buying it from f' instead. The vertical-decay hypothesis is what makes the realising function's transform* vertically integrable*, the second half of 9.3's domain condition; the first half, that the transform is defined on the line, is automatic once h is continuous and bounded near* 0 *and decaying.* ***The bound is re-derived, not quoted.*** *The route below gives $|\wideparen{\partial_t u(t,\cdot)}(c+1+i\tau)| \le (\|f\|_\infty t^c/c) \cdot |c+i\tau| \cdot |m(c+1+i\tau)|$ for fixed t; only the* shape *— a* *polynomial factor in τ against m — matters for vertical integrability, not the exact power of t, so nothing above depends on having isolated the sharpest constant.* ***Verified against every family the article names****, in the remark below: Gamma, stable ($\alpha < 1$) and Bessel-K all satisfy the hypothesis, by the same exponential Stirling decay in each case; the pure delay fails it, correctly, that family's operator being local (10.5) rather than reached by this route at all.)*

Proof. By 9.6, $u(\cdot, x)$ is the primitive of the field of f': $u(t, x) = \int_0^t \mathbb{E}[f'(\rho - xT_1)]\, d\rho$ for every t. The integrand $\rho \mapsto \mathbb{E}[f'(\rho - xT_1)]$ is continuous, by dominated convergence (f' bounded and continuous, $|f'| \leq \|f'\|_\infty$ an integrable dominating constant on any compact ρ-range), so the fundamental theorem of calculus gives $t \mapsto u(t, x)$ a genuine pointwise derivative at *every* t, equal to $g(x) := \mathbb{E}[f'(t - xT_1)]$ — continuous in x by the same dominated convergence argument, run in x instead of ρ. Set $h(x) := x\, g(x) = x\, \partial_t u(t, x)$, continuous.

$\tilde{H}(z)(I^{z-1}f)(t) = \widetilde{\partial_t u(t, \cdot)}(z)$ for $0 < \operatorname{Re} z < z_*$ (9.6, the derivative clause, extended strip since $f \in \mathcal{D}$ is bounded), and the weight x shifts the Mellin argument by one, so $\tilde{h}(z) = \widetilde{\partial_t u(t, \cdot)}(z + 1)$. By 9.16's recursion $B(-z) = \tilde{H}(z+1)/\tilde{H}(z)$ and 9.8(2)'s transform identity shifted by one, $\tilde{h}(z) = B(-z)\, \widetilde{u(t, \cdot)}(z)$ for $0 < \operatorname{Re} z < z_* - 1$, off the isolated zeros of $\tilde{H}(z)$: the realising-function condition of 9.3, with $u(t, \cdot)$ in the role of its g, at symbol B, height c.

For the vertical-integrability bound, write $z = c + 1 + i\tau$. By definition of I, $\Gamma(z - 1)(I^{z-1}f)(t) = \int_0^t (t - r)^{z-2} f(r)\, dr$, and $f \in \mathcal{D}$ is bounded ($\|f\|_\infty \leq \|f'\|_1$, A.18), so

$$\left|\Gamma(z-1)(I^{z-1}f)(t)\right| \;\leq\; \|f\|_\infty \int_0^t (t-r)^{\operatorname{Re} z - 2}\, dr \;=\; \|f\|_\infty \frac{t^{\operatorname{Re} z - 1}}{\operatorname{Re} z - 1} \;=\; \|f\|_\infty \frac{t^c}{c}$$

(using $\operatorname{Re} z = c + 1 > 1$). Since $\Gamma(z) = (z-1)\Gamma(z-1)$, $|\Gamma(z)(I^{z-1}f)(t)| \leq |z - 1|\, \|f\|_\infty\, t^c/c$, and $\widetilde{\partial_t u(t, \cdot)}(z) = m(z)\, \Gamma(z)(I^{z-1}f)(t)$ (9.2), so

$$\left|\tilde{h}(c + i\tau)\right| \;=\; \left|\widetilde{\partial_t u(t, \cdot)}(c + 1 + i\tau)\right| \;\leq\; \frac{\|f\|_\infty\, t^c}{c}\, |c + i\tau|\, \left|m(c + 1 + i\tau)\right|.$$

Since $|c + i\tau| \leq (c+1)(1 + |\tau|)$, the displayed hypothesis makes the right side integrable in τ over $\mathbb{R}$: $\tilde{h}$ is vertically integrable at height c.

So $u(t, \cdot)$ lies in the domain of A at height c with realising function h, which is continuous; 9.18 gives $A[u(t, \cdot)](x) = h(x)/x = g(x) = \partial_t u(t, x)$, at every $x > 0$. □

Remark 9.30 (which families are classical)**.** 9.29's hypothesis is a statement about $m(z) = \mathbb{E}[T_1^{-z}]$'s decay along one vertical line, checked against every family the article names.

Gamma ($F(s) = \gamma \log(1 + s)$, $\gamma > 1$, 9.12): $m(z) = \Gamma(\gamma - z)/\Gamma(\gamma)$, analytic for $\operatorname{Re} z < \gamma$. For $c \in (0, \gamma - 1)$, $\gamma - (c + 1) \in (0, \gamma - 1)$ avoids Γ's poles, and the classical vertical-line decay $|\Gamma(\sigma + i\tau)| \sim \sqrt{2\pi}\, |\tau|^{\sigma - 1/2} e^{-\pi|\tau|/2}$ (already used at 9.15) gives $|m(c + 1 + i\tau)|$ exponential decay in $|\tau|$: the hypothesis holds.

Stable, $\alpha < 1$ ($F(s) = s^\alpha$, 9.11): $m(z) = \Gamma(z/\alpha)/(\alpha\Gamma(z))$. The same asymptotic applied to numerator and denominator gives $|m(c{+}1{+}i\tau)|$ decaying like $e^{-(\pi/2)(1/\alpha - 1)|\tau|}$ up to a polynomial factor, exponential since $\alpha < 1$: the hypothesis holds.

Bessel-K ($T_1 = 1/(2\gamma_a)$, 8.3; $m(z) = 2^z\Gamma(a + z)/\Gamma(a)$, 9.13): $|2^{c+1+i\tau}| = 2^{c+1}$ is constant in τ, so the decay of $|m(c + 1 + i\tau)|$ is exactly that of $|\Gamma(a + c + 1 + i\tau)|$ — exponential, by the same asymptotic as the Gamma family. **Checked directly, not assumed**: the closed form $2^z\Gamma(a + z)$ looks, read along the real axis, like it grows without bound, but $2^{i\tau}$ has modulus 1 and contributes nothing to the decay along a *vertical* line; the family is classical.

Pure delay ($F(s) = b_0 s$, $b_0 > 0$): $T_1 \equiv b_0$ a.s., so $m(z) = b_0^{-z}$, and $|m(c{+}1{+}i\tau)| = b_0^{-(c+1)}$ is constant in τ: not integrable against $(1 + |\tau|)\, d\tau$. The hypothesis fails, correctly. The transport operator's eigenfunctions are the constant-modulus exponentials, whose Mellin data does not decay, and this family's inversion is instead 10.5's territory — a polynomial (here, first-order) symbol, local by construction, not reached by 9.29's route at all.

Corollary 9.31 (the signaling problem is well posed)**.** Assume (H), work in the canonical gauge, and let $f \in \mathcal{D}$, with $u(t,x) := (\Phi_{0,x} f)(t)$ the scale-space field. Then the signaling problem for f satisfies all three of Hadamard's demands — existence, uniqueness, continuous dependence on the data — and is stable under perturbation of the exponent besides:

1. *(Existence.)* u is a Mellin solution of the signaling problem for f (9.26), with profile-dominated modes (9.28(1)).

2. *(Uniqueness.)* It is the unique Mellin solution for f with profile-dominated modes (9.28(2)).

3. *(Continuous dependence on the data.)* For $g \in \mathcal{D}$ with field $v(t,x) := (\Phi_{0,x} g)(t)$,

$$\sup_{x>0} \|u(\cdot,x) - v(\cdot,x)\|_1 \;\le\; \|f - g\|_1,$$

   and along the evolution,

$$\|u(\cdot,y) - v(\cdot,y)\|_1 \;\le\; \|u(\cdot,x) - v(\cdot,x)\|_1 \qquad (0 < x < y).$$

4. *(Stability under perturbation of the exponent.)* Let F_n be admissible exponents with $F_n(s) \to F(s)$ for every $s > 0$, and let $u_n(t,x) := (\mu_{0,x}^{(n)} * f)(t)$ be the corresponding fields, $\mu_{0,x}^{(n)}$ the canonical-gauge kernel of F_n — the law of $x\,T_1^{(n)}$ with $\mathbb{E}[e^{-sT_1^{(n)}}] = e^{-F_n(s)}$. Then for every $x > 0$,

$$u_n(\cdot,x) \;\to\; u(\cdot,x) \quad \text{in } X_0.$$

   When every F_n satisfies (H), the left side is, by (1)–(2) applied to F_n, the unique profile-dominated Mellin solution of the perturbed problem: the solutions of the perturbed problems converge, at every scale, to the solution of the limit problem.

▷ **[T]** proved-target *(**Not tagged.** Assembly, and priced as such: clauses (1)–(2) are 9.28 restated; clause (3) is two applications of Tonelli through 7.3's representation; clause (4) is 3.5 (ledger ledger A5) read on the kernels, upgraded to L^1 convergence of the fields by a Scheffé argument proved inline — the only estimate here that is not a restatement, and it is elementary and self-contained. The node exists so that the chapter's headline word —* well posed *— is carried by a stated and proved result rather than by arc prose; 9.32 marks the boundary of the claim. (H) enters only through clauses (1)–(2): clauses (3) and (4) use only the representation and hold for every admissible family and all data in X.)*

Proof. (1) and (2) are 9.28, restated clause for clause.

(3) By 7.3 the solution operator at scale x is convolution with the probability measure $\mu_{0,x}$, so $u(\cdot,x) - v(\cdot,x) = \mu_{0,x} * (f-g)$; and convolution with a probability measure μ carried by $[0,\infty)$ is nonexpansive on X: by Tonelli, $\int |(\mu * h)(t)|\,dt \le \iint |h(t-y)|\,d\mu(y)\,dt = \mu([0,\infty))\,\|h\|_1 = \|h\|_1$. That gives the first display, uniformly in x. For $0 < x < y$ the hemigroup factorizes $\Phi_{0,y} = \Phi_{x,y}\Phi_{0,x}$, so $u(\cdot,y) - v(\cdot,y) = \mu_{x,y} * \big(u(\cdot,x) - v(\cdot,x)\big)$ with $\mu_{x,y}$ again a probability measure, and the same bound applies.

(4) Two steps: the kernels converge weakly, and weak convergence convolves into X_0-convergence.

The kernels. Fix $x > 0$. The Laplace transform of $\mu_{0,x}^{(n)}$ at s is $e^{-F_n(xs)}$, which converges to $e^{-F(xs)}$, the transform of $\mu_{0,x}$, for every $s > 0$; so $\mu_{0,x}^{(n)} \to \mu_{0,x}$ weakly by 3.5. (The same argument runs on any increment: the transform of $\mu_{x,y}^{(n)}$ is $e^{-(F_n(ys) - F_n(xs))}$, so every $\mu_{x,y}^{(n)} \to \mu_{x,y}$ weakly as well.)

The fields. We show that $\mu_n \to \mu$ weakly, for probability measures on $[0,\infty)$, forces $\mu_n * h \to \mu * h$ in X for every $h \in X$, and apply this at $\mu_n = \mu_{0,x}^{(n)}$, $h = f$. First let g be continuous with compact support and $g \geq 0$. Then $(\mu_n * g)(t) = \int g(t-y)\, d\mu_n(y) \to (\mu * g)(t)$ at every t, the integrand being bounded and continuous in y; and the integrals are all equal: by Tonelli, $\int(\mu_n * g) = \mu_n([0,\infty)) \int g = \int g$, and likewise for μ. Pointwise convergence of nonnegative integrable functions whose integrals converge to the limit's integral upgrades to L^1 convergence — Scheffé's lemma, and here it is one line: writing $a_+ := \max(a, 0)$,

$$\int \left|\mu_n * g - \mu * g\right| \;=\; \int (\mu_n * g - \mu * g) \;+\; 2\int \big(\mu * g - \mu_n * g\big)_+,$$

the first term zero as just computed, and the second tending to zero by dominated convergence, since $0 \leq (\mu * g - \mu_n * g)_+ \leq \mu * g \in L^1$ and the integrand tends to zero pointwise. A general continuous compactly supported g splits as $g_+ - g_-$, both again continuous with compact support. Finally, for $h \in X$ and $\varepsilon > 0$ pick such a g with $\|h - g\|_1 < \varepsilon$ — continuous functions of compact support are dense in L^1 — and the contraction of (3) closes the estimate:

$$\|\mu_n * h - \mu * h\|_1 \;\leq\; \|\mu_n * (h-g)\|_1 + \|\mu_n * g - \mu * g\|_1 + \|\mu * (g-h)\|_1 \;\leq\; 2\varepsilon + \|\mu_n * g - \mu * g\|_1,$$

whose limit superior over n is at most 2ε, for every $\varepsilon > 0$. Applied at $\mu_n = \mu_{0,x}^{(n)}$ and $h = f$ this gives $u_n(\cdot, x) \to u(\cdot, x)$ in X; every field here is causal, f and the kernels being carried by $[0, \infty)$, so the convergence is in X_0. The closing sentence of (4) is (1)–(2) applied to F_n, whose hypotheses — (H) for the exponent, $f \in \mathcal{D}$ for the data — are exactly what is assumed. □

Remark 9.32 (what the stability rests on; what is not claimed)**.** The stability clauses of 9.31 come from the representation rather than from the equation: the solution operator at each scale is convolution with a probability measure (7.3), and the exponent stability is the continuity theorem for Laplace transforms (3.5) read on the kernels. Neither uses (H), which enters only through the solution concept of clauses (1)–(2); both therefore hold for every admissible family and every signal in X, not only on the class the signaling problem is posed on. In the scale-Cauchy reading of A.21, the second contraction of clause (3) says that a perturbation of the record at one scale does not grow at any coarser one. What is *not* claimed is a stability estimate in a topology adapted to A itself — continuity of the solution in the Mellin-class data, or convergence of the operators behind clause (4). The conclusion there is convergence of the solution fields, never of the symbols, and the distinction is real: $F(s) + s/n \to F(s)$ pointwise with every perturbed exponent carrying drift, so $z_* = \infty$ along the whole sequence (9.25(2)) while the limit's z_* may be finite, and the strips the symbols live on do not converge to the limit's. Whether the signaling problem is well posed in a topology adapted to A is a question this article leaves open.

Chapter 10

The locality theorem: Theorem 5′

Theorem 5 of `@fagerstrom2005temporal` singles out $\alpha = \frac{1}{2}$ — the heat signaling equation — as the unique member of the stable family whose memory-line evolution is a local, positivity-respecting differential equation. This chapter proves the hemigroup analogue.

The structure is genuinely two-layered: locality *alone* opens a ladder of admissible local media of every order, each rung an $(n-1)$-parameter family, and it is the positive maximum principle — the requirement that the medium be a physical transport medium — that closes the ladder at order two, where it leaves the Bessel family together with the degenerate transport member of case (1). Throughout we assume (H), work in the canonical gauge, write $m(z) := \mathbb{E}[T_1^{-z}]$, and take A and B as in Chapter 9.

Definition 10.1 (locality; positive maximum principle)**.** The inversion operator A is *local of order* n if it agrees with a differential expression $\sum_{j=0}^{n} c_j(x)\,\partial_x^j$, with $c_j \in C((0,\infty))$ and $c_n \not\equiv 0$, on $C_c^\infty((0,\infty))$ *and on the profiles* $H(s\,\cdot)$*,* $s > 0$. It satisfies the *positive maximum principle* (PMP) if $(Ag)(x_0) \le 0$ whenever $g \in C_c^\infty((0,\infty))$ attains a nonnegative maximum at x_0.

The profiles are part of the test class because the ($\Rightarrow$) direction of 10.2 must reach the symbol, and $C_c^\infty((0,\infty))$ is disjoint from the only class on which B's behavior is controlled; the maximum principle stays on $C_c^\infty((0,\infty))$, where a profile, having no interior maximum, would say nothing. ▷ [**T**] proved-target *(**Formalised**, as `Hemigroup.SelfDecomposableExponent.IsLocalOfOrder` and `SatisfiesPMP` in `LocalOperator.lean`. **The profiles were added to the test class after formalisation**, and the reason is exactly the one given above: without them 10.2's ($\Rightarrow$) direction cannot reach the symbol, since $C_c^\infty((0,\infty))$ is disjoint from the only class on which B's behavior is controlled. Three readings had to be fixed, and each is a choice rather than a transcription. **Locality is bundled as data, not as an existential.** "Agrees with* some *differential expression" would make the coefficients inaccessible, and 10.2's conclusion is a statement* about *them — that $c_j(x) = \gamma_j x^{j-1}$. As a structure with a `coeff` field the lemma says what it says here; as a `Prop` every use would re-obtain coefficients with no way to say they are the same ones. The price is that locality is then data, so the ($\Leftarrow$) direction of 10.2 concludes `Nonempty (…)` — which is the propositional reading, and the one 10.5 quantifies over. **The two clauses are separate structures, because they cost differently.** `IsLocalOfOrderCore` is the C_c^∞ clause and is what a polynomial symbol supplies cheaply — it is what 10.2's ($\Leftarrow$) direction proves. `IsLocalOfOrder` extends it with the profile clause. That clause is true for the same reason, but by a different computation: $\partial_x^j H(sx) = \int (-st)^j e^{-sxt}\,\mu(dt)$, followed by the Gamma-integral hinge of 9.2. **The differentiation step is done**, as `Hemigroup.SelfDecomposableExponent.iteratedDeriv_profile`, and it cost almost nothing: $H(u) = \mathbb{E}\,e^{-uT_1}$* is *Mathlib's moment generating function at a negative argument, and the library already carries every derivative of that on the interior of the set where the exponential moment converges — which for a law on $[0,\infty)$ contains the whole negative axis. The dominated-convergence induction this looked like it needed was already in place. (The development had met the same coincidence once before, on the other transform: chapter*

9 identifies `negMomentC` *with Mathlib's* `complexMGF`*.)* ***And the Fubini-and-Gamma computation that remained is now done too****, as* `mellin_pow_mul_iteratedDeriv_profile`*: it cost no new analysis either, because $x \mapsto \int t^j e^{-xt}\mu(dt)$ is the Laplace transform of $t^j\mu(dt)$ and 9.2 — restated for a measure carried by $(0,\infty)$, which is all its proofs used — applies to it unchanged.* ***Both clauses are now proved in both directions****, so the separation has served its purpose and is kept for what it records: the profile clause cost the engine and one estimate (the polynomial decay of Γ) that the C_c^∞ clause got for free, and a structure that hid the difference would have hidden where the work was. $C_c^\infty((0,\infty))$* ***means support inside the open half-line****, not merely compact support in $\mathbb{R}$. The distinction is load-bearing rather than pedantic: it is exactly what kills the boundary term at the origin in the integrations by parts below, where the weight x^{z-1} is singular.* ***The maximum principle needs a real-valued reading.*** *A is complex-valued here, and $(Ag)(x_0) \le 0$ is not a statement about a complex number. The Lean version quantifies over real-valued test functions and asserts* $\mathrm{Re}(Ag)(x_0) \le 0$ *— weaker than first proving that A preserves realness, and therefore the safer of the two.)*

Lemma 10.2 (covariant local = polynomial symbol)**.** A is local of order n if and only if B is a polynomial of degree n; in that case the covariance built into the Mellin class (Definition 9.4) forces the coefficients

$$c_j(x) \;=\; \gamma_j\, x^{j-1}, \qquad A \;=\; \frac{1}{x}\sum_{j=0}^{n} \gamma_j\, \theta(\theta-1)\cdots(\theta-j+1) \;=\; \frac{1}{x}\, B(\theta).$$

▷ **[T]** proved-target *(****Machine-checked as an equivalence****, and carried by three declarations rather than one, since the two directions need different things and only one of them needs covariance.* `nonempty_isLocalOfOrder_iff_symbol_eq` *is the equivalence, and is a* bundle *in the sense 7.3 and 9.8 are: it necessarily depends on everything both halves do, so the per-half lines of* `CIAxiomGuard.lean` *remain the load-bearing ones. It also says* less *than its halves — the coefficient form $c_j(x) = \gamma_j x^{j-1}$ displayed above cannot be stated in an equivalence of this shape, and lives in* `exists_symbol_eq_of_isLocalOfOrder`*.* ***($\Leftarrow$) is machine-checked****, as* `Hemigroup.SelfDecomposableExponent.isLocalOfOrderCore_of_symbol_eq`*, on A17 and nothing else.* ***Its symbol hypothesis is asked for off the zeros of $\tilde{H}$, and that is what made it take (H).*** *The two are one decision. $B = \tilde{H}(z{+}1)/\tilde{H}(z)$ is a quotient, so at a zero of the denominator no identity with a polynomial holds — in Lean the value there is the junk* 0 *— and a hypothesis asserting one at* every *point of the strip is one that ($\Rightarrow$) cannot supply, so the two halves would not compose. Restricting it costs nothing analytically: A is an inverse Mellin transform, an integral* over *the line rather than an evaluation on it, so a null set is discarded (*`mellinInv_congr_line_ae`*). What it costs is that the zeros are null only by 9.16, which needs (H). This node previously recorded that ($\Leftarrow$) uses (H) nowhere in its argument, and that remains true — the Euler factors, the vertical decay and the inversion are facts about test functions and say nothing about F. (H) enters as the price of a hypothesis the other direction can meet, not as a step of the proof; and it was in any case already making the statement non-vacuous, the strip $0 < c < z_* - 1$ being inhabited only when $z_* > 1$. Three roles a hypothesis can play, kept apart.* ***The ($\Leftarrow$) direction needed one analytic fact, and it is now proved****: the Mellin symbol of the Euler operator, $\mathcal{M}\big[x^j g^{(j)}\big](z) = \big(\prod_{i<j}(-z-i)\big)\,\tilde{g}(z)$ for test functions g, which is j integrations by parts against the weight x^{z-1} with no boundary terms. Mathlib carries no Mellin/derivative interface at all —* `MellinTransform.lean` *and* `MellinInversion.lean` *between them prove nothing about derivatives — so it is written from scratch, as* `Hemigroup.mellin_pow_mul_iteratedDeriv`*, and* `#print axioms` *gives Lean core alone. Everything else in that direction is linearity.* ***Where the support hypothesis is spent****, since it is the reason the statement is true: Mathlib's integration by parts on the whole line asks for differentiability of each factor only on the* other's *support. The weight $x \mapsto x^w$ is differentiable away from the origin, and $C_c^\infty((0,\infty))$ says exactly that the origin is not*

in the region where its derivative is needed. Compact support alone would leave a boundary term at 0*, where the weight is singular, and the identity would be false as stated — which is why 10.1's test class is rendered with support inside* $(0,\infty)$ *and not with compact support in* $\mathbb{R}$*. The factor is isolated as* `mellinEulerFactor` *rather than inlined, because the sign convention is where this chapter's bookkeeping errors would live: Lean's* `inversionSymbol z` *is this chapter's* $B(-z)$*.* ***Of the (⇒) direction, the coefficient half is machine-checked*** *as* `coeff_eq_of_isLocalOfOrder`*: covariance forces* $c_m(\sigma) = c_m(1)\sigma^{m-1}$*, so the homogeneous form the statement above asserts is not imposed but derived. It rests on two things, both proved here. The covariance* $A\Delta_\sigma = \sigma^{-1}\Delta_\sigma A$ *itself is* `inversionOperator_lineDilate`*. What licenses "comparing coefficients" at all — that a differential expression is determined by its coefficients — is* `exists_isTestFunction_jet`*: at every point of* $(0,\infty)$ *and every order* m*, a test function whose jet there is the* m*-th basis vector, built as a monomial cut off by a bump of outer radius* $x_0/2$*, so that its support sits inside* $(0,\infty)$ *— the same constraint the integrations by parts need, arrived at from the other side. Feeding one such jet into both readings of* $A\Delta_\sigma g$ *at the point* σ *collapses both sums to a single term; and since* $x=\sigma$ *sends* x/σ *to* 1*, a single jet at* 1 *settles every* σ *at once.* ***The remaining clause is now machine-checked, and formalising it showed that the proof above skips a step.*** *The clause is that the symbol is then the corresponding polynomial, which the proof gets by "Mellin-transforming on a line and using injectivity". What that argument delivers is that* $B\tilde{g}$ *and* $P\tilde{g}$ *have the same inverse transform for every* test function g*. Two things separate it from* $B = P$*. The first is routine: concluding equality of the symbols needs the forward-after-inverse direction of Mellin inversion, which Mathlib lacks (it has the Fourier analogue, and the two are bridged by its own* `mellin_eq_fourier`*). The second is not routine.* $P\tilde{g}$ *is vertically integrable, since* $E_j(z)\tilde{g}(z)$ *is itself the transform of a test function; but* $B\tilde{g}$ *need not be,* $B = \tilde{H}(z+1)/\tilde{H}(z)$ *carrying no growth bound — which is precisely why 9.3 carries integrability as a hypothesis rather than deriving it. So the comparison has to be routed through the class where* B*'s behavior is known, namely the profiles* $H(s\,\cdot)$ *of 9.5, and those are* not *test functions: they are neither compactly supported nor supported away from the origin. Passing from one class to the other is an approximation argument, and it is not made above. That is why 10.1 now tests locality on the profiles as well, and the correction is not a formalisation detail — the printed proof needs it too, and the proof below is written the corrected way.* ***The route on the profiles needs no integration by parts.*** *The engine above,* $\mathcal{M}[x^j g^{(j)}](w) = E_j(w)\tilde{g}(w)$*, was proved for test functions by* j *integrations by parts; on the profiles the boundary terms would have to be argued, and they do not have to be, because* $\partial_x^j H(sx) = (-s)^j \int t^j e^{-sxt}\,\mu(dt)$ *turns the derivative into an integral before the transform is taken. What is then left is three moves and no estimate: the weight* x^j *is a Mellin shift, so it never enters a Fubini; the dilation is a factor; and* $x \mapsto \int t^j e^{-xt}\,\mu(dt)$ *is the Laplace transform of the* weighted *measure* $t^j\mu(dt)$*, so 9.2 applies to it verbatim. Nothing in that lemma had to be restated for a weight — its three steps are now stated for a measure carried by* $(0,\infty)$ *with a finite negative moment, which is all their proofs ever used. The bookkeeping* $E_j(w) = (-1)^j\Gamma(w+j)/\Gamma(w)$ *is the functional equation* j *times.* ***"$B = P$ on the strip" has to say which equality it means, and the answer is not the pointwise one.*** *What the argument delivers at every point of the strip, with no side condition, is the* recursion $\tilde{H}(z+1) = P(z)\tilde{H}(z)$*;* B *is that identity divided by* $\tilde{H}(z)$*. Where* $\tilde{H}$ *vanishes the division is not available, and in Lean the quotient* $\tilde{H}(z+1)/0$ *takes the junk value* 0 *while* $P(z)$ *is a polynomial value — so the unconditional pointwise reading is false as stated. Three statements are therefore proved, and they are genuinely different: the recursion, unconditionally (which is what 10.3 consumes, its clause (2) arguing from the identity between analytic functions and never from a value of* B*);* $B = P$ *as meromorphic functions, unconditionally, which is 9.17's* `EventuallyEq`*; and* $B(z) = P(z)$ *as values, exactly off the zeros of* $\tilde{H}$*. Nothing available rules those zeros out:* $\tilde{H} = \Gamma\cdot m$ *is known nonzero only at the* real *points of the strip (9.16), which is all chapter 9 needed, and excluding complex zeros would be a theorem about self-decomposable* *laws that this article does not have. This is the third time the same question has been answered in this development — 9.3's realising identity and 9.17's conclusion are the other two — and each time it was writing the statement that raised it.* ***(⇐) was weakened to match**, so that its symbol hypothesis is exactly what (⇒) delivers; see above for what that cost.* ***(⇐) also has to be run twice, because 10.1 tests two classes**, and the profile run is where this node's last piece of analysis was. Applying a polynomial symbol to $H(s\,\cdot)$ gives $h(x) = \sum_j \gamma_j x^j \partial_x^j H(sx)$ with $\tilde{h}(z) = P(z)s^{-z}\tilde{H}(z)$, and recovering h from it by Mellin inversion needs that transform vertically integrable. On a test function that is free — $E_j(z)\tilde{g}(z)$ is itself the transform of a test function. On a profile it is the estimate $\int |\tau|^n\, |\Gamma(c+i\tau)|\, d\tau < \infty$ for $n = \deg P$, of which 9.15 needed only the case $n = 0$; and it is the same two lines, since $\Gamma(z+k)/\Gamma(z)$ is a product of k factors each of imaginary part τ, giving $|\Gamma(c+i\tau)|\,|\tau|^k \le \Gamma(c+k)$, with the binomial theorem reducing a polynomial in $|\tau|$ to a finite sum of those. No asymptotics — which is the same lesson ledger A12 taught, now at a fifth reading of it.* ***So nothing is left between the halves, and the node is an equivalence.**)*

Proof. (⇐) $x^j\partial_x^j = \theta(\theta-1)\cdots(\theta-j+1)$, so a polynomial B, expanded in falling factorials, gives $x^{-1}B(\theta) = \sum \gamma_j x^{j-1}\partial_x^j$, a differential expression of order $\deg B$. This has to be checked on both classes of 10.1. On $C_c^\infty((0,\infty))$ it is the Euler-factor identity and the linearity of $\mathcal{M}^{-1}$ along the line. On a profile $H(s\,\cdot)$ the same argument applies to $h(x) = \sum_j \gamma_j x^j \partial_x^j H(sx)$, whose transform is $P(z)s^{-z}\tilde{H}(z)$ by the identity used in (⇒) below; that transform is absolutely integrable on the line because $|\Gamma(c+i\tau)|$ decays faster than every power of τ, so $\mathcal{M}^{-1}\mathcal{M}h = h$ and $x^{-1}h$ is $(A\,H(s\,\cdot))$.

(⇒) Every operator of the covariant Mellin class (9.4) satisfies the covariance $A\Delta_\sigma = \sigma^{-1}\Delta_\sigma A$, where $(\Delta_\sigma g)(x) = g(x/\sigma)$ — machine-checked as `Hemigroup.inversionOperator_lineDilate`, the mechanism being entirely Mellin-side: dilation multiplies the transform by σ^z, and multiplying the symbol by σ^z translates the inverse transform's argument, both reducing to $x^{-(c+i\tau)}\sigma^{c+i\tau} = (x/\sigma)^{-(c+i\tau)}$. Imposing this on $\sum c_j(x)\partial_x^j$ and comparing coefficients of $g^{(j)}(x/\sigma)$ gives $c_j(x)\sigma^{-j} = \sigma^{-1}c_j(x/\sigma)$, i.e. $c_j(x) = c_j(1)x^{j-1}$. Rewriting via falling factorials, A agrees with $x^{-1}P(\theta)$ for the polynomial $P(w) = \sum_j \gamma_j E_j(w)$, where $E_j(w) = \prod_{i<j}(-w-i)$.

It remains to identify P with B, and the comparison is made on the profiles rather than on $C_c^\infty((0,\infty))$ — the class 10.1 was widened to include, and the only one on which B's behavior is known. Fix $s > 0$ and put $g = H(s\,\cdot)$. Locality gives $(Ag)(x) = \sum_j \gamma_j x^{j-1}\partial_x^j H(sx)$, while 9.8(1)'s eigenfunction relation gives $(Ag)(x) = sH(sx)$; equating them and clearing the weight,

$$s\,x\,H(sx) \;=\; \sum_{j=0}^{n} \gamma_j\, x^j\, \partial_x^j H(sx), \qquad x > 0.$$

The left side is the realising function h of 9.3 for g, with $\tilde{h}(w) = s^{-w}\tilde{H}(w+1)$. On the right, $\partial_x^j H(sx) = (-s)^j \int t^j e^{-sxt}\mu(dt)$, so $x \mapsto x^j\partial_x^j H(sx)$ has, by the Mellin shift x^j, the dilation factor, and 9.2 applied to the measure $t^j\mu(dt)$, the transform $E_j(w)\tilde{g}(w)$ throughout $0 < \operatorname{Re} w < z_*$. Each term converges there, so the sum may be transformed term by term, giving $\tilde{h}(w) = P(w)\tilde{g}(w)$ — which is exactly the hypothesis of 9.5 with P in the place of a candidate symbol. Hence B and P agree as meromorphic functions on $0 < \operatorname{Re} z < z_* - 1$; equivalently, since $\tilde{g}(w) = s^{-w}\tilde{H}(w)$ never loses its nonvanishing factor,

$$\tilde{H}(z+1) \;=\; P(z)\,\tilde{H}(z)$$

at every point of the strip, and $B(z) = P(z)$ at every point where $\tilde{H}(z) \neq 0$. No injectivity of the inverse Mellin transform is used, and none is available. □

Lemma 10.3 (the moment recursion)**.** Suppose A is local of order n, i.e. B is a polynomial of degree n. Then:

1. $B(0) = 0$; write $B(-z) = z\,Q(z)$ with Q a polynomial of degree $n-1$;
2. $z_* = \infty$: all negative moments of T_1 are finite;
3. $m(z+1) = Q(z)\,m(z)$ for all $z > 0$, and $Q > 0$ on $(0,\infty)$.

$\triangleright$ [**A**] analytic interface ledger A13 *(the singularity of a transform with monotonic integrand at the real point of its axis of convergence; `@widder1941laplace`, Ch. II, §5, Thm. 5b, pp. 58–59. Cited, not reproven here.* ***Assignment.*** *ledger A13 carries clause (2) alone, and only its final step: that a finite z_* would be a singularity of $\tilde H$, contradicting the analytic continuation the recursion supplies. Clauses (1) and (3) are* ***[T]*** *and* ***both are now machine-checked****, split off so that a node spending A13 on clause (2) does not appear to spend it on them: the vanishing $B(0) = 0$ is 10.10, which also records that the polynomial hypothesis plays no part in it, and the factorisation together with the recursion is 10.12.* ***One thing does not survive the split, and it is the range.*** *Clause (3) is stated here on $(0,\infty)$, and the split-off node proves it on $0 < z < z_* - 1$; the difference is exactly clause (2), so writing it on the whole half-line is already using A13. On the locality hypothesis the two ranges coincide once A13 is spent, which is why the chapter may go on using the wider form — but the machine-checked statement is the narrower one, and saying so is the point of the split. Two cautions recorded in the ledger. First, the anchor is the theorem number: Widder's own footnote credits the result to Hamburger (1921), and "Pringsheim–Landau" — the name the draft uses, and the received one — is attached to the Dirichlet- and power-series forms rather than to this page. Second, the hypothesis is monotonicity of the integrand, which holds here because $H = e^{-F}$ is the transform of a positive random variable; Widder gives on the same page the counterexample showing a general Laplace integral need have no singularity on its axis.)*

Proof. (1) By 9.16, $B(-z) = z\,m(z+1)/m(z)$ on the strip. As $z \downarrow 0$, $m(z) \to 1$ by dominated convergence and $m(z+1) \to m(1) < \infty$ by (H), so $B(0) = 0$; since B is a polynomial, $Q(z) := B(-z)/z$ is one of degree $n-1$.

(2) On $0 < z < z_* - 1$ we have the identity $\tilde H(z+1) = B(-z)\tilde H(z)$ between functions analytic on the strip $0 < \operatorname{Re} z < z_*$. If $z_* < \infty$, the right side is analytic at $z = z_* - 1 + \varepsilon$ for small ε, which extends $\tilde H$ analytically through the real point z_*. But $\tilde H(z) = \int_0^\infty x^{z-1} H(x)\,dx$ has nonnegative integrand — $H(x) = \mathbb{E}\,e^{-xT_1} \geq 0$ — which is exactly the hypothesis ledger A13 is anchored on, and that theorem makes the real endpoint of the convergence strip a singularity when finite. Hence $z_* = \infty$.

(3) The recursion is now 9.16 on all of $(0,\infty)$, and positivity of Q follows from $m > 0$. $\square$

Lemma 10.4 (log-convexity)**.** $z \mapsto m(z)$ is log-convex on $(0,\infty)$. $\triangleright$ [**T**] proved-target *(****Machine-checked****, as `Hemigroup.SelfDecomposableExponent.convexOn_log_negMoment`, together with the $[0,\infty]$-valued form `negMoment_le_rpow_mul_rpow` from which it is read off. `#print axioms` shows A17 and nothing else — A17 only because T_1 is a kernel of the construction, so every statement about it inherits the entry. None of this chapter's own cited interfaces is touched.* ***Two readings, and the difference matters.*** *The $[0,\infty]$ form is* unconditional*: no standing hypothesis, no finiteness, no strip, since divergence is a value there rather than a failure. The real-valued form stated above needs m finite and positive, so the formal statement carries the domain $(0,z_*)$ rather than $(0,\infty)$ — which is the same interval once $z_* = \infty$, but that is 10.3(2), the one clause of that lemma ledger A13 carries, and this node does not presuppose it. Positivity of m is where (H)'s first clause enters: the integral runs over $(0,\infty)$, so an atom at the origin would be invisible to it.)*

Proof. By Cauchy–Schwarz, $m\big(\frac{z_1+z_2}{2}\big) = \mathbb{E}\big[T_1^{-z_1/2}\,T_1^{-z_2/2}\big] \leq m(z_1)^{1/2} m(z_2)^{1/2}$, which is midpoint log-convexity; with continuity of m this gives log-convexity.

The appeal to continuity is avoidable, and the formalisation does without it: Hölder at the exponent pair $\left(\frac{1}{\theta}, \frac{1}{1-\theta}\right)$, applied to $T_1^{-z_1}$ and $T_1^{-z_2}$, gives

$$m\big(\theta z_1 + (1-\theta) z_2\big) \;=\; \mathbb{E}\big[(T_1^{-z_1})^{\theta}\,(T_1^{-z_2})^{1-\theta}\big] \;\leq\; m(z_1)^{\theta}\, m(z_2)^{1-\theta}$$

at every $\theta \in [0,1]$ at once, Cauchy–Schwarz being the case $\theta = \frac{1}{2}$. Since that inequality is the definition of log-convexity, nothing remains to upgrade. □

Theorem 10.5 (Theorem 5′: the locality theorem)**.** Assume (H). The inversion operator A of 9.8 is local *and* satisfies the positive maximum principle **iff** one of the following holds.

1. *(Order 1; pure transport, degenerate.)* $B(\theta) = -c'\theta$ with $c' > 0$: then $A = -c'\,\partial_x$, the kernels are the deterministic delays $\mu_{0,x} = \delta_{x/c'}$, and the signaling problem is the transport equation. This member has no smoothing and is excluded by requiring absolutely continuous kernels (7.5).

2. *(Order 2; the Bessel family.)* After a normalization of the unit of time: $B(\theta) = 2\,\theta(\theta - a)$ for some $a > 0$; the kernels are the inverse-gamma family $T_1 \overset{d}{=} 1/(2\gamma_a)$ of 8.3; and in the parabolic gauge ξ (6.4) the signaling problem is the classical degenerate-parabolic PDE

$$\partial_t u \;=\; \tfrac{1}{2}\,\partial_\xi^2 u + \frac{\beta}{\xi}\,\partial_\xi u, \qquad \beta = \tfrac{1}{2} - a, \qquad u(t,0) = f(t), \quad u(0,\xi) = 0.$$

Within case (2), the depth-homogeneous (semigroup) members are exactly $a = \frac{1}{2}$, i.e. $\beta = 0$ — the heat signaling equation — recovering `@fagerstrom2005temporal`, Thm. 5, as the zero-drift point of the family. The kernel moments in case (2) satisfy $\mathbb{E}\,T_x^n < \infty$ iff $n < a$. ▷ [**A**] analytic interface ledger A14 *(Courrège's classification of operators satisfying the positive maximum principle, `@courrege1966sur`, Corollaire 2, p. 2-10. Cited, not reproven here.* ***Assignment.*** *ledger A14 carries one step of the (⇒) direction and nothing else: the order bound, that a local operator satisfying the PMP is a pure second-order diffusion operator with no jump part — the locality clause specifically, not the general integro-differential structure theorem. Note its scan paginates by exposé, so the anchor reads p. 2-10 and not p. 10.* ***ledger A15 used to appear here and no longer does.*** *It carried the uniqueness of the log-convex solution of the recursion, which is what pins m to the gamma form. In the order-2 case the recursion has a* single *Gamma factor, and uniqueness is then the classical Bohr–Mollerup theorem rather than the general Krull–Webster theorem; that case is now proved in-repo as 10.11, so this node is [**T**] at that step. The general entry stays alive at 10.6, whose order-n ladder has $n-1$ Gamma factors and for which Bohr–Mollerup is not enough. The order-1 case never used the entry: there* $\log m$ *is convex with constant unit increments, hence affine, which is elementary and given below. Everything in the (⇐) direction is [**T**]: admissibility of the Bessel family is 8.3, locality is 9.13, and the PMP verification is 10.13,* ***now machine-checked*** *— split off for the same reason the other [T] parts of this chapter were, since ledger A14 sits on the* opposite *use of the maximum principle and a node reporting both under one status would say the principle costs an interface when half of it costs nothing. The homogeneity claim is [**T**] through 7.4.* ***The order-2 branch of (⇒) is now machine-checked****, as* `exists_gamma_form_of_isLocalOfOrder_two`*: a local operator of order* 2 *forces* $m(z) = q_1^z\Gamma(a+z)/\Gamma(a)$ *on* $(0,\infty)$ *with* $q_1, a > 0$*, which is* $T_1 \overset{d}{=} 1/(q_1\gamma_a)$ *(the Lean declaration names this witness* `c_2`*).* ***Both citations enter as hypotheses rather than as axioms****, phrased so either can be demoted the day it is proved: clause (2) of 10.3 is the named hypothesis* `AllNegMomentsFinite`*, and the order bound is simply the order* 2. `#print axioms` *accordingly shows A17 and* neither *ledger A13 nor ledger A14 — the interfaces sit in the statement, where a reader sees them, and not in the trust base.* ***Where that step's work actually is****: not in the Gamma form, which is one application of 10.11, but in* $a > 0$*. Writing* $Q(z) = q_1 z + a_0$*, positivity of* Q *on the open half-line gives only* $a_0 \geq 0$

— a linear function positive on $(0,\infty)$ may vanish at the origin — so $a_0 = Q(0) = m(1) > 0$ has to be taken as a limit, and that needs $m(z+1) \to m(1)$, which is 10.10's dominated-convergence argument one unit to the right, with dominating function $t^{-2}+1$. It is available exactly because $2 < z_*$, i.e. because clause (2) is in force — and sayable at all only because z_* is now valued in $[0,\infty]$; see 9.1. ***The order-1 branch is machine-checked too****, as* `exists_pow_form_of_isLocalOfOrder_one`*: $m(z) = c'^{\,z}$ with $c' > 0$, i.e. $T_1 = 1/c'$ almost surely. It uses neither Krull–Webster nor Bohr–Mollerup, exactly as this node records — at order one Q is the constant $-\gamma_1$, and what is needed instead is that a convex function with constant unit increments is affine.* ***Mathlib does not carry that****, and the usual proof — a periodic function is bounded, a convex function bounded above on $(0,\infty)$ is nonincreasing, nonincreasing and periodic is constant — goes through continuity of a convex function on an open interval. None of it is needed: convexity at the three points $x < y < y+1$, writing $y = \lambda x + (1-\lambda)(y+1)$ with $\lambda = 1/(y+1-x)$, gives $g(y) \le \lambda g(x) + (1-\lambda)g(y)$ by periodicity alone, and cancelling leaves $g(y) \le g(x)$. Three points and one inequality, with no limit and no continuity; the only limit in the branch is at the end, fixing the constant from $m(z) \to 1$.* ***So both branches of (⇒) are now proved****, and what this node still cites is exactly its two interfaces and nothing around them.* ***What keeps this node itself untagged, named.*** *It carries no* `\lean` *tag and appears white in the dependency graph with green children, and that is the accurate reading rather than an oversight — the theorem asserts strictly more than is proved, in four places, and it is worth saying which.* (i) *What (⇒) delivers is the* moment *classification, 10.14. Case (2) goes on to identify the kernels as the inverse-gamma family with the Bessel-K profile, and to read the signaling problem in the parabolic gauge as $\partial_t u = \frac{1}{2}\partial_x^2 u + (\beta/x)\partial_x u$. Those are statements about* densities *and about a PDE, they run through 9.13, and through it they need the Bessel function K — which Mathlib does not carry. This is the one genuinely upstream obstruction in the chapter, and it blocks the ladder's Meijer-G densities in 10.6 for the same reason.* (ii) *In (⇐), admissibility of the family is 8.3, itself an* ***[A]*** *node; its locality clause is 9.13 again. Of that direction only the maximum-principle verification is machine-checked, as 10.13.* (iii) *The homogeneity claim runs through 7.4, not formalized.* (iv) *And the order bound is ledger A14 itself. So the reachable content sits in the four split-off nodes — 10.10, 10.12, 10.13, 10.14 — and the theorem node stays white until Bessel K exists upstream and ledger A14 and 8.3 are discharged. Splitting rather than tagging is what keeps that difference visible; a* `\leanok` *here would claim the densities and the PDE along with the moments.)*

Proof. (⇒) 10.1's pair of hypotheses — locality (agreement with a differential expression on $C_c^\infty((0,\infty))$) together with the positive maximum principle there — is exactly Courrège's *local* positive maximum principle, whose Corollaire 2 (p. 2-10), ledger A14, bounds the order of such an operator by 2: it is a pure second-order diffusion operator with no jump part. By 10.2, $\deg B \le 2$, and $B(0) = 0$ by 10.3(1) — equivalently, there is no killing, which is (A5).

Order 1. Here $Q \equiv c'$ is constant, so $m(z+1) = c' m(z)$ with $m(0) = 1$. By 10.4, $\varphi := \log m$ is convex on $(0,\infty)$ and satisfies $\varphi(z+1) - \varphi(z) = \log c'$ for every $z > 0$. The difference quotients of a convex function are nondecreasing, and on each unit interval they average to $\log c'$; a nondecreasing function whose average over every unit interval is the same constant equals that constant identically, so φ is affine. Hence $m(z) = c'^{\,z}$, the constant fixed by $m(0^+) = 1$. Thus $T_1 = 1/c'$ a.s., $B(-z) = c'z$, i.e. $B(\theta) = -c'\theta$ and $A = -c' x^{-1}\theta = -c'\partial_x$; positivity of Q gives $c' > 0$.

Order 2. Here $Q(z) = q_1 z + q_0$ with $q_1 > 0$, by positivity of Q as $z \to \infty$ in 10.3(3), and $q_0 = Q(0) = m(1) > 0$; write $Q(z) = q_1(z+a)$ with $a := q_0/q_1 > 0$. The recursion $m(z+1) = q_1(z+a)\,m(z)$ with $m(0+) = 1$ — the normalization being 10.10's first limit — has a *unique* log-convex solution by 10.11, namely

$$m(z) \;=\; q_1^z\,\frac{\Gamma(a+z)}{\Gamma(a)} \;=\; \mathbb{E}\Big[(q_1\,\gamma_a)^z\Big],$$

and m is log-convex by 10.4, hence equals it. So $m(z) = \mathbb{E}[(q_1\gamma_a)^z]$ for every real $z > 0$. Both sides are, as functions of z, the moment functions of $\log(1/T_1)$ and of $\log(q_1\gamma_a)$: they are analytic for $\operatorname{Re} z > 0$ ($m = \tilde{H}/\Gamma$ by 9.16 and Γ's own analyticity, the right side being entire), and, since $|T_1^{-z}|$ and $|(q_1\gamma_a)^z|$ are bounded by 1 on the line $\operatorname{Re} z = 0$, each is continuous up to it as well. The identity therefore extends, first by the identity theorem across the open half-plane, then by continuity to the line $\operatorname{Re} z = 0$, where it says that $\log(1/T_1)$ and $\log(q_1\gamma_a)$ have the same Fourier transform; Fourier uniqueness then identifies the two laws, giving $T_1 \overset{d}{=} 1/(q_1\gamma_a)$. (This is the one place in this proof that spends transform uniqueness on the scale variable itself, rather than on moment data alone.) The time normalization $q_1 = 2$ gives 8.3, the profile is the Bessel-K function, and the operator is computed in 9.13: $B(\theta) = 2\theta(\theta - a)$, with the parabolic-gauge form as displayed.

($\Leftarrow$) *Case (1).* Admissibility is the pure-drift member $F(s) = s/c'$ of (7.1), and a first-order operator satisfies the PMP trivially, since $g'(x_0) = 0$ at an interior maximum. *Case (2).* Admissibility of the inverse-gamma family is 8.3; locality is 9.13; and, in the parabolic gauge ξ, a second-order operator $\frac{1}{2}\partial_\xi^2 + \frac{\beta}{\xi}\partial_\xi$ with nonnegative leading coefficient satisfies the PMP, because at an interior nonnegative maximum $g'(\xi_0) = 0$ and $g''(\xi_0) \le 0$, so $(Ag)(\xi_0) = \frac{1}{2}g''(\xi_0) \le 0$.

The homogeneity claim. By 7.4, depth-homogeneous members have $F(s) = c\,s^\alpha$; within the inverse-gamma family the exponent is a pure power only for $a = \frac{1}{2}$. Indeed, by 8.3, $e^{-F(s)} = \frac{2^{1-a}}{\Gamma(a)}(\sqrt{2s})^a K_a(\sqrt{2s})$, and the large-argument asymptotic $K_a(w) \sim \sqrt{\pi/(2w)}\,e^{-w}$ as $w \to \infty$ gives

$$F(s) \;=\; \sqrt{2s} \;-\; \Big(\frac{a}{2} - \frac{1}{4}\Big)\log s \;+\; O(1), \qquad s \to \infty,$$

which is a pure power of s only when the logarithmic coefficient vanishes, i.e. $a = \frac{1}{2}$ — where $K_{1/2}$ is elementary and $H(y) = e^{-\sqrt{2y}}$ exactly, the $\frac{1}{2}$-stable, i.e. heat, case. The moment statement is 8.3. $\square$

Proposition 10.6 (the ladder: locality without the maximum principle)**.** Fix $n \ge 2$, $c' > 0$ and $a_1, \dots, a_{n-1} > 0$, and let

$$T_1 \overset{d}{=} \frac{1}{c'\,\gamma_{a_1}\gamma_{a_2}\cdots\gamma_{a_{n-1}}} \quad \text{(independent factors)}, \qquad m(z) = c'^{\,z}\prod_{i=1}^{n-1}\frac{\Gamma(a_i + z)}{\Gamma(a_i)}.$$

Then the family is admissible in the sense of 7.3 and satisfies (H), and its inversion operator is *local of order n*, with polynomial symbol

$$B(-z) \;=\; c'\,z\prod_{i=1}^{n-1}(z + a_i);$$

the kernels are Meijer G-function densities. Conversely, every local admissible family of order n has m equal to the unique log-convex solution of $m(z{+}1) = Q(z)m(z)$ for its polynomial Q; when the roots of Q are real — necessarily ≤ 0, giving $Q(z) = c'\prod(z + a_i)$ with $a_i > 0$ — this is the family above, and the laws are identified from m as in the proof of 10.5. $\triangleright$ [**A**] analytic interface ledger A16 ledger A15 *(the HCM tower, `@bondesson1992generalized`, Thm. 5.1.1 p. 69 and Thm. 4.3.1 p. 57 with its hypotheses on p. 56; and Krull–Webster, `@webster1997log`, Thm. 3.1. Cited, not reproven here.* ***Assignment.*** *ledger A16 carries admissibility of the displayed family, in two steps: that* HCM *is closed under products of independent variables and under reciprocals, and that* HCM $\subset$ GGC $\subset$ SD*. The first inclusion,* HCM $\subset$ GGC*, is Theorem 4.3.1, p. 57, whose regularity hypotheses (a)–(c) on p. 56 Bondesson removes on p. 68, so it applies to the Gamma densities used here without a separate check. A caution the ledger records and the proof must respect:* HCM *is* not *closed under convolution, so only the multiplicative structure is available here — and the second inclusion,* GGC $\subset$ SD*, is asserted by*

Bondesson in a single sentence on p. 30 rather than as a numbered theorem, which makes it the thinnest link in this node. ledger A15 carries the converse direction only, exactly as in 10.5; the two hypotheses of Webster's theorem it must discharge are handled in the proof below, for the general degree-$(n-1)$ Q. ***Two sentences moved out of the statement (2026-08-24).*** *The Meijer G identification is now the proof's closing paragraph, and the complex-conjugate-roots aside is now 10.7; neither reordering changes what is claimed. The Meijer G identification is carried by neither ledger entry and needs neither: nothing later uses it, and it is cited to a stated theorem plus two printed transformation rules. Two things about it are deliberate. It takes* two *sources because `@springer1970distribution` covers products of gamma variables but not reciprocals of them — its only quotient result is for two factors — so the inversion routes through `@samko1993fractional`. And the G-parameters of T_1 itself are not written out: Springer–Thompson use the density convention $M\{f\}(s) = \mathbb{E}[x^{s-1}]$, under which the parameters are the shapes minus one, whereas the m of this proposition is the moment convention $\mathbb{E}[X^z]$, under which they are the shapes; and the reciprocal shifts them again. The order stated above is for the* product*, where it is unambiguous. The conjecture about complex-conjugate roots is flagged as open and is claimed by neither entry.)*

Proof. Admissibility. Gamma densities are hyperbolically completely monotone, and by ledger A16 the HCM class is closed under products of independent variables and under reciprocals, while HCM laws are generalized gamma convolutions and hence self-decomposable. So $-\log \mathbb{E}e^{-sT_1}$ is of the form (7.1), and (H) holds because all negative moments are finite and $\mathbb{P}(T_1 = 0) = 0$.

The symbol. $B(-z) = z\, m(z+1)/m(z) = c' z \prod(z + a_i)$, a polynomial, so A is local of order n by 10.2.

The converse. This is 10.3 and 10.4 together with ledger A15, as in the proof of 10.5, now for the general polynomial Q of degree $n-1$. Two hypotheses of Webster's theorem must be discharged, not assumed. The asymptotic condition is automatic: Q is a polynomial, so $Q(z+w)/Q(z) \to 1$ as $z \to \infty$ for each fixed $w > 0$. The normalization is not automatic — Webster pins the solution with $f(1) = 1$, where this proposition's normalization is $m(0^+) = 1$ — so apply Webster to $\hat{m} := m/m(1)$, which satisfies $\hat{m}(1) = 1$ and the same recursion; this gives uniqueness of $\hat{m}$ and hence of m, the constant $m(1) = Q(0)\, m(0^+) = Q(0)$ being read off the recursion at $z = 0$. A real root $-a_i \geq 0$ of Q would contradict $Q > 0$ on $(0, \infty)$ together with $Q(0) = m(1) > 0$, so real roots give $a_i > 0$, and the Krull–Webster solution is the displayed gamma product.

The Meijer G identification. The product $\gamma_{a_1} \cdots \gamma_{a_{n-1}}$ has a $G^{n-1,0}_{0,n-1}$ density, for arbitrary distinct shapes, by `@springer1970distribution`, Thm. 1, p. 722; the reciprocal and the scale factor c' are then absorbed by the G-function transformation rules (`@samko1993fractional`, eqs. (1.96) and (1.97), p. 22). $\square$

Remark 10.7 (complex roots)**.** Whether complex-conjugate root pairs can support an admissible family is left open; we conjecture not. The question is taken up in a companion note (in preparation), where spectrally leading pairs are excluded and order three is settled.

Remark 10.8 (the stable slice; Gauss multiplication)**.** Depth-homogeneity intersects the ladder at the equally spaced parameters $a_j = j/n$ and $c' = n^n$: by the Gauss multiplication formula,

$$m(z) \;=\; n^{nz} \prod_{j=1}^{n-1} \frac{\Gamma(j/n + z)}{\Gamma(j/n)} \;=\; n\,\frac{\Gamma(nz)}{\Gamma(z)}, \qquad \tilde{H}(z) = n\,\Gamma(nz), \qquad H(y) = e^{-y^{1/n}},$$

the $\frac{1}{n}$-stable laws — recovering the classical identity that represents the $\frac{1}{n}$-stable subordinator as a scaled reciprocal product of $n-1$ gamma variables. Thus the local cases $\alpha = 1/n$ of `@fagerstrom2005temporal` are precisely the homogeneous slice of the ladder, and each unfolds under the hemigroup axioms into an $(n-1)$-parameter inhomogeneous family of local media.

The maximum principle eliminates $n \geq 3$ in both settings; what remains at $n = 2$ is the single point $\alpha = \frac{1}{2}$ there, and the Bessel line here.

Remark 10.9 (what the theorem settles)**.** 10.5 completes the resolution of the practicality problem of 8.5 *within the local class*: the drift weight $\beta = \frac{1}{2} - a$ is a design parameter, and any prescribed number of finite kernel moments is achievable by a local, positivity-respecting, continuously scale-covariant, time-causal medium, namely $n < a$. The residual trade-off is the tail: inverse-gamma kernels have polynomial tails t^{-1-a} for every finite a, so *all* moments — exponential localization — still require leaving the local class for the Gamma-type corner of 8.2.

Homogeneity, locality and moment control thus interact as follows: homogeneity together with covariance forces stable kernels, which have no mean; locality together with covariance and the PMP allows any finite number of moments but polynomial tails; full moment control requires nonlocal delay accrual.

Lemma 10.10 (the symbol vanishes at the origin)**.** Assume (H). Then, along the reals,

$$m(z) \longrightarrow 1 \qquad \text{and} \qquad B(-z) \;=\; \frac{z\, m(z+1)}{m(z)} \;\longrightarrow\; 0 \qquad (z \downarrow 0).$$

Consequently $B(0) = 0$ for any B continuous at the origin — in particular whenever B is a polynomial, which is 10.3(1). ▷ **[T]** proved-target *(**Machine-checked**, as `Hemigroup.SelfDecomposableExponent.tendsto_inversionSymbol_nhdsGT_zero`, with the first limit as `tendsto_negMoment_nhdsGT_zero`. `#print axioms` shows A17 and nothing else; ledger A13 is not touched. **Why this is separate from 10.3.** That lemma states clause (1) under the hypothesis that B is a polynomial of degree n, and gets $B(0) = 0$ from this limit by continuity. **The polynomial hypothesis is inert in the argument**: what the proof uses is that $m(z) \to 1$, that $m(z+1)$ stays bounded, and that $z \to 0$ — none of which mentions B's form. Stating the limit without it is what makes the machine-checked content visible, and keeps a node that spends ledger A13 on clause (2) from appearing to spend it on clause (1) as well. This is the same distinction the article draws for 9.15: what a proof cites is an upper bound on what its statement needs. **Where (H) enters, in both clauses and differently.** Its first clause, no atom at zero delay, is what makes every $m(z)$ positive, the integral defining m running over $(0, \infty)$; its second, $z_* > 1$, is what makes the dominating function of the first limit integrable and the interval $(1, z_*)$ — from which the bound on $m(z+1)$ is taken — nonempty.)*

Proof. The first limit. Dominated convergence. For $t > 0$ the integrand $t^{-z} = e^{-z \log t}$ tends to 1 as $z \to 0$; for $0 < z \leq 1$ it is dominated by $1 + t^{-1}$, which is $\mu_{0,1}$-integrable because (H) places 1 strictly below z_*. Hence $m(z) = \mathbb{E}\, T_1^{-z} \to \mathbb{E}\, 1 = 1$.

The second. Fix $\zeta \in (1, z_*)$, nonempty by (H). For $0 < z \leq \zeta - 1$ we have $z + 1 \leq \zeta$, and splitting the integral at $t = 1$ — below it $t^{-(z+1)} \leq t^{-\zeta}$, above it $t^{-(z+1)} \leq 1$ — gives $m(z+1) \leq m(\zeta) + 1$, a bound independent of z. So $z\, m(z+1) \to 0$, while $m(z) \to 1 \neq 0$ by the first limit, and the quotient tends to 0. ☐

Lemma 10.11 (the shifted Gamma recursion; Krull–Webster at a linear Q)**.** Let $c, a > 0$ and let $m : (0, \infty) \to (0, \infty)$ be log-convex and satisfy

$$m(z+1) \;=\; c\,(z+a)\, m(z) \quad (z > 0), \qquad m(z) \to 1 \;\; (z \downarrow 0).$$

Then

$$m(z) \;=\; c^{z}\, \frac{\Gamma(a+z)}{\Gamma(a)} \qquad (z > 0).$$

▷ **[T]** proved-target *(**Machine-checked**, as `Hemigroup.eq_gamma_form_of_logConvex_of_recursion`. `#print axioms` gives **Lean core alone** — not even A17, since the statement names no kernel. That is the check* *that the ledger entry is discharged rather than relocated.* ***What this replaces.*** *10.5's order-2 case applied ledger A15, the Krull–Webster uniqueness theorem, to exactly this recursion. With Q linear the recursion has a single Gamma factor, and Krull–Webster is then the classical Bohr–Mollerup theorem, which Mathlib carries. 10.6 is a different matter: its order-n ladder has $n-1$ Gamma factors, log-convexity alone does not reduce to Bohr–Mollerup there, and ledger A15 remains.* ***The one point of technique.*** *Bohr–Mollerup pins a function on $(0,\infty)$ satisfying $f(y+1) = yf(y)$; ours carries the shift a. Substituting $f(y) = m(y-a)\cdots$ would need m below the interval it is given on, and extending it downward through the recursion is an induction on $\lceil a \rceil$ that also leaves the domain where log-convexity was established. Shifting the other way costs nothing, which is the proof below.* ***An honest note on the trade.*** *This is* longer *than the citation it removes — unlike the five substitutions recorded after 3.3, where the elementary route was also the shorter one. What is bought is the trust base, not the exposition.)*

Proof. Choose an integer $n \geq a$ and put $d := n - a \geq 0$. Define, for $y > 0$,

$$f(y) \;:=\; \frac{m(y+d)\;c^{-(y+d)}}{P(y)}, \qquad P(y) := \prod_{j=0}^{n-1}(y+j),$$

which uses m only on $(d,\infty) \subseteq (0,\infty)$. Two shifts cancel: the recursion gives $m(y+1+d) = c\,(y+n)\,m(y+d)$, since $d + a = n$, while $P(y+1)\,y = P(y)\,(y+n)$; together with $c^{-(y+1+d)} = c^{-(y+d)}c^{-1}$ this leaves

$$f(y+1) \;=\; y\,f(y) \qquad (y>0).$$

f is positive, and $\log f(y) = \log m(y+d) - (y+d)\log c - \log P(y)$ is convex on $(0,\infty)$: the first term is 10.4 precomposed with a translation, the second is affine, and the third is convex because $-\log(y+j)$ is, log being concave. Bohr–Mollerup applied to $f/f(1)$ therefore gives $f = f(1)\,\Gamma$ on $(0,\infty)$.

Write $C := f(1)$ and $G(z) := c^z\Gamma(z+a)$. Unwinding the definition of f at $y = z - d$ and using $P(y)\,\Gamma(y) = \Gamma(y+n)$ with $y + n = z + a$ gives $m(z) = C\,G(z)$ for every $z > d$. Both m and G satisfy the same recursion on $(0,\infty)$, so $m(z+k)\,G(z) = m(z)\,G(z+k)$ for every $k \in \mathbb{N}$; choosing k with $z+k > d$ and cancelling $G(z+k) > 0$ extends $m = C\,G$ to all of $(0,\infty)$. Finally $z \downarrow 0$ gives $C\,\Gamma(a) = 1$, since $c^z \to 1$ and Γ is continuous at $a > 0$. $\square$

Lemma 10.12 (the symbol's linear factor and the moment recursion)**.** Assume (H), and let A be local of order n at a height $0 < c < z_* - 1$, so that by 10.2 its symbol is $P(z) = \sum_{j=0}^{n}\gamma_j E_j(z)$ with $\gamma_j = c_j(1)$. Then $\gamma_0 = 0$, and with

$$Q(z) \;:=\; -\sum_{k=0}^{n-1}\gamma_{k+1}\,E_k(z+1)$$

one has $P(z) = z\,Q(z)$ for every z, and for every real z with $0 < z < z_* - 1$ the value $Q(z)$ is a positive real satisfying

$$m(z+1) \;=\; Q(z)\,m(z).$$

▷ [**T**] proved-target ***(Machine-checked,*** *as* `exists_symbolQuotient_of_isLocalOfOrder`*, on A17 and nothing else — in particular not on ledger A13, which is the whole reason for the split.* ***The factorisation is exact and contains no analysis.*** *$E_{j+1}(z) = -z\,E_j(z+1)$ is the Euler factors' own recursion, so removing the factor z shifts the argument by one and drops the index by one, and Q comes out in the same falling-factorial basis one degree lower. What the analysis is spent on is $\gamma_0 = 0$, and only in one step: 10.10 gives the limit $B(-c) \to 0$ without any hypothesis on B's form, and the polynomial hypothesis enters exactly here, to turn a limit into a value.* ***The recursion*** *is 9.2 read at a real point.* $\tilde{H}$ *does not vanish at the real points of the strip (9.16), so the symbol identity of 10.2 — which is conditioned on exactly that — holds pointwise there, and cancelling* z *against the closed form* $B(-z) = z\,m(z+1)/m(z)$ *leaves the recursion. Positivity of* Q *is then not a separate argument:* $Q(z)$ *is a ratio of two positive moments, so it inherits both its realness and its sign.* ***The range is*** $0 < z < z_* - 1$***, and that is the honest one.*** *10.3(3) states the recursion on all of* $(0, \infty)$*, which presupposes* $z_* = \infty$ *— clause (2), the one clause ledger A13 carries. Stating the range A13 is not needed for is what makes the split mean something.* ***Why the degree of*** Q ***is not stated as a degree.*** *The blueprint calls* Q *a polynomial of degree* $n-1$*; the Lean statement gives the expansion above instead, which is that claim in the falling-factorial basis, the basis being triangular.* `Polynomial` *does not appear anywhere in this development —* `mellinEulerFactor` *is a plain function — and introducing it for one adjective would propagate through every statement below. At the order the chapter actually consumes,* `symbolQuotient_two` *makes the claim literal:* $Q(z) = \gamma_2 z + (\gamma_2 - \gamma_1)$*, linear with leading coefficient* $\gamma_2 \neq 0$*, which is the shape 10.11 takes.)*

Proof. $\gamma_0 = 0$. $E_j(0) = \prod_{i<j}(-i)$ vanishes for $j \geq 1$, its $i = 0$ factor being 0, so $P(0) = \gamma_0$. At a real c with $0 < c < z_* - 1$ the transform $\tilde{H}(c)$ is nonzero (9.16), so 10.2 gives $B(-c) = P(c)$ there. Letting $c \downarrow 0$, the left side tends to 0 by 10.10 and the right side to $P(0) = \gamma_0$ by continuity; limits in $\mathbb{C}$ are unique, so $\gamma_0 = 0$.

The factorisation. With $\gamma_0 = 0$ the $j = 0$ term drops, and reindexing $j = k+1$,

$$P(z) \;=\; \sum_{k=0}^{n-1} \gamma_{k+1} E_{k+1}(z) \;=\; \sum_{k=0}^{n-1} \gamma_{k+1}\bigl(-z\,E_k(z+1)\bigr) \;=\; z\,Q(z),$$

by $E_{j+1}(z) = -z\,E_j(z+1)$, which is the definition of E with its first factor peeled.

The recursion. Fix a real z with $0 < z < z_* - 1$. Then $\tilde{H}(z) \neq 0$, so $B(-z) = P(z) = z\,Q(z)$; and 9.16's closed form gives $B(-z) = z\,m(z+1)/m(z)$. Both $m(z)$ and $m(z+1)$ are finite and strictly positive — finite because $z + 1 < z_*$, positive because T_1 charges only $(0, \infty)$ — so $m(z) \neq 0$ and, cancelling the factor $z \neq 0$, $Q(z) = m(z+1)/m(z)$. That exhibits $Q(z)$ as a quotient of positive reals, which is both the recursion and the positivity. □

Lemma 10.13 (the maximum principle on a local operator of order at most two)**.** Let A agree on $C_c^\infty((0,\infty))$ with $\sum_{j=0}^n c_j(x)\partial_x^j$, where $n \leq 2$. If $\operatorname{Re} c_0 \leq 0$ and $\operatorname{Re} c_2 \geq 0$ on $(0, \infty)$, then A satisfies the positive maximum principle. In particular, if the symbol is the polynomial $\sum_{j \leq n} \gamma_j E_j$ with $n \leq 2$, $\gamma_0 = 0$ and $\operatorname{Re}\gamma_2 \geq 0$, then the PMP holds — which covers both cases of 10.5, the first-order one vacuously in the leading hypothesis and the second-order one with $\gamma_2 = 2$. ▷ [**T**] proved-target *(**Machine-checked**, as* `satisfiesPMP_of_symbol_eq`*, on A17 — through* `inversionOperator`*, which quantifies over the construction's kernels — and* ***not on ledger A14****. Keeping that visible is the reason for the split.* ***10.5 uses the maximum principle twice, in opposite directions and at opposite costs****, and the ledger entry sits on one of them alone. The (⇒) use is the* order bound*: that a local operator satisfying the PMP is a pure second-order diffusion with no jump part. That is Courrège's classification, ledger A14, cited and staying cited. The (⇐) use is the* verification *above, and it is elementary. A node that reported both under one status would say the maximum principle costs an interface, when half of it costs nothing.* ***Mathlib has Fermat's rule and only its first half.*** `IsLocalMax.deriv_eq_zero` *is there; nothing says the second derivative is nonpositive at a local maximum, and there is no second-derivative test at an extremum anywhere in the library. So* `deriv_deriv_nonpos_of_isLocalMax` *is written from scratch, by the classical argument: were* $f''(a)$ *positive, the slope of* f' *at* a *would be eventually positive, hence* $f' > 0$ *just to the right of* a*, hence* f *strictly increasing there — and* a *is a maximum. It mentions nothing of this development and reduces to Lean core.* ***The first-order coefficient is unconstrained****, and that is the whole content of the step:* $g'(x_0) = 0$ *at an interior maximum* *kills the $j = 1$ term whatever c_1 is, which is why 10.5's case (1) needs no hypothesis beyond the absence of killing. And the absence of killing, $\gamma_0 = 0$, is not an extra assumption either — it is 10.12, holding for every local operator of the class.* ***Where the real-valued reading is paid for.*** *10.1 asserts* $\mathrm{Re}(Ag)(x_0) \le 0$*, A being $\mathbb{C}$-valued, so the differential expression has to be pushed through* Re *and the derivatives of $x \mapsto (g(x) : \mathbb{C})$ identified with those of g. That identification is `iteratedDeriv_ofReal_comp`, proved for the profile in this chapter's differentiation step and reused here on all of $\mathbb{R}$, where a test function is smooth. It is the one place this argument is longer than two lines.* ***R5, closed.*** *Under `isLocalOfOrderCoreOfSymbolEq`'s own hypotheses — the ones this lemma is stated from, via `satisfiesPMP_of_symbol_eq` — plus reality of the symbol's witness coefficients γ_j (an extra hypothesis `hsymbol`'s γ does not itself carry), `Hemigroup.SelfDecomposableExponent.im_inversionOperator_eq_zero_of_symbol_eq` shows $(Ag)(x_0)$ has zero imaginary part on a real test function, and `eq_ofReal_re_inversionOperator_of_symbol_e` packages that as $(Ag)(x_0) = ((Ag)(x_0)$'s real part, cast back) — so under that extra hypothesis the article's real-number reading $(Ag)(x_0) \le 0$ and this lemma's $\mathrm{Re}(Ag)(x_0) \le 0$ are the same statement, not merely the second weaker than the first. Both print A17 through `inversionOperator`, plus nothing else. Not folded into `satisfiesPMP_of_symbol_eq` itself, because that lemma's own γ is exactly 10.5's two cases' coefficients (real by inspection at each use) and the general statement is honest that realness of γ is not free at the level this lemma is stated.)*

Proof. Let $g \in C_c^\infty((0,\infty))$ be real-valued and attain a nonnegative maximum over $(0,\infty)$ at $x_0 > 0$. Since $(0,\infty)$ is open, x_0 is a local maximum of g on $\mathbb{R}$, so $g'(x_0) = 0$ by Fermat's rule and $g''(x_0) \le 0$: were $g''(x_0) > 0$, the slope $\big(g'(y) - g'(x_0)\big)/(y - x_0) = g'(y)/(y - x_0)$ would be positive for all y near x_0, so $g' > 0$ on an interval (x_0, b), so g would be strictly increasing on $[x_0, y_1]$ for $x_0 < y_1 < b$ and $g(y_1) > g(x_0)$, contradicting the maximum.

Hence, taking real parts in $(Ag)(x_0) = \sum_{j \le n} c_j(x_0) g^{(j)}(x_0)$ and using that each $g^{(j)}(x_0)$ is real,

$$\mathrm{Re}(Ag)(x_0) \;=\; \mathrm{Re}\, c_0(x_0)\, g(x_0) \;+\; \mathrm{Re}\, c_1(x_0) \cdot 0 \;+\; \mathrm{Re}\, c_2(x_0)\, g''(x_0) \;\le\; 0,$$

the first term being a nonpositive number times a nonnegative one and the third a nonnegative times a nonpositive; terms with $j > n$ do not occur, as $n \le 2$. For the symbol form, 10.2 gives $c_j(x) = \gamma_j x^{j-1}$, so $\mathrm{Re}\, c_0(x) = \mathrm{Re}(\gamma_0 x^{-1}) = 0$ and $\mathrm{Re}\, c_2(x) = x\, \mathrm{Re}\, \gamma_2 \ge 0$ for $x > 0$. □

Lemma 10.14 (the moment classification of a local operator)**.** Assume (H), that every negative moment of T_1 is finite (10.3(2)), and that A is local of order $n \le 2$ (10.5's order bound). Then exactly one of

$$m(z) \;=\; c'^{\,z} \qquad \text{or} \qquad m(z) \;=\; q_1^z\, \frac{\Gamma(a+z)}{\Gamma(a)} \qquad (z > 0)$$

holds, with $c' > 0$ in the first case and $q_1, a > 0$ in the second — that is, $T_1 = 1/c'$ almost surely, or $T_1 \overset{d}{=} 1/(q_1 \gamma_a)$. ▷ [**T**] proved-target *(**Machine-checked**, as `exists_moment_form_of_isLocalOfOrder`, on A17 and* neither *ledger A13 nor ledger A14: both are hypotheses of the statement rather than axioms, phrased as ledger A17 is so that either demotes to a lemma without touching anything below.* ***Order* 0 *is impossible, and cheaply****, which is what makes the case list a dichotomy rather than a trichotomy settled by inspection. 10.1 asks the leading coefficient not to vanish identically; 10.3(1) gives $\gamma_0 = 0$; and covariance gives $c_0(x) = \gamma_0 x^{-1}$, identically zero. So the definition's two clauses are already incompatible at order* 0*, with no analysis in between, and ledger A13 is not needed for it either. That is `not_isLocalOfOrder_zero`.* ***The maximum principle does not appear in the statement****, and its absence is the honest reading. 10.5's ($\Rightarrow$) direction uses the PMP* only *through ledger A14, whose conclusion is the order bound; a collation carrying `SatisfiesPMP` as an unused hypothesis would misreport where the principle is spent. Where the principle* is used*, in the other direction, is 10.13.* ***What this does not say*** *is that the kernels are the Bessel-K densities of 8.3: that identification is a statement about densities and Mathlib carries no Bessel*

K. The moment content is the whole of what is reachable, and it is also the whole of what the rest of the chapter consumes.)

Proof. Order 0 is excluded as above. At order 1, $Q \equiv -\gamma_1 =: c'$ is constant and positive, so $\log m$ is convex with constant unit increments, hence affine (10.5's Order-1 step), and $m(z) = c'^{\,z}$. At order 2, $Q(z) = q_1 z + a_0$ with $q_1, a_0 > 0$, and 10.11 applied to $Q(z) = q_1(z + a_0/q_1)$ gives the Gamma form. Orders above 2 are excluded by hypothesis. $\square$

Chapter 11

Implementation

This chapter records how the scale spaces of 7.3 are computed. The structural fact that organizes everything: the scale direction never needs to be discretized approximately.

Proposition 11.1 (exactness at the knots)**.** Let $0 = x_0 < x_1 < \cdots < x_M$ be any finite set of scale knots and define the discrete cascade

$$u_0 := f, \qquad u_{k+1} := \mu_{x_k,\, x_{k+1}} * u_k.$$

Then $u_k = \Phi_{0,x_k} f$ *exactly*, for every knot set; and refining the knot set changes nothing at the old knots. ▷ [**T**] proved-target *(One line of hemigroup algebra; the content is that the cascade is not an approximation scheme for the family but the family itself, sampled — the only discretization error in a digital implementation is the time discretization of the increment filters.)*

Proof. Immediate induction from the hemigroup (A6): $\mu_{0,x_{k+1}} = \mu_{x_k,x_{k+1}} * \mu_{0,x_k}$. □

Proposition 11.2 (the Gamma cascade)**.** For the Gamma family with integer γ, the increment filter is rational:

$$\widehat{\mu}_{x_k,x_{k+1}}(s) \;=\; \Big(\frac{1 + x_k\, s}{1 + x_{k+1}\, s}\Big)^{\gamma},$$

a cascade of γ *identical* first-order pole–zero sections (pole at $-1/x_{k+1}$, zero at $-1/x_k$). Each section admits the one-state realization, free of input differentiation,

$$\dot{w} = \frac{y - w}{x_{k+1}}, \qquad v \;=\; \frac{x_k}{x_{k+1}}\, y \;+\; \Big(1 - \frac{x_k}{x_{k+1}}\Big)\, w,$$

(y input, v output), whose transfer is $(1+x_k s)/(1+x_{k+1} s)$. The output is a *convex combination* of input and state. ▷ [**T**] proved-target *(Transform algebra; the convex combination is what makes positivity and unit mass manifest at the realization level. In discrete time, with sample interval Δ and sampled-and-held input, each section admits the exact exponential-integrator update $w_{n+1} = e^{-\Delta/x_{k+1}} w_n + (1 - e^{-\Delta/x_{k+1}}) y_n$ — unconditionally stable and free of overshoot; the cascade is not exact end-to-end, the input to the next section being the previous section's response rather than a held signal, so time discretization is a genuine approximation, while positivity, unit mass and stability are preserved exactly by the convex update.)*

Proof. The increment transfer is the ratio of the endpoint transforms: by the hemigroup (A6), $\widehat{\mu}_{0,x_{k+1}} = \widehat{\mu}_{x_k,x_{k+1}}\, \widehat{\mu}_{0,x_k}$, and $\widehat{\mu}_{0,x}(s) = (1 + xs)^{-\gamma}$ is nonvanishing for $\operatorname{Re} s \geq 0$, so dividing gives the display. The transfer identity of the section: $v = \frac{x_k}{x_{k+1}} y + \big(1 - \frac{x_k}{x_{k+1}}\big) \frac{y}{1 + x_{k+1} s} = y\, \frac{1 + x_k s}{1 + x_{k+1} s}$; the convex-combination clause is $0 < x_k / x_{k+1} < 1$. □

Remark 11.3 (covariance in practice)**.** With geometric knots $x_k = q^k x_0$ the *continuum* cascade is carried to itself by every rescaling $\sigma \in q^{\mathbb{Z}}$, as a shift of the ladder; in the sampled system the time constants are tied to the sample interval Δ, so the shift is exact on the knot set but not on the sampled filters. Because the underlying family exists at *every* scale, an arbitrary σ is handled by instantiating the same family at the shifted knots — a new filter bank with new time constants and re-initialized state, not a relabelling of the running one; the operation is principled exactly because the kernels between knots belong to the family. This is the operational content of continuous covariance, and the point of the pole–zero structure: a pure-pole cascade (truncated exponentials with distinct time constants), the finite form of the time-causal limit-kernel construction of `@lindeberg2016time`, also marches a hemigroup, but its kernels are hypoexponential laws whose family is *not* dilation-closed, so only the discrete covariance of the ladder itself survives. The zeros are what the hemigroup increments of the Sato–Gamma process add, and they are exactly what buys the continuum.

Remark 11.4 (implementing the local corner)**.** The Bessel/inverse-gamma increments have transcendental transforms (ratios of Bessel-K profiles), hence *no* finite-dimensional time-recursive realization; the same holds for the stable family ($e^{-(x_{k+1}^\alpha - x_k^\alpha)\, s^\alpha}$ in the homogeneous gauge). The implementation options are: (i) direct convolution against tabulated increment kernels per knot — burdened by the polynomial tails t^{-1-a}, so truncation error decays only polynomially in the window length; (ii) marching the memory-line PDE, in the parabolic gauge ξ of 6.4, $\partial_t u = \frac{1}{2} u_{\xi\xi} + \frac{\beta}{\xi} u_\xi$ in time (implicit in ξ, with care at the degenerate boundary), which reintroduces scale-direction discretization error; or (iii) rational approximation of the irrational symbol, in the manner standard for fractional-order elements in filter and control design (`@oustaloup2000frequency`); 11.6 keeps that route inside the axiom class, and its state-count economy is a heuristic, with no error bound claimed. All three are workable; none matches the cascade of 11.2 in simplicity or exactness. This is the practical face of 10.5: locality of the medium is bought at the price of heavy-tailed, non-rational kernels.

Remark 11.5 (finite differences on the memory line)**.** The explicit finite-difference scheme of `@fagerstrom2005temporal`, §§6–7, for the heat signaling equation extends verbatim to the Bessel family. Working in the parabolic gauge ξ of 6.4, on the logarithmic grid $\eta = \log \xi$ the operator is constant-coefficient up to a rate factor,

$$\tfrac{1}{2}\,\partial_\xi^2 + \frac{\beta}{\xi}\,\partial_\xi \;=\; e^{-2\eta}\Big[\tfrac{1}{2}\,\partial_\eta^2 + \big(\beta - \tfrac{1}{2}\big)\,\partial_\eta\Big],$$

so the stencil is uniform along the scale ladder and only the update rate varies with position — the discrete form of the Lamperti time change (`@lamperti1972semi`). With the first-order term upwinded, the explicit update matrix is sub-stochastic, so positivity and non-amplification are automatic, and the discrete scheme is itself a hemigroup of sub-probability kernels: a random-walk medium whose absorption-time laws are discrete analogues of the inverse-gamma kernels. It is recorded for continuity with the 2005 paper but not recommended: the explicit step is constrained by the finest cell, to steps Δt of order $\xi_{\min}^2\,\Delta\eta^2$, while the coarse-scale dynamics live on horizons $\sim \xi_{\max}^2$, so the cost of covering the ladder grows like $(\xi_{\max}/\xi_{\min})^2$ (an implicit march trades this for per-sample band solves plus additional discretization error). The finite-moment regime $a > 1$, i.e. $\beta < -\frac{1}{2}$ — the very reason to prefer the Bessel family — is advection-dominated near the singular boundary and numerically the least pleasant. And, decisively, any marching of the signaling equation approximates the scale direction, which 11.1 shows can be sampled exactly. The principled numerics for the local corner is instead the following.

Proposition 11.6 (the Thorin subclass; gamma cascades approximate the local corner)**.** For an admissible F (7.3) the following are equivalent:

1. the memory kernel k is completely monotone — the classical Sonine case of A.7(2);

2. F has a *Thorin representation*

$$F(s) \;=\; b_0\, s + \int_{(0,\infty)} \log\Big(1+\frac{s}{\tau}\Big)\, U(d\tau), \qquad U \ge 0,\ \int_{(0,1]} \log\tfrac{1}{\tau}\, U(d\tau) + \int_{(1,\infty)} \frac{U(d\tau)}{\tau} < \infty;$$

3. the kernels are generalized gamma convolutions (GGC).

In that case, for any atomic $U_N = \sum_{i=1}^{N} w_i\, \delta_{\tau_i}$ such that $F_N(s) := b_0 s + \sum_i w_i \log(1 + s/\tau_i)$ converges to F pointwise — as it does for the standard quadratures of U, the weights carrying the two integrability conditions — the exponents F_N are *themselves admissible*, finite multi-Gamma members of 7.3, and the kernel families converge weakly at every scale; positivity and unit mass hold *exactly at every truncation level*, the approximation staying inside the axiom class rather than merely near it. For integer weights the increments over any knot ladder are rational of order $\sum_i w_i$ up to the pure delay $e^{-b_0(x_{k+1}-x_k)s}$ contributed by the drift — itself a pure time shift — realized by the cascade of 11.2 applied per Thorin node, with time constants x/τ_i. ▷ [**A**] analytic interface ledger A1 ledger A21 *(Thorin's characterization of the GGC class, plus Bernstein–Widder.* ***Assignment.*** *ledger A21 carries the equivalence (2) ⇔ (3): a law on* $[0, \infty)$ *is GGC iff it is infinitely divisible with Lévy density* ℓ *such that* $y\,\ell(y)$ *is completely monotone (*`@bondesson1992generalized`*, Thm. 3.1.1, p. 30 — clause (1) verbatim,* $y\,\ell(y)$ *being this development's* k*), and, exponent-side, the Thorin–Bernstein form with its unique representing measure and exactly the displayed integrability condition (*`@schilling2012bernstein`*, Thm. 8.2, p. 109, (i) ⇔ (iii)). ledger A1, in its general-measure form, carries the step* k *completely monotone* $\Rightarrow k(t) = \int e^{-\tau t}\, U(d\tau)$ *inside (1) ⇒ (2). Everything else is* ***[T]*** *in prose: the integration of* F' *against A.1, the three-regime finiteness bookkeeping, the admissibility of the truncations through 7.1 and 7.3, and the weak convergence through 3.5. The graded evidentiary statuses of the substrate claim — in-class approximation proved at any positive weights, exact rational realization proved at integer weights, real-weight fitting a heuristic — are the paper's discussion, not part of this statement.)*

Proof. (1) ⇔ (2): by Bernstein–Widder (ledger A1; in the working source, `@schilling2012bernstein`, Thm. 1.4, p. 3), k completely monotone means $k(t) = \int e^{-\tau t}\, U(d\tau)$ for a positive measure U on $[0, \infty)$, finite on compacts since the integral converges; an atom of U at $\tau = 0$ would put a constant component into k, making $\int_1^\infty k(t)\, t^{-1} dt$ — hence F — infinite, so U lives on $(0, \infty)$. By A.1, $F'(s) = b_0 + \int U(d\tau)/(s + \tau)$, and integrating in s from 0 gives the Thorin form, the integrability condition being finiteness of $F(1)$ — three regimes: $\log(1 + 1/\tau) \sim \log(1/\tau)$ as $\tau \downarrow 0$, $\sim 1/\tau$ as $\tau \to \infty$, and bounded on the compact middle, where U is finite; conversely differentiate. (2) ⇔ (3) is Thorin's characterization of the GGC class (ledger A21; `@thorin1977pareto`, `@thorin1977lognormal`). In the Lévy-data form — GGC iff the Lévy density ℓ has $y\,\ell(y)$ completely monotone, which is clause (1) verbatim, $y\,\ell(y)$ being this development's k — it is `@bondesson1992generalized`, Thm. 3.1.1, p. 30; the exponent-side form is the Thorin–Bernstein class of `@schilling2012bernstein`, Thm. 8.2, p. 109. Admissibility of F_N: $sF_N'(s) = b_0 s + \sum_i w_i\, s/(s + \tau_i)$ is a Bernstein function, so 7.1 applies and F_N is of the form (7.1); $F_N(0+) = 0$, the integrability conditions are trivial for a finite sum, and 7.3 makes the family admissible. Pointwise convergence of F_N gives convergence of the transforms $e^{-F_N(xs)}$, hence weak convergence of the kernels by the continuity theorem (3.5). □

Example 11.7 (numerical illustration: the tail contrast)**.** Compare the kernels of the three corners at matched delay: Gamma kernels with $\gamma \in \{1, 2, 4\}$ and scale $x = 1/\gamma$ (mean delay 1); the Bessel/inverse-gamma kernel with $a = 2$, normalized to mean delay 1; and the $\frac{1}{2}$-stable kernel of `@fagerstrom2005temporal` normalized to median delay 1, its mean being infinite. At matched mean the Gamma kernels are the most concentrated, sharpening with γ (mode $(\gamma - 1)/\gamma \to$ mean), while the inverse-gamma kernel buys its early sharp peak by exporting mass into the tail, and the stable kernel does so to the point of losing the mean altogether;

the Gamma tails fall exponentially, the inverse-gamma and stable tails follow the powers t^{-1-a} and $t^{-3/2}$ of 10.9. The practical consequence is quantified by the window length T_p capturing a fraction p of the kernel mass ($\int_0^{T_p} \phi = p$), the quantity governing the truncation error of route (i) in 11.4. At matched delay 1, the pairs ($T_{0.99}$, $T_{0.999}$) are:

- $\Gamma(\gamma{=}1)$: $(4.61,\ 6.91)$; $\Gamma(\gamma{=}2)$: $(3.32,\ 4.62)$; $\Gamma(\gamma{=}4)$: $(2.51,\ 3.27)$;
- inverse-gamma, $a = 2$: $(6.73,\ 22.0)$;
- $\frac{1}{2}$-stable: $(2.90 \times 10^3,\ 2.90 \times 10^5)$.

For the Gamma family each extra decade of captured mass costs a fixed additive window increment (exponential tails); for the inverse-gamma with $a = 2$ it costs a factor $10^{1/a}$ (about 3.2); for the $\frac{1}{2}$-stable, of index $\alpha = \frac{1}{2}$, a factor $10^{1/\alpha} = 100$ — capturing 99.9% of its mass requires a window five orders of magnitude longer than its median delay. Since the stable mean is infinite, the delay proxy of the influence-curve discussion of `@fagerstrom2005temporal`, §8, is the *mode*; repeating the comparison at matched mode delay 1 ($x = 1/(\gamma - 1)$ for the Gamma kernels, the inverse-gamma and stable normalized likewise) gives, as triples (mean, $T_{0.99}$, $T_{0.999}$):

- $\Gamma(\gamma{=}2)$: $(2,\ 6.64,\ 9.23)$; $\Gamma(\gamma{=}4)$: $(4/3,\ 3.35,\ 4.35)$;
- inverse-gamma, $a = 2$: $(3,\ 20.2,\ 66.1)$;
- $\frac{1}{2}$-stable: $(\infty,\ 1.91 \times 10^4,\ 1.91 \times 10^6)$

— at matched mode, capturing 99.9% of the $\frac{1}{2}$-stable kernel requires a window six orders of magnitude beyond the peak. This is the quantitative content of the statement that the 2005 kernels are too heavy-tailed for practical measurement, and of its resolution within the hemigroup class.

Chapter 12

The temporal N-jet: the memory embodies its derivatives

The jet vocabulary iterates the operator core: for $n \geq 1$ let

$$\mathcal{D}^n \;:=\; \big\{\, f \in X_0 : \text{there are } f^{(1)}, \dots, f^{(n)} \in X_0 \text{ with } f^{(j)} = \textstyle\int_0^{\cdot} f^{(j+1)},\ 0 \leq j < n \,\big\},$$

so that $\mathcal{D}^1 = \mathcal{D}$: each derivative up to order n is again a causal L^1 function that is the primitive of the next, and the vanishing boundary values are exactly what turns $I^z f^{(n)}$ into $I^{z-n} f$ below.

Definition 12.1. The *temporal N-jet* of the scale space at (t, x) is $\{\partial_t^n u(t, x)\}_{0 \leq n \leq N}$, and the *scale-normalized jet* is $\{x^n \partial_t^n u(t, x)\}_{n \leq N}$, which in the canonical gauge transforms under the joint dilation $(t, x) \mapsto (\sigma t, \sigma x)$ like the field itself. ▷ **[T]** proved-target *(A definition; its one claim — the transformation law of the scale-normalized jet — is the chain rule under the joint dilation, read off the canonical-gauge covariance of 6.3, each ∂_t contributing a factor σ^{-1} that the weight x^n restores.)*

Theorem 12.2 (the embodied jet)**.** Assume (H), canonical gauge, $f \in \mathcal{D}^n$, and let $u(\cdot, x) = \Phi_{0,x} f$. Then:

1. For every $t > 0$ and $1 < \operatorname{Re} z < z_*$,

   $$\widetilde{\partial_t^n u(t, \cdot)}(z) \;=\; \tilde{H}(z)\,\big(I^z f^{(n)}\big)(t) \;=\; \tilde{H}(z)\,\big(I^{\,z-n} f\big)(t),$$

   the second equality holding as a Riemann–Liouville integral for $\operatorname{Re} z > n$ and by analytic continuation — a fractional derivative of f — below.

2. Consequently, whenever $z_* > n + 1$, the n-th temporal derivative is the n-th power of the inversion operator applied to the *present* memory state: in the Mellin sense on $n + 1 < \operatorname{Re} z < z_*$,

   $$\partial_t^n u(t, \cdot) \;=\; A^n\, u(t, \cdot), \qquad A^n = \frac{1}{x^n}\, B^{(n)}(\theta), \qquad B^{(n)}(-z) := \frac{\tilde{H}(z+n)}{\tilde{H}(z)} = \prod_{j=0}^{n-1} B(-z-j)$$

   for $0 < \operatorname{Re} z < z_* - n$ — the identity evaluating the symbol at the shifted argument, as at order one.

In particular the temporal jet at time t, up to every order $n < z_* - 1$, is an instantaneous linear functional of the single profile $x \mapsto u(t, x)$; no record of past states is required. The identity is Mellin-level, with the pointwise reading subject to the same pole discipline as order one (9.10). ▷ **[T]** proved-target *(The embodiment principle as theorem content: differentiation is an instantaneous spatial readout of the present memory state, no second memory recording the field's history. The proof is an induction through 9.6's derivative clause plus a telescoping of 9.3's shift relation — chapter 9's apparatus, nothing new taken on trust.)*

Proof. (1) By induction on the order: for $j < n$, $f^{(j)}$ is the primitive of $f^{(j+1)} \in X_0$, so 9.6's derivative clause applies at each step — $\partial_t \mathbb{E}[f^{(j)}(\cdot - xT_1)] = \mathbb{E}[f^{(j+1)}(\cdot - xT_1)]$ as X_0-derivatives, the boundary conditions of $\mathcal{D}^n$ licensing each step — and n steps give $\partial_t^n u(t,x) = \mathbb{E}\big[f^{(n)}(t - xT_1)\big]$. The lemma applied to the causal $f^{(n)} \in X_0$ then gives the first equality; the second is $I^z f^{(n)} = I^{z-n} I^n f^{(n)} = I^{z-n} f$, $I^n f^{(n)} = f$ being the primitive structure of $\mathcal{D}^n$ again. (2) Telescoping the shift relation of 9.3 n times, $\mathcal{M}[A^n g](z) = \prod_{j=1}^n B(j-z)\,\tilde g(z-n) = \frac{\tilde H(z)}{\tilde H(z-n)}\,\tilde g(z-n)$; insert $\tilde g = \widetilde{u(t,\cdot)}$ from 9.6 and compare with (1). □

Remark 12.3 (differentiation is re-indexing)**.** Combined with 9.9, 12.2(1) says that the memory line stores, once and simultaneously, the analytic family $\{(I^z f)(t)\}$ of fractional integrals of the past — and the jet of every order is a *re-reading* of that same stored family at shifted index, order n living at Mellin depth $\operatorname{Re} z > n$ of the strip. Nothing new is stored when one differentiates; the embodiment of the signal already contains its jet, at every order below $z_* - 1$.

Proposition 12.4 (the Gamma family: jets by subtraction along the shape column)**.** For the Gamma family write $u_m(\cdot, x)$ for the field of shape m ($u_0 := f$). From $s(1+xs)^{-m} = x^{-1}\big[(1+xs)^{-(m-1)} - (1+xs)^{-m}\big]$,

$$x\,\partial_t\,u_m \;=\; u_{m-1} - u_m \;=\; -\,(\nabla u)_m, \qquad (\nabla u)_m := u_m - u_{m-1},$$

and hence, iterating at fixed x,

$$x^n\,\partial_t^n\,u_\gamma \;=\; (-\nabla)^n u_\gamma \;=\; \sum_{j=0}^{n} (-1)^{n-j}\binom{n}{j}\,u_{\gamma-j}(t,x), \qquad 0 \le n \le \gamma.$$

The column $\{u_m(\cdot,x)\}_{m=0}^{\gamma}$ is realized by γ identical leaky integrators in series with time constant x, $u_m = (1 + x\,\partial_t)^{-1} u_{m-1}$ — the gamma memory of `@devries1992gamma` — so the scale-normalized jet at scale x is a binomial-weighted *subtraction of coexisting filter states*, with no temporal differencing anywhere in the architecture, γ states per scale, and jets available up to order γ. ▷ **[T]** proved-target *(Transform algebra on 11.2's sections. The identity is exact for all $n \le \gamma$, extending beyond 12.2's strip condition — the transforms are rational and continue meromorphically — consistently with $z_* = \gamma$ marking the strip reading; the relaxation of (H) by meromorphic continuation is among the open problems the paper records.)*

Proof. The transform identity is $(1+xs)^{-m}\big[(1+xs) - 1\big] = xs(1+xs)^{-m}$; induction in n at fixed x; the realization is 11.2 at $x_k = 0$, where the section degenerates to the pure pole $(1 + x_{k+1}s)^{-1}$. □

Remark 12.5 (receptive fields; Laguerre structure)**.** By Rodrigues' formula the jet kernels are associated Laguerre: with $\phi_x^{(m)}$ the Gamma density of shape m at scale x and $\phi_x = \phi_x^{(\gamma)}$,

$$x^n\,\partial_t^n\,\phi_x \;=\; \frac{n!\,\Gamma(\gamma-n)}{\Gamma(\gamma)}\,\phi_x^{(\gamma-n)}\;L_n^{(\gamma-n-1)}\big(t/x\big), \qquad 0 \le n < \gamma,$$

a Gamma density of shape $\gamma - n$ times an associated Laguerre polynomial — the shape dropping as the order rises, each derivative spending one of the kernel's poles. And the shape column spans the order-γ rational function space with the single pole at $-1/x$: the classical Laguerre filter space at time constant x, exponentials times polynomials of degree $< \gamma$, whose orthogonal basis is the Laguerre functions. Both clauses are computations; the orthogonality belongs to the basis of the space, not to the jet kernels themselves, whose polynomial parts all have degree $\gamma - 1$. The comparison the clauses support is structural: the temporal receptive-field family stands to the causal scale space as the Hermite/derivative-of-Gaussian family stands to the Gaussian one — a graded basis, read off the embodied state by subtraction. The difference is that the grading here is by the shape index rather than by differentiation order alone.

Remark 12.6 (the local corner: derivatives as curvatures of the memory trace)**.** For the Bessel family, $z_* = \infty$ (10.3(2)), so every order is available; and in the parabolic gauge ξ of 6.4, 12.2(2) is the n-th power of the local operator of 9.13:

$$\partial_t^n u \;=\; \Big(\tfrac{1}{2}\,\partial_\xi^2 + \frac{\beta}{\xi}\,\partial_\xi\Big)^n u,$$

the jet read from spatial stencils of order $2n$ on the instantaneous memory profile. In the heat case $\beta = 0$ this is the statement that the observer reads temporal derivatives as curvatures, and iterated curvatures, of its memory trace — the embodiment reading already implicit in `@fagerstrom2005temporal`, Thm. 4, here extended from the local corner to the whole class by the Mellin form.

Appendix A

Memory kernels, Sonine conservation, and the scale-Cauchy problem: Theorem 3′

The representation (7.1) attaches to every admissible family a nonincreasing function k. This appendix shows that k is the *memory kernel* of the evolution across scale; that it forms a Sonine pair with the associated potential kernel, through the conservation identity $\kappa * \ell \equiv 1$; and that the kernels of 7.3 satisfy a linear Volterra integral equation from which they can be computed, without Laplace inversion. This half of the appendix is self-contained: the scale-evolution result it needs is stated and proved as A.2.

Throughout we work in the canonical gauge, so that $\mu_{0,x}$ is the law of $x\,T_1$. Throughout, F is an admissible exponent as in (7.1), with drift b_0 and kernel k, and $F \not\equiv 0$ (equivalently (ND); 5.3). A function $f \in X$ is *causal* if it vanishes a.e. on $(-\infty, 0)$, in which case $\hat{f}(s) = \int_0^\infty e^{-st} f(t)\,dt$ is defined for $s > 0$ and satisfies $|\hat{f}(s)| \le \|f\|_1$.

Lemma A.1 (the kernel behind F')**.** Let F be as in (7.1). Then

$$F'(s) \;=\; b_0 \;+\; \int_0^\infty e^{-st}\,k(t)\,dt, \qquad s > 0.$$

▷ **[T]** proved-target *(**Machine-checked**, as `Hemigroup.SelfDecomposableExponent.hasDerivAt_toRealExponent`; `#print axioms` reduces to Lean core. Differentiation under the integral sign, which Mathlib has; all the work is in the side conditions, and* none of them is a hypothesis*. The dominating function $e^{-(s/2)t}k(t)$ is integrable because 8.8 extracts both integrability facts from the class membership itself, so this lemma applies to every admissible F with no regularity assumed of k beyond the definition. That is worth recording because the proof below reads as though $\int_0^1 k < \infty$ and boundedness at infinity were extra conditions to be checked per family; they are not, and 8.8 is why.*

Two small design points the formalisation fixed. The neighbourhood on which the domination holds is $(s/2, \infty)$ rather than a ball, which is what makes the bound uniform without a case split. And the exponent must first be read as a real-valued Bochner integral — the development carries it in $[0, \infty]$ so that no integrability hypothesis travels with the definition — which is `toRealExponent_eq`, valid for $s \ge 0$ by the structure's `ne_top` field, and it is that identity, not the derivative, that the two readings of the exponent meet in.)

Proof. Differentiating the integrand of (7.1) in s gives $e^{-st}k(t)$, which for $s \ge s_0 > 0$ is dominated by $e^{-s_0 t}k(t)$. That dominating function is integrable near 0 because $\int_0^1 k < \infty$, and at infinity because k is bounded there, being nonincreasing; so differentiation under the integral sign is justified. □

Thus the memory kernel of the scale flow is the self-decomposability function k itself, in L^1-normalized dilation to the current scale. The monotonicity of k, which entered abstractly as condition (3) of 7.1, is precisely the admissibility of $\kappa^{(x)}$ as a Lévy tail. In the extremal stable case $k(t) = \alpha t^{-\alpha}/\Gamma(1-\alpha)$ and the operator $\partial_t(\kappa^{(x)} * \cdot)$ below is the Riemann–Liouville derivative $D^{\alpha}_{t,+}$ of `@fagerstrom2005temporal`; in the Gamma family $k(t) = \gamma e^{-t}$, a bounded kernel.

Proposition A.2 (scale evolution in memory-kernel form)**.** Let $(\Phi_{x,y})$ be as in 7.3, in the canonical gauge, let $f \in X$ be causal, and set $u(\cdot, x) := \Phi_{0,x} f = \mu_{0,x} * f$. Then, with the transform equation $\partial_x \hat{u} = -sF'(xs)\hat{u}$ of A.15 in hand:

For each $x > 0$ the causal function $w(\cdot, x) := \kappa^{(x)} * u(\cdot, x)$ lies in L^1_{loc}, and for all $0 \le a \le b$,

$$u(\cdot, b) - u(\cdot, a) \;=\; -\,\partial_t \int_a^b \big(\kappa^{(x)} * u(\cdot, x)\big)\,dx \qquad \text{in } \mathcal{D}'(\mathbb{R}).$$

Differentiating in x, u satisfies, in the distributional sense on $\mathbb{R}_t \times (0,\infty)_x$ and with boundary value $u(\cdot, 0) = f$,

$$\partial_x u \;=\; -\,\partial_t\big(\kappa^{(x)} * u\big) \;=\; -\,b_0\,\partial_t u \;-\; \partial_t \int_0^t \tfrac{1}{x}\,k\big(\tfrac{t-r}{x}\big)\,u(r,x)\,dr.$$

Since $u(\cdot, x)$ is causal, the outer ∂_t may equivalently be moved onto u inside the convolution: the Riemann–Liouville and Caputo forms of the operator coincide on this family. ▷
[**T**] proved-target *(the scalar transform equation this rests on was split off as A.15. Both halves are **[T]** and no ledger entry is involved either way, so the split is not about cost — it is about* reach*: the transform equation is a chain rule and is formalisable today, while the identity here is distributional, and Mathlib has no distribution theory to state it in. Keeping them in one node would have made a provable result unreachable behind an unprovable one.*

Corrected 2026-08-11. *That last clause was true when written and is not now:* `Analysis/Distribution/` *carries test functions, the space $\mathcal{D}'(\Omega, F)$, the Dirac distribution, and a distributional directional derivative as a continuous linear map. It is still not enough for this node, whose statement needs a locally integrable function read as a distribution and a distribution convolved with a measure, and neither is there yet — the file's own docstring says it "contains very few mathematical statements". The correction is recorded because this is a moving target: the right habit is to re-check when Mathlib is bumped, not to treat the node as permanently out of reach.)*

Proof. Local integrability. For $T > 0$,

$$\int_0^T |w(t,x)|\,dt \le \kappa^{(x)}([0,T])\,\|u(\cdot,x)\|_1 \le \Big(b_0 + \int_0^{T/x} k\Big)\|f\|_1 < \infty,$$

locally uniformly in $x > 0$; joint measurability follows from (A7). By Tonelli, for $s > 0$, $\hat{w}(s,x) = \widehat{\kappa^{(x)}}(s)\,\hat{u}(s,x) = F'(xs)\,\hat{u}(s,x)$.

Integration in x. Integrate A.15 over $x \in [a,b]$. The exchange of $\int_a^b dx$ with the Laplace integral is justified by Tonelli, since $\int_a^b s\,F'(xs)\,\|u(\cdot,x)\|_1\,dx \le \|f\|_1 \int_a^b sF'(xs)\,dx = \|f\|_1\big(F(bs) - F(as)\big) < \infty$, finite also for $a = 0$. This gives

$$\hat{u}(s,b) - \hat{u}(s,a) \;=\; -\,s\int_a^b \hat{w}(s,x)\,dx \;=\; -\,s\,\widehat{W}(s), \qquad W := \int_a^b w(\cdot,x)\,dx,$$

with $W \in L^1_{\mathrm{loc}}$ and causal.

Inversion. The causal, locally integrable function $V(t) := \int_0^t \big(u(r,b) - u(r,a)\big)\,dr + W(t)$ has Laplace transform $\big(\hat{u}(s,b) - \hat{u}(s,a)\big)/s + \widehat{W}(s) = 0$ for every $s > 0$, so $V = 0$ a.e. by 3.9. That is the displayed identity. The final statement holds because $u(\cdot, x)$ vanishes on $(-\infty, 0)$, so no boundary term arises when moving ∂_t through the convolution. □

Remark A.3 (reading: a general fractional relaxation equation)**.** A.2 is the memory-kernel form of the per-scale generator; Theorem 10.4 below upgrades it to a classical scale-Cauchy problem $\partial_x u = -\varphi_x(\partial_t)u$ with $\varphi_x(s) = sF'(xs)$, on an explicit operator core. It identifies the scale evolution as a *general fractional* relaxation equation in the sense of `@kochubei2011general` and `@luchko2021general`, with scale-dilated kernel; the case $k(t) = \alpha t^{-\alpha}/\Gamma(1-\alpha)$ recovers the fractional scale-Cauchy problem of `@fagerstrom2005temporal`, Thm. 3.

Lemma A.4 (potential kernel)**.** For $x > 0$ set $\varphi_x(s) := sF'(xs)$. Then φ_x is a nonzero Bernstein function, positive on $(0,\infty)$, and there is a unique positive, locally finite measure $\ell^{(x)}$ on $[0,\infty)$ — the *potential kernel* — with

$$\widehat{\ell^{(x)}}(s) \;=\; \frac{1}{\varphi_x(s)} \;=\; \frac{1}{s\,F'(xs)}, \qquad s > 0.$$

▷ [**T**] proved-target ***(Machine-checked**, as `Hemigroup.SelfDecomposableExponent.existsUnique_potentialKernel`; `#print axioms` gives ledger A17 and nothing else. Route B — construct the measure as the subordinator's potential $U = \int_0^\infty \mu_t\,dt$ rather than represent it via Bernstein–Widder — is what is proved: causality, the transform by Tonelli, local finiteness from convergence at one point, and uniqueness from Laplace injectivity for measures that are not finite; the two sub-lemmas that were once open — that φ_x is a Lévy exponent with its triple exhibited (`exists_levyTriple_symbol`: drift b_0, Lévy measure the Stieltjes measure $-d[k(\cdot/x)/x]$), and that the laws μ_t form a* measurable *family (`exists_subordinatorFamily`) — are proved. The statement's first sentence, that φ_x is a nonzero Bernstein function positive on $(0,\infty)$, is carried by those two declarations and by `symbol_pos` rather than by the $\exists!$ theorem's conjuncts (fidelity review R21).*

***That second clause is the finding, and the work order had omitted it.** Forming $\int_0^\infty \mu_t\,dt$ reads as bookkeeping and is not: ledger A17 supplies μ_t for each t* by choice, indepen-dently*, so nothing connects the choices across t and the integral is not a measure at all. The route to it is where the* subordinator *structure finally does work rather than being merely named: $\mu_{t+t'} = \mu_t * \mu_{t'}$ and causality make $t \mapsto \mu_t(Iic\,r)$ antitone, hence measurable; the intervals form a π-system generating the Borel sets and the μ_t are finite, so Dynkin extends measurability to every Borel set. The increasing paths, which the prose treats as intuition, are exactly what make the potential measure* exist*. Same shape as the `IsCausal` clause that writing A.5 found: a hypothesis that reads as a formality and carries the reachability of the statement.)*

Proof. With $B(s) := sF'(s) \in \mathrm{BF}\setminus\{0\}$ by 7.1(2), we have $\varphi_x(s) = B(xs)/x \in \mathrm{BF}\setminus\{0\}$, and $\varphi_x > 0$ on $(0,\infty)$ by 3.4.

For existence, *construct* the measure rather than representing it. Since φ_x is a Bernstein function without killing term it is the exponent of a subordinator: let $(\mu_\tau^{(x)})_{\tau\ge 0}$ be its laws, $\widehat{\mu_\tau^{(x)}}(s) = e^{-\tau\varphi_x(s)}$, and set

$$\ell^{(x)} \;:=\; \int_0^\infty \mu_\tau^{(x)}\,d\tau,$$

the subordinator's *potential measure*. By Tonelli its transform is $\int_0^\infty e^{-\tau\varphi_x(s)}\,d\tau = 1/\varphi_x(s)$ for every $s > 0$, which is finite, so $\ell^{(x)}$ is locally finite; and uniqueness is uniqueness of the Laplace transform for locally finite measures whose transform converges somewhere (3.9).

Two remarks on this route. It uses no property of CM: neither $\mathrm{CM}\circ\mathrm{BF} \subseteq \mathrm{CM}$ (3.3(2), ledger A2) nor Bernstein–Widder in its general form (3.3(1), ledger A1), both of which the older argument spent. What it does use is the existence of the subordinator, ledger A17, which the main theorem has already spent — so the exchange is two interfaces for none. And the integral $\int_0^\infty \mu_\tau^{(x)}\,d\tau$ is a measure only because $\tau \mapsto \mu_\tau^{(x)}$ is measurable, which is not automatic and is where the subordinator's increasing paths are actually used: $\mu_{\tau+\tau'}^{(x)} = \mu_\tau^{(x)} * \mu_{\tau'}^{(x)}$ with causality makes $\tau \mapsto \mu_\tau^{(x)}\big((-\infty, r]\big)$ nonincreasing, hence measurable, and a Dynkin argument on the generating π-system $\{(-\infty, r]\}$ extends this to every Borel set.

The scaling law follows by comparing transforms: $\widehat{\ell^{(x)}}(s) = x/B(xs) = x\,\widehat{\ell^{(1)}}(xs)$. $\square$

Theorem A.5 (Sonine conservation)**.** For every $x > 0$,

$$\kappa^{(x)} * \ell^{(x)} \;=\; \mathrm{Leb}_{[0,\infty)}, \qquad \text{i.e.} \qquad \int_{[0,t]} \ell^{(x)}\big([0,\, t-r]\big)\, \kappa^{(x)}(dr) \;=\; t \quad \text{for all } t \geq 0.$$

▷ **[T]** proved-target *(**Machine-checked**, as `Hemigroup.SelfDecomposableExponent.sonine_conservation`; `#print axioms` reduces to Lean core, so this half's headline crosses no interface at all. The tagged declaration states the identity restricted to $[0,\infty)$; `sonine_conservation'` is the unrestricted equality of measures on $\mathbb{R}$ displayed above, the two agreeing because both factors — $\kappa^{(x)} * \ell^{(x)}$ and $\mathrm{Leb}_{[0,\infty)}$ — are causal, i.e. vanish on $(-\infty, 0)$ (fidelity review R23).*

It does not wait on A.4. *The Lean statement quantifies over any causal ℓ meeting that lemma's specification rather than naming the $\ell^{(x)}$ it produces, so the identity is independent of how existence is discharged — which matters here, because existence is the one open question of this half and the one place a third trust-boundary entry could enter. Only A.13 needs it. Decoupling the two is also what exposed three hypotheses this statement had been borrowing silently: that ℓ is causal, that its transform is prescribed in $[0,\infty]$ rather than as a Bochner integral (which reads 0, not ∞, when the integrand fails to be integrable), and axiom (ND) — without which φ_x vanishes, $\kappa^{(x)} = 0$, and the identity is* false. *All three are inherited in the paper by taking $(\Phi_{x,y})$ from 7.3; a statement that stands on its own has to say them.*

Everything the proof needs was built for other reasons, which is the one respect in which this node was cheap: transforms multiply over convolution (chapter 4), the first factor is A.16, the second is the hypothesis, and the conclusion is 3.9 — the extension of ledger A6 to measures that are not finite, which was proved in advance precisely because $\kappa^{(x)}$, $\ell^{(x)}$ and Lebesgue measure are none of them finite and this identity compares them.)

Proof. Both sides are locally finite measures on $[0,\infty)$ with finite Laplace transforms, and those transforms agree:

$$\widehat{\kappa^{(x)}}(s)\,\widehat{\ell^{(x)}}(s) = \frac{\varphi_x(s)}{s} \cdot \frac{1}{\varphi_x(s)} = \frac{1}{s} = \widehat{\mathrm{Leb}}(s),$$

using A.1 for the first factor and A.4 for the second. 3.9 concludes. $\square$

Note the scaling structure: $\kappa^{(x)}$ dilates with L^1 normalization and $\ell^{(x)}$ with sup normalization (A.4), and it is exactly this mismatch that renders the identity scale-invariant — the pair $(\kappa^{(x)}, \ell^{(x)})$ is itself scale-covariant.

Corollary A.6 (exact invertibility; conservation of information)**.** Define, on causal $v \in X$, the generalized derivative and integral

$$D^{(x)}v := \partial_t\big(\kappa^{(x)} * v\big), \qquad I^{(x)}v := \ell^{(x)} * v.$$

Then $D^{(x)}I^{(x)}v = v$ in $\mathcal{D}'$, hence a.e. ▷ **[T]** proved-target

Proof. By A.5 and associativity of convolution — Tonelli on causal supports — $\kappa^{(x)} * \ell^{(x)} * v = \mathrm{Leb} * v = \int_0^{\cdot} v(r)\,dr$, and $\partial_t \int_0^t v = v$. $\square$

A.6 is the conservation content of the axioms in Volterra form: the infinitesimal smoothing applied at each scale is exactly invertible, so no single step of the cascade destroys information — it only redistributes it, consistently with (A5), which fixes the mass. Probabilistically, $\ell^{(x)}$ is the potential (renewal) measure of the subordinator with exponent φ_x, $\kappa^{(x)}$ is its drift-plus-Lévy-tail, and A.5 is the renewal identity balancing expected occupation against tail decay.

Proposition A.7 (regularity of the pair)**.** Write $\kappa^{(x)} = b_0\,\delta_0 + \kappa_c^{(x)}$, with $\kappa_c^{(x)}(dt) := \frac{1}{x}\,k\big(\frac{t}{x}\big)\,dt$ the absolutely continuous part of A.16 — an unconditional decomposition, k being nonincreasing. The two clauses below record when $\ell^{(x)}$ admits an analogous decomposition $\ell^{(x)} = c\,\delta_0 + \ell_c^{(x)}$, for a constant $c \geq 0$ not in general equal to b_0, and describe the regularity of $\kappa_c^{(x)}$ and $\ell_c^{(x)}$:

1. $\ell^{(x)}$ admits such a decomposition with $\ell_c^{(x)}$ a nonincreasing density if and only if $1/F'$ is a Bernstein function — equivalently, φ_x is a *special* Bernstein function; by dilation invariance the condition does not depend on x.

2. $\kappa_c^{(x)}$ is completely monotone iff k is completely monotone, which holds iff F' belongs to the Stieltjes class $\mathcal{S} := \big\{\, h(s) = \frac{a}{s} + b + \int_{(0,\infty)} \frac{\sigma(d\tau)}{s+\tau} \,\big\}$. In that case $\ell^{(x)}$ also admits the decomposition above with $\ell_c^{(x)}$ completely monotone, i.e. the absolutely continuous parts $(\kappa_c^{(x)}, \ell_c^{(x)})$ form a classical Sonine pair of CM kernels; moreover k completely monotone implies F is a complete Bernstein function.

▷ [**A**] analytic interface ledger A9 *(special Bernstein functions, the Stieltjes class and complete Bernstein functions; @schilling2012bernstein, Thm. 11.3, Thm. 7.3 and Thm. 6.2, with $\mathcal{S}$ defined in Ch. 2. Cited, not reproven here.* ***Assignment.*** *Clause (1) is carried by ledger A9* entire*: the characterization of the potential measures of special subordinators is SSV Thm. 11.3 and is not proved here. Within clause (2) the ledger carries the two structural inputs — that* $1/h \in \mathrm{CBF}$ *for* $h \in \mathcal{S}\setminus\{0\}$ *(Thm. 7.3), and that a complete Bernstein function has a completely monotone Lévy density (Thm. 6.2) — while the computations joining them are [**T**] and given below. The existence clause, which needs no ledger entry at all, was split off as A.13; before the split this node reported A9's cost for a restatement of three proved results. Note the edition hazard recorded in the ledger: SSV's Thm. 11.3 is Thm. 10.3 in the first edition, a chapter having been inserted at position 10 in the second.)*

Proof. (1) is the cited characterization of potential measures of special subordinators.

(2) Dilation preserves complete monotonicity, so the first equivalence reduces to a statement about k. If k is completely monotone then $k(t) = \int e^{-\tau t}\sigma(d\tau)$ for a positive measure σ, and A.1 gives $F'(s) = b_0 + \int \sigma(d\tau)/(s+\tau) \in \mathcal{S}$. Conversely, an $\mathcal{S}$-representation of F' must have $a = 0$ — an a/s term corresponds to an additive constant in k, contradicting $\int_1^\infty k(t)t^{-1}dt < \infty$ — and then $k(t) = \int e^{-\tau t}\sigma(d\tau)$ is completely monotone.

For the second assertion, $1/h \in \mathrm{CBF}$ for $h \in \mathcal{S}\setminus\{0\}$, so $g := 1/F'(x\,\cdot) \in \mathrm{CBF}$ and $\widehat{\ell^{(x)}}(s) = g(s)/s$. Writing the Lévy–Khintchine triple of g as (a_g, b_g, ν_g), with completely monotone Lévy density because $g \in \mathrm{CBF}$, gives $g(s)/s = \frac{a_g}{s} + b_g + \widehat{\bar\nu_g}(s)$, that is $\ell^{(x)} = b_g\,\delta_0 + \ell_c^{(x)}$ with $\ell_c^{(x)}(t) := a_g + \bar\nu_g(t)$; and $a_g + \bar\nu_g(t) = a_g + \int_t^\infty$(completely monotone) is completely monotone.

Finally, k completely monotone implies the Lévy density $k(t)/t$ of F is completely monotone, being a product of completely monotone functions, hence $F \in \mathrm{CBF}$. □

Part (2) delineates exactly where the classical Sonine calculus with completely monotone kernel pairs — the standing hypothesis of most of the general-fractional-calculus literature — applies: on the complete-Bernstein subfamily, which Chapter 10 identifies with the locality corner of the memory-line realization. For general admissible F the pair, and A.5, live at the level of measures.

Proposition A.8 (Volterra equation for the kernels)**.** Let F be as in (7.1), let $x > 0$, and set $\mu_x := \mu_{0,x}$. Then, as measures on $[0,\infty)$,

$$t\,\mu_x(dt) \;=\; \big(\theta_x * \mu_x\big)(dt), \qquad \theta_x(dt) := b_0\,x\,\delta_0(dt) + k\big(\tfrac{t}{x}\big)\,dt. \tag{V.1}$$

▷ [**T**] proved-target *(a transform computation — no citation is needed and none is made. The descent from this measure identity to the density equation*

(V.2), which is what needs ledger A10, was split off as A.14; before the split this node reported that entry's cost for results that do not incur it. The uniqueness clause, which is the scalar linear ODE rather than the transform computation, was split off on 2026-08-15 as A.9. ***Proved in Lean*** *(2026-08-15),* `Hemigroup/Volterra.lean`*, on ledger A17 alone. θ_x* ***is x times the memory kernel*** *— the atom scales from b_0 to $b_0 x$ and the density from $k(t/x)/x$ to $k(t/x)$ — so $\widehat{\theta}_x(s) = xF'(xs)$ is A.16 scaled and needs no computation of its own. The one new step is $\mathcal{L}[t\mu](s) = -\mathcal{L}[\mu]'(s)$, and it is dominated convergence with a* constant *bound: $te^{-xt} \le e^{-1}/x$ uniformly in t, the maximum of ue^{-u}, which a finite measure integrates with no side condition. That estimate turns out to need no hypothesis on t at all, being free to the left of the origin where its left-hand side is negative.)*

Proof. Taking Laplace transforms, $\mathcal{L}[t\,\mu_x](s) = -\frac{d}{ds}e^{-F(xs)} = xF'(xs)\,e^{-F(xs)}$, while by A.1 with the substitution $\tau = t/x$, $\widehat{\theta}_x(s) = b_0 x + \int_0^\infty e^{-st}k(t/x)\,dt = xF'(xs)$. So both sides of (V.1) have the finite transform $xF'(xs)e^{-F(xs)}$ and coincide by 3.9. ☐

Proposition A.9 (the Volterra equation determines the kernel)**.** μ_x is the *unique* probability measure on $[0,\infty)$ satisfying (V.1). ▷

[T] proved-target *(a scalar linear ODE in the Laplace variable, and no citation. Separated from A.8 because the two halves use different machinery: that one is a transform identity, this one solves a differential equation.* ***Proved in Lean*** *(2026-08-15),* `Hemigroup/Volterra.lean`*, on ledger A17 alone — and* ***the estimate written when the two were separated held exactly****. It said the pieces were in place: $\hat{\nu}$ differentiable on $(0,\infty)$ is* `hasDerivAt_laplace`*, which had been stated for an* arbitrary *causal finite measure precisely so a competitor could use it; the remaining step is antiderivative uniqueness, the shape chapter 8's closed forms already run on; the one extra ingredient is $\hat{\nu} > 0$. All three were what the proof needed and no more. The one thing the estimate did not foresee is that the antiderivative-uniqueness lemma had been written in the narrow form its first consumer needed — comparing an exponent with a candidate — and $-\log\hat{\nu}$ versus $F(x\cdot)$ is not an instance of that. Generalising it to two arbitrary functions is four lines and made the old statement a corollary, which is the sequence this development treats as routine when a second consumer appears.)*

Proof. If ν is a probability measure on $[0,\infty)$ with $t\,\nu(dt) = \theta_x * \nu$, then $\hat{\nu}$ is differentiable on $(0,\infty)$ with $-\hat{\nu}'(s) = \mathcal{L}[t\nu](s) = xF'(xs)\,\hat{\nu}(s)$. Solving this linear ODE with $\hat{\nu}(0+) = 1$ gives $\hat{\nu}(s) = e^{-F(xs)}$, hence $\nu = \mu_x$ by 3.6. ☐

Equation (V.2) is the constructive counterpart of the inverse Laplace transform in 7.3: the kernels are *derived* from the memory kernel k by Volterra integration. We record here, without a formal statement, why this determines ϕ_x rather than proving it: on $(b_0 x, \infty)$, where ϕ_x is supported, (V.2) is linear and homogeneous, and a *regular* equation of that form — continuous coefficient, bounded away from zero — has the trivial solution only; it is the vanishing of the coefficient $t - b_0 x$ exactly at the left endpoint of that interval that admits a nontrivial family instead, from which the unit-area axiom (A5) selects ϕ_x. Positivity of that choice is the property Picard iteration of (V.2) preserves term by term, starting from a nonnegative seed, on the interior where the equation is regular.

Example A.10 (verification: the Gamma family)**.** For $F(s) = \gamma\log(1+s)$ we have $b_0 = 0$, $k(u) = \gamma e^{-u}$, and $\phi_x(t) = t^{\gamma-1}e^{-t/x}/(\Gamma(\gamma)x^\gamma)$ by 8.2. Indeed, since $e^{-u/x}e^{-(t-u)/x} = e^{-t/x}$,

$$\int_0^t \gamma e^{-u/x}\,\frac{(t-u)^{\gamma-1}e^{-(t-u)/x}}{\Gamma(\gamma)\,x^\gamma}\,du \;=\; \frac{\gamma\,e^{-t/x}}{\Gamma(\gamma)\,x^\gamma}\cdot\frac{t^\gamma}{\gamma} \;=\; t\,\phi_x(t),$$

which is (V.2) for this family, as required.

Remark A.11 (computation; relation to the 2005 series)**.** For $F(s) = s^\alpha$, i.e. $k(u) = \alpha u^{-\alpha}/\Gamma(1-\alpha)$, Picard iteration of (V.2) regenerates term by term the series expansion following Theorem 2 of `@fagerstrom2005temporal`; the Volterra route thus contains the original series as its homogeneous special case. Numerically, (V.2) is a weakly singular Volterra equation of the second kind, amenable to standard product-integration quadrature, and it provides an inversion-free forward scheme for the kernels for arbitrary admissible k.

Remark A.12 (literature)**.** Kernel pairs satisfying $\kappa * \ell \equiv 1$ go back to `@sonine1884sur`; the operator calculus built on them is the general fractional calculus of `@kochubei2011general` and `@luchko2021general`, usually developed under completely monotone or Sonine-class kernel hypotheses, which by A.7(2) delineate the complete-Bernstein subfamily. In Volterra-equation theory the kernels $\kappa^{(x)}$ are the *completely positive* kernels of `@clement1981asymptotic`; the resolvent theory of `@pruss1993evolutionary` then yields well-posedness and positivity of the evolution in A.2 by purely deterministic arguments. The integral equation (V.1) is the classical characterization of infinitely divisible laws on the half-line of `@steutel2004infinite`.

Proposition A.13 (the Sonine pair exists)**.** For every admissible F and every $x > 0$ the pair $(\kappa^{(x)}, \ell^{(x)})$ exists at the level of positive measures on $[0, \infty)$ and satisfies $\kappa^{(x)} * \ell^{(x)} = \mathrm{Leb}_{[0,\infty)}$. ▷ [**T**] proved-target *(a restatement of A.1, A.4 and A.5, carrying nothing of its own. It is stated separately because the regularity clauses it used to share a node with are ledger A9, and existence is not: at the level of measures the pair is unconditional, and only the question of whether those measures have* densities of a given monotonicity class *needs the citation. That distinction is the content of A.7, and is what its opening phrase about the drift atoms is there to signal.*

Proved from A.4 2026-08-11*, so it now waits on exactly what that node waits on — a single open sub-lemma, that φ_x is a Lévy exponent with its triple exhibited. The one thing it needed on its own account was σ-finiteness of $\ell^{(x)}$: A.5 is stated for an* `SFinite` *measure because it convolves, while A.4 delivers* local *finiteness. The two are the same thing for a causal measure — exhaust $[0, \infty)$ by the $[0, n]$ and note everything below the origin is null — which is the converse of the implication this appendix already needed in the other direction, that a convergent transform forces local finiteness.)*

Proof. A.1 constructs $\kappa^{(x)}$, A.4 constructs $\ell^{(x)}$ and gives its uniqueness, and A.5 is the identity. □

Proposition A.14 (the Volterra equation for the density)**.** Let F be as in (7.1), let $x > 0$, and set $\mu_x := \mu_{0,x}$. If $k \not\equiv 0$ then μ_x is absolutely continuous, and its density ϕ_x solves the linear Volterra equation

$$\big(t - b_0 x\big)\, \phi_x(t) \;=\; \int_0^t k\big(\tfrac{u}{x}\big)\, \phi_x(t-u)\, du \qquad \text{for a.e. } t > 0. \tag{V.2}$$

Away from its left endpoint $t = b_0 x$ — where the *coefficient* $t - b_0 x$ vanishes; ϕ_x vanishes below it, the drift shifting every kernel by $b_0 x$ (7.5), but need not vanish at it — this is a Volterra equation of the second kind. ▷ [**A**] analytic interface ledger A10 *(absolute continuity of nondegenerate self-decomposable laws;* `@sato1999levy`*, Thm. 27.13. Cited, not reproven here.* ***Assignment.*** *ledger A10 carries absolute continuity and nothing else. Given it, the descent from the measure identity (V.1) of A.8 to the density equation (V.2) is restriction to densities and is [**T**]. (V.2) is the point of the pair of propositions, being the inversion-free forward scheme; the hypothesis $k \not\equiv 0$ is nondegeneracy, and excluding the degenerate case — the pure drift $F(s) = b_0 s$, whose kernel is the atom $\delta_{b_0 x}$ — is ours, not the citation's.)*

Proof. When $k \not\equiv 0$, μ_x is a nondegenerate self-decomposable law by 7.2, and ledger A10 makes it absolutely continuous. Restricting the identity (V.1) of A.8 to densities yields (V.2). □

Lemma A.15 (the transform equation)**.** Let $(\Phi_{x,y})$ be as in 7.3, in the canonical gauge, let $f \in X$ be causal, and set $u(\cdot, x) := \Phi_{0,x} f$. Then for each $s > 0$ the function $\hat{u}(s,x) = e^{-F(xs)} \hat{f}(s)$ is continuously differentiable in x, with

$$\partial_x \hat{u}(s,x) \;=\; -\,s\,F'(xs)\,\hat{u}(s,x).$$

▷ **[T]** proved-target *(the chain rule, F being smooth on $(0,\infty)$ by 3.3(3). Numbered 9.14, a number the draft does not use. It is stated on its own because it is the one part of A.2 that can be formalised without a theory of distributions: everything else in that proposition is an identity in $\mathcal{D}'(\mathbb{R})$.)*

Proof. F is smooth on $(0,\infty)$ by 3.3(3), so $x \mapsto e^{-F(xs)}$ is continuously differentiable there with derivative $-sF'(xs)e^{-F(xs)}$; multiply by the constant $\hat{f}(s)$. □

Lemma A.16 (the memory kernel's transform)**.** For $x > 0$ define the locally finite measure

$$\kappa^{(x)}(dt) \;:=\; b_0\,\delta_0(dt) \;+\; \tfrac{1}{x}\,k\big(\tfrac{t}{x}\big)\,dt.$$

Then $\partial_x F(xs) = s\,F'(xs) = s\,\widehat{\kappa^{(x)}}(s)$, so the symbol $\varphi_x(s) := sF'(xs)$ is s times the transform of the memory kernel. ▷ **[T]** proved-target *(**Machine-checked**, as `Hemigroup.SelfDecomposableExponent.laplace_memoryKernel`; `#print axioms` reduces to Lean core. Split from A.1 because the two clauses cost differently, and the split was worth making even though both are now proved: the derivative formula is differentiation under the integral sign, while this is a change of variables under a measure that is neither finite nor a probability measure.*

$\kappa^{(x)}$ is the first such measure in the development, and that is what the clause costs. The mathematics is one substitution, $\tau = t/x$, and the L^1 normalisation $\frac{1}{x}k(\frac{t}{x})$ is exactly what makes it an equality rather than a proportionality. Everything else is bookkeeping the paper does not have to do: the transform here is a Bochner integral, so the exponential's integrability must be established separately against the atom and against the density, and the density is only a.e. *measurable, k being only monotone. It is the first factor of the Sonine identity, which is why it came next.)*

Proof. The substitution $\tau = t/x$ in $F'(xs) = b_0 + \int_0^\infty e^{-xs\tau} k(\tau)\,d\tau$, which is A.1 at xs, turns the integral into $\int_0^\infty e^{-st}\,\frac{1}{x}k(\frac{t}{x})\,dt$; the drift b_0 is the transform of $b_0\delta_0$, which is constant in s. □

Lemma A.17 (the potential kernel scales)**.** For $x > 0$, $\ell^{(x)}$ is x times the pushforward of $\ell^{(1)}$ under $t \mapsto xt$; in density form, $\ell^{(x)}(t) = \ell^{(1)}(t/x)$. ▷ **[T]** proved-target *(a comparison of transforms and nothing else: $\widehat{\ell^{(x)}}(s) = 1/(sF'(xs)) = x/\varphi_1(xs) = x\,\widehat{\ell^{(1)}}(xs)$, and A.4's uniqueness clause does the rest. **Proved in Lean** (2026-08-15), `Hemigroup/PotentialScaling.lean`, and **Lean core** — it does not even reach ledger A17, being about measures meeting a specification rather than about the constructed family. **The blocker this node recorded was not one.** It had said the clause presupposes a* named $\ell^{(x)}$*, since the existence-and-uniqueness statement means the object must be chosen before a scaling law can be predicated of it, and that choice is a definition the development has not needed elsewhere. It need not be chosen: the law is stated against* any two *measures meeting the specification — exactly as A.5 and A.13 already quantify over their ℓ, and as the scale-Cauchy half of this appendix quantifies over ν_1. The uniqueness half of A.4 is what makes the hypothesis pin the objects down, so quantifying over them loses nothing and introduces no definition without a consumer.* A specification is a hypothesis, and a hypothesis can be quantified over*; what cannot be is a construction, which is why this appendix built one for the tail measure and needs none here.)*

Proof. Both measures are causal and locally finite, and their transforms agree on $(0,\infty)$; apply the uniqueness clause of A.4. □

The second half of the appendix upgrades A.2 from a distributional identity to a classical Cauchy problem in the scale variable, on an explicit operator core. Throughout we work in the canonical gauge and write

$$X_0 := L^1(\mathbb{R}_+),$$

identified with the causal elements of X; by (A3) every $\Phi_{x,y}$ restricts to X_0. On X_0 let $(\tau_r)_{r\ge 0}$ denote the *delay semigroup*, $(\tau_r f)(t) = f(t-r)\mathbf{1}_{t\ge r}$, a strongly continuous semigroup of positive isometries, and define the core

$$\mathcal{D} \;:=\; \bigl\{\, f \in X_0 :\; f \text{ absolutely continuous},\; f' \in X_0,\; f(0) = 0 \,\bigr\}.$$

Lemma A.18 (delay derivative)**.** $\mathcal{D}$ is dense in X_0 and invariant under every τ_r and every $\Phi_{x,y}$, and for $f \in \mathcal{D}$,

$$\lim_{h\downarrow 0} \tfrac{1}{h}\bigl(\tau_h f - f\bigr) = -\,f' \quad \text{in } X_0, \qquad \bigl\|\tau_r f - f\bigr\|_1 \le \min\bigl(2\|f\|_1,\; r\,\|f'\|_1\bigr).$$

▷ **[T]** proved-target *(The setting this lemma is about — X_0, the delay semigroup and $\mathcal{D}$ — is defined in `Hemigroup/DelayCore.lean`. X_0 is modelled as a predicate on $X = L^1(\mathbb{R})$ rather than as a subtype, on the same grounds as the measure side's causality predicate; $\mathcal{D}$ is modelled as a predicate on genuine functions, because Chapters 9–10 quantify over its consequences* pointwise *and an L^1 class has no pointwise values; and $\mathcal{D}$ is defined by the primitive $f = \int_0^{\cdot} f'$ rather than by absolute continuity, since the passage from the latter to the former is the Lebesgue fundamental theorem, which Mathlib does not carry — absolute continuity is recovered as a consequence. That $\mathcal{D}$ so defined supplies exactly the hypotheses 9.8 takes about its signal — neither fewer nor more — is a machine-checked equivalence. Two steps of the proof below are reached differently in Lean, neither changing the result. The estimate needs no X_0-valued Bochner integral: bounding $|f(t-r) - f(t)|$ pointwise by $\int_{(0,r]} |f'(t-u)|\,du$ and exchanging in $[0,\infty]$ gives the same constant with no integrability side condition. And the limit needs no separate treatment of $[0,h)$: the identity $h^{-1}(\tau_h f - f) + f' = -h^{-1}\int_0^h (\tau_u f' - f')\,du$ is exact on all of $\mathbb{R}$, so the quotient is an* average *of translation defects and cannot exceed their supremum, with $f(0) = 0$ spent once in making the identity hold rather than in repairing it at an endpoint.)*

Proof. Density is standard.

Invariance under τ_r. $\tau_r f$ is absolutely continuous, vanishes on $[0,r]$, and $(\tau_r f)' = \tau_r f'$.

Invariance under $\Phi_{x,y} = \mu_{x,y} * \cdot$. From $f = \mathbf{1}_{[0,\infty)} * f'$ we get $\mu * f = \mathbf{1}_{[0,\infty)} * (\mu * f')$, so $\mu * f$ is absolutely continuous with derivative $\mu * f' \in X_0$ and value 0 at 0.

The limit. On $[h,\infty)$, $\frac{1}{h}\bigl(f(t-h) - f(t)\bigr) + f'(t) = -\frac{1}{h}\int_{t-h}^{t}\bigl(f'(\rho) - f'(t)\bigr)d\rho$, whose L^1-norm tends to 0 by continuity of translation in L^1; on $[0,h)$ the contribution is $\frac{1}{h}\int_0^h |f| + \int_0^h |f'| \le \sup_{[0,h]} |f| + o(1) \to |f(0)| = 0$.

The estimate. $\tau_r f - f = -\int_0^r \tau_\rho f'\,d\rho$ as a Bochner integral — the integrated form of the limit — together with $\|\tau_\rho f'\|_1 = \|f'\|_1$ and the trivial bound $\|\tau_r f - f\|_1 \le 2\|f\|_1$. □

Definition A.19 (per-scale generator; Phillips form)**.** Let ν_1 be the Lebesgue–Stieltjes measure $-dk$ on $(0,\infty)$, with k taken right-continuous and $k(\infty) = 0$, so that $\nu_1\bigl((r,\infty)\bigr) = k(r)$; and for $x > 0$ put $\nu_x := x^{-1}$ times the pushforward of ν_1 under $r \mapsto xr$, i.e. $\nu_x\bigl((r,\infty)\bigr) = k(r/x)/x$. For $f \in \mathcal{D}$ define

$$\varphi_x(\partial_t)\, f \;:=\; b_0\, f' \;+\; \int_0^\infty \bigl(f - \tau_r f\bigr)\, \nu_x(dr) \;=\; b_0\, f' \;+\; \frac{1}{x}\int_0^\infty \bigl(f - \tau_{xr} f\bigr)\, \nu_1(dr).$$

▷ **[T]** proved-target ***(Defined in Lean** (2026-08-14), in `Hemigroup/PhillipsGenerator.lean`, and **not blocked by generation theory** — the re-check that found so is recorded at A.20. Everything this definition names already existed: $\mathcal{D}$*

and (τ_r) *are A.18's, and* ν_1 *is this appendix's* `exists_tailMeasure` *— the quantile construction made for A.4 precisely because* `StieltjesFunction` *does not apply to a Lévy tail.* ***The*** X_0***-valued Bochner integral needed no new theory****, against a plan that had recorded it as the expensive part of the scale-Cauchy development:* $X = L^1(\mathbb{R})$ *is a complete normed real space,* τ_r *is a continuous linear map on it, and* $r \mapsto \tau_r f$ *is continuous, so the integrand is strongly measurable and the definition is* total *— clause (1) below is what says it means something on* $\mathcal{D}$*, not what makes it well formed. Note the contrast with A.18, where the blueprint named a Bochner integral the obligation did not need; here the integral is in the statement rather than in a proof, so it has to be built, and building it costs nothing.* ***The right-continuity normalisation is neither available nor needed.*** *A* k *that is only nonincreasing has no right-continuous representative nameable here, and* `exists_tailMeasure` *accordingly delivers* $\nu_1\big((r,\infty)\big) = k(r)$ *for* almost every r *rather than every* r*. That is exactly enough, because every use of the tail below sits under an integral in* r *— the same accounting already made above for the potential kernel. Accordingly* ν_1 *enters the Lean statements as a* parameter *meeting that specification rather than as a construction, so that the development here does not wait on an existence half;* `exists_hasLevyTail` *closes the gap, and the two displayed forms above are one change of variables apart (*`phillipsGenerator_eq_smul_integral`*).)*

Lemma A.20 (properties of the generator)**.** For $f \in \mathcal{D}$ and $x > 0$:

1. The integral converges absolutely in X_0, with
$$\|\varphi_x(\partial_t) f\|_1 \;\le\; b_0\|f'\|_1 + \int_0^\infty \min\big(2\|f\|_1,\, xr\,\|f'\|_1\big)\, x^{-1}\nu_1(dr) \;<\; \infty.$$
2. $\mathcal{L}\big[\varphi_x(\partial_t) f\big](s) = \varphi_x(s)\,\hat{f}(s)$ for $s > 0$, where $\varphi_x(s) = sF'(xs)$.
3. $\varphi_x(\partial_t)$ commutes with every $\Phi_{y,z}$ on $\mathcal{D}$.
4. $x \mapsto \varphi_x(\partial_t) f$ is continuous from $(0,\infty)$ to X_0.
5. $\kappa^{(x)} * f$ is a.e. equal to an absolutely continuous function with $\big(\kappa^{(x)} * f\big)' = \varphi_x(\partial_t) f$ a.e.; that is, the Phillips form coincides on $\mathcal{D}$ with the memory-kernel operator of A.16.

▷ **[T]** proved-target *(**Proved in Lean** (2026-08-15), all five clauses, and **Lean core** — the whole of this lemma is interface-free, and the sweeping "blocked on* C_0*-semigroup theory" under which it sat was wrong: no clause mentions a generator's domain, a resolvent, or a generation theorem, and the "strong continuity of* τ*" that (4) invokes is chapter 4's continuity of translation.* `Hemigroup/PhillipsGenerator.lean`*; the statements quantify over any* ν_1 *meeting the tail specification rather than over the constructed family, which is what keeps them off the ledger.* ***The whole lemma rests on one convergence fact****,* $\int(1 \wedge r)\,\nu_1(dr) < \infty$*: it bounds the Bochner integral in (1), dominates the limit in (4), and justifies the exchange in (5). And that fact is* one layer cake*,* $\int(1\wedge r)\nu_1 = \int_0^1 \nu_1\big((u,\infty)\big)du = \int_0^1 k$*, not the integration by parts the proof above names — the* $k(1) < \infty$ *half of that argument's hypothesis is unused, and the tail identity is consumed under an integral in* u*, which is where* ν_1*'s almost-everywhere specification is paid for and found free. Worth noting where* $f \in \mathcal{D}$ *is spent:* $\|\tau_r f - f\|_1 \le r\|f'\|_1$ *makes the integrand small at the origin, while* $\|\tau_r f - f\|_1 \le 2\|f\|_1$ *holds for every* $f \in X$*, so the core hypothesis buys only the first — the half* ν_1*'s mass at the origin needs.* ***Three of the five clauses are pass-throughs of a bounded functional****: (3) pushes* $\Phi_{y,z}$ *through the Bochner integral, (2) pushes* $\mathcal{L}$*, and (5) pushes* $g \mapsto \int_0^t g$*, each turning a statement about the vector-valued integral into one about scalars. That* $\mathcal{L}$ *is available this way is not obvious and was first got wrong here: it is unbounded on* $L^1(\mathbb{R})$*, but the article's transform integrates over* $(0,\infty)$ *only, so its weight* $\mathbf{1}_{(0,\infty)}(t)e^{-st}$ *is bounded by* 1 *and the functional exists. There is no Fubini–Tonelli argument in (2) at all, and none of the three needs the Phillips integral to be shown causal.*

Four departures from the proof above, none changing the content. *(2)'s $\mathcal{L}[f'] = s\hat{f}$ is not an integration by parts: $f(0) = 0$ is already inside the definition of $\mathcal{D}$, so applying the transform to A.18's difference-quotient limit and reading $(e^{-sh} - 1)/h \to -s$ off a derivative at the origin suffices, with no boundary term to make vanish. (3) needs neither causality nor normalisation of Φ — it holds for any finite measure, being (A2) for the L^1 convolution plus linearity. (4) forms no compact set: $\min(2\|f\|_1, xr\|f'\|_1)$ increases in x, so one bound at the top of a neighbourhood covers it, compactness having stood in for monotonicity. And **(5) needs no uniqueness theorem**: the primitive of $\varphi_x(\partial_t)f$ is* evaluated *rather than characterised, so 3.9 is not spent, and the conclusion is equality at every t rather than almost every — with the trailing $(\kappa^{(x)} * f)(0+) = 0$ free, a primitive vanishing at the origin by construction. What replaces the uniqueness step is one signed exchange, got from the layer cake by splitting the integrand at zero; splitting f' instead, which an earlier reading proposed, would not work, the primitive of $(f')^+$ being nondecreasing and so generally not in L^1. Finally, (5) is stated in the primitive vocabulary $\mathcal{D}$ is defined in, and the Lean ν_1 meets an* almost-everywhere *tail specification, a k that is only nonincreasing having no right-continuous representative nameable there.)*

Proof. (1) Combine A.18 with $\int(1\wedge r)\,\nu_x(dr) < \infty$, which follows from $\int_0^1 k < \infty$ and $k(1) < \infty$ by integration by parts.

(2) Termwise: $\mathcal{L}[f'] = s\hat{f}$, using $f(0) = 0$, and $\mathcal{L}[f - \tau_r f] = (1 - e^{-sr})\hat{f}$. Fubini is justified by (1), and $b_0 s + \int(1 - e^{-sr})\nu_x(dr) = \varphi_x(s)$ by A.1 — the two Bernstein representations of φ_x agree.

(3) Convolutions commute; pull $\Phi_{y,z}$ through the Bochner integral.

(4) In the dilated form of A.19 the integrand $\frac{1}{x}(f - \tau_{xr}f)$ is, for x in a compact subset of $(0,\infty)$, continuous in x for each r by strong continuity of τ, and dominated by $C\min(\|f\|_1,\, r\|f'\|_1)$, which is ν_1-integrable. Apply dominated convergence; the drift term is trivial.

(5) Both $\varphi_x(\partial_t)f \in L^1$ and $\kappa^{(x)} * f \in L^1_{\mathrm{loc}}$ are causal with finite Laplace transforms. By (2) and A.1 the causal function $V(t) := \int_0^t \big(\varphi_x(\partial_t)f\big)(\rho)\,d\rho - (\kappa^{(x)} * f)(t)$ has transform $\varphi_x(s)\hat{f}(s)/s - F'(xs)\hat{f}(s) = 0$ for all $s > 0$, so $V = 0$ a.e. by 3.9. That is the claim; the value $(\kappa^{(x)} * f)(0+) = 0$ follows from $\kappa^{(x)}([0,t]) \to b_0$ and $\sup_{[0,t]}|f| \to 0$. $\square$

Theorem A.21 (Theorem 3′: the scale-Cauchy problem)**.** Let $(\Phi_{x,y})$ be as in 7.3, in the canonical gauge, and let $f \in \mathcal{D}$. Set $u(\cdot, x) := \Phi_{0,x}f$. Then:

1. $u(\cdot, x) \in \mathcal{D}$ for every $x \geq 0$;
2. $x \mapsto u(\cdot, x)$ belongs to $C\big([0,\infty); X_0\big) \cap C^1\big((0,\infty); X_0\big)$ and satisfies

$$\partial_x u(\cdot, x) \;=\; -\,\varphi_x(\partial_t)\,u(\cdot, x) \quad (x > 0), \qquad u(\cdot, 0) = f,$$

   with ∂_x a norm-derivative in X_0;

3. u is the *unique* solution in the class of $v \in C([0,\infty); X_0) \cap C^1((0,\infty); X_0)$ with $v(\cdot, x) \in \mathcal{D}$ for $x > 0$, satisfying the equation for $x > 0$ and $v(\cdot, 0) = f$.

▷ **[T]** proved-target *(the hemigroup analogue of `@fagerstrom2005temporal`, Thm. 3, which is recovered as the homogeneous case in A.24. Proved here, not cited.)*

Proof. (1) is the Φ-invariance of $\mathcal{D}$ (A.18), and continuity at every $x \geq 0$ is (A7).

(2) Write $v(\cdot, x) := \varphi_x(\partial_t)\,u(\cdot, x)$. By A.20(3), $v(\cdot, x) = \Phi_{0,x}\,\varphi_x(\partial_t)f$, so by A.20(4), contractivity of $\Phi_{0,x}$, and (A7),

$$\|v(\cdot, x) - v(\cdot, x')\|_1 \;\leq\; \big\|\varphi_x(\partial_t)f - \varphi_{x'}(\partial_t)f\big\|_1 + \big\|(\Phi_{0,x} - \Phi_{0,x'})\,\varphi_{x'}(\partial_t)f\big\|_1 \;\longrightarrow\; 0,$$

so $x \mapsto v(\cdot, x)$ is continuous from $(0,\infty)$ to X_0.

By A.20(5) applied to $u(\cdot,x)\in\mathcal{D}$, each $\kappa^{(x)}*u(\cdot,x)$ is absolutely continuous with derivative $v(\cdot,x)$, vanishing at 0; hence for $0<a\le b$, Fubini gives

$$\int_a^b \big(\kappa^{(x)}*u(\cdot,x)\big)(t)\,dx \;=\; \int_0^t\int_a^b v(\rho,x)\,dx\,d\rho,$$

so the distributional ∂_t in A.2(2) evaluates to the Bochner integral $\int_a^b v(\cdot,x)\,dx$:

$$u(\cdot,b)-u(\cdot,a) \;=\; -\int_a^b v(\cdot,x)\,dx.$$

Since the integrand is continuous in x, the fundamental theorem of calculus for Bochner integrals yields $u\in C^1((0,\infty);X_0)$ with $\partial_x u=-v$, which is the equation.

(3) Let v be any solution in the stated class and fix $s>0$. Then $\eta(x):=\hat v(s,x)$ is continuous on $[0,\infty)$ and C^1 on $(0,\infty)$, with

$$\eta'(x)=\mathcal{L}[\partial_x v](s)=-\mathcal{L}[\varphi_x(\partial_t)v](s)=-\varphi_x(s)\,\eta(x)$$

by A.20(2); differentiation and $\mathcal{L}$ exchange because $|\hat g(s)|\le\|g\|_1$. Solving this scalar linear ODE on $[\varepsilon,x]$ and letting $\varepsilon\downarrow 0$, using continuity at 0, gives $\eta(x)=\hat f(s)\,e^{-F(xs)}$. Thus $\hat v(s,x)=\hat u(s,x)$ for all $s>0$, and $v=u$ by 3.6. □

Proposition A.22 (fixed-scale semigroups; core property)**.** For fixed $x>0$, the operator $-\varphi_x(\partial_t)$ with domain $\mathcal{D}$ is closable, and its closure generates the strongly continuous contraction semigroup on X_0 of convolutions by the probability measures with Laplace transforms $e^{-\tau\varphi_x(s)}$, $\tau\ge 0$ — the semigroup subordinate to the delay semigroup via φ_x. Moreover $\mathcal{D}$ is a core for that generator. ▷ [**A**] analytic interface ledger A11 *(Phillips' subordination theorem, `@phillips1952generation`, Thm. 4.3 with the hypotheses in its equation (40); and the core criterion, `@engel2000one`, Prop. II.1.7. Cited, not reproven here.* ***Assignment.*** *ledger A11 carries two complementary halves and neither suffices alone. Phillips gives the form of the subordinate generator; he asserts that its domain* contains $\mathcal{D}$, *not that $\mathcal{D}$ is a core for it, which is precisely the gap Engel–Nagel closes. Two things the citation does not carry, both [**T**]: that $\mathcal{D}$ is dense and semigroup-invariant, proved in A.18, and that φ_x is a Bernstein function without killing term, proved in A.4. A notational caution recorded in the ledger: Phillips writes neither $-\varphi(-A)$ nor anything in that notation; his displayed form becomes it only after setting the drift and killing parameters to zero. Note finally that A.21 does* not *use this proposition — the Cauchy problem is solved by a scalar ODE, not by generation theory, so this node adds a reading rather than a dependency.)*

Proof. φ_x is a Bernstein function without killing term by A.4, so Phillips' subordination theorem applies to the delay semigroup, whose generator extends $f\mapsto -f'$ on $\mathcal{D}$ (A.18): the subordinate generator is the closure of the Phillips form on the generator's domain. That $\mathcal{D}$ is a core follows from the cited criterion, since $\mathcal{D}$ is dense and invariant under the subordinate semigroup, which consists of convolutions (A.18). □

Example A.23 (the Gamma family: time-recursive form)**.** For $F(s)=\gamma\log(1+s)$ we have $\nu_x(d\rho)=\gamma x^{-2}e^{-\rho/x}\,d\rho$, an exponential memory, and $\varphi_x(s)=\gamma s/(1+xs)$, so

$$\varphi_x(\partial_t)\,u \;=\; \gamma\,\partial_t\big(1+x\,\partial_t\big)^{-1}u \;=\; \gamma\,w', \qquad \text{where } w \text{ solves } w+x\,w'=u,\quad w(0)=0.$$

The auxiliary variable w is computed by one causal first-order ODE, so the scale step is realized by a single recursive low-pass filter — the strongest possible form of the time-recursivity requirement of `@fagerstrom2005temporal`, §5: the observer embodies its past in one first-order state per scale.

Remark A.24 (recovering Theorem 3 of 2005)**.** For $F(s) = s^\alpha$ with $0 < \alpha < 1$ we have $\nu_x\big((\rho,\infty)\big) = \alpha x^{\alpha-1}\rho^{-\alpha}/\Gamma(1-\alpha)$ and $b_0 = 0$, so $\varphi_x(\partial_t) = \alpha x^{\alpha-1} D^\alpha_{t,+}$ with $D^\alpha_{t,+}$ the Riemann–Liouville/Marchaud derivative, and A.21 reads $\partial_x u = -\alpha x^{\alpha-1} D^\alpha_{t,+} u$. Under the reparametrization $\tau := x^\alpha$ this is exactly

$$\partial_\tau u \;=\; -\,D^\alpha_{t,+}\,u,$$

Theorem 3 of `@fagerstrom2005temporal` verbatim — identifying the scale parameter τ of that paper as the α-gauge of 6.4 applied to the canonical scale coordinate.

Corollary A.25 (non-enhancement at past-dominating extrema)**.** Let $f \in \mathcal{D}$ with f' bounded and continuous, and $u(t,x) := (\Phi_{0,x} f)(t)$. Then u, $\partial_t u$ and $\partial_x u$ are jointly continuous on $\mathbb{R} \times (0,\infty)$, the scale-Cauchy equation of A.21 holds pointwise,

$$\partial_x u(t,x) \;=\; -\,b_0\,\partial_t u(t,x) \;-\; \frac{1}{x}\int_0^\infty \big[\,u(t,x) - u(t-xr,\,x)\,\big]\,\nu_1(dr),$$

the integral absolutely convergent, and at every *past-dominating* point — a t with $u(t,x) \geq u(t',x)$ for all $t' \leq t$ — at which additionally $b_0 = 0$ or $\partial_t u(t,x) = 0$,

$$\partial_x u(t,x) \;\leq\; 0,$$

with the dual inequality at past-dominated points. In particular, along any differentiable path $x \mapsto t_m(x)$ of temporal critical points, $\frac{d}{dx}\,u\big(t_m(x),x\big) = \partial_x u\big(t_m(x),x\big) \leq 0$ wherever the tracked maximum dominates its past — whatever the path's speed. ▷

[**T**] proved-target *(The pointwise form of non-enhancement that survives causality: the kernels delay what they smooth, so the fixed-time form fails for every causal family, and what holds is decay along the feature's own delay path, conditional on past-domination. The proviso is sharp — a secondary maximum in the shadow of a larger earlier peak has brackets of both signs, the echo can enhance it — and the unconditional statement is the counting form of 8.13, the two being complementary.)*

Proof. Pointwise regularity. In the canonical gauge $u(t,x) = \mathbb{E}\,f(t - xT_1)$ and $\partial_t u(t,x) = \mathbb{E}\,f'(t - xT_1)$, the latter by dominated convergence (f' bounded); both are bounded by $\|f\|_\infty$, $\|f'\|_\infty$ and jointly continuous, again by dominated convergence, f and f' being bounded and continuous (f is absolutely continuous with $f' \in L^1$, and f' is continuous by hypothesis).

The pointwise identity. Set $g(t,x) := b_0\,\partial_t u(t,x) + \frac{1}{x}\int_0^\infty [u(t,x) - u(t-xr,x)]\,\nu_1(dr)$. The integral converges absolutely, the integrand being bounded by $\min\big(2\|f\|_\infty,\, xr\,\|f'\|_\infty\big)$ and $\int(1 \wedge r)\,\nu_1(dr)$ finite as in A.20(1); by the same domination, g is jointly continuous. Its Laplace transform is $\varphi_x(s)\,\hat{u}(s,x)$ — the computation of A.20(2), run pointwise — so $g(\cdot,x)$ is the continuous representative of $\varphi_x(\partial_t)\,u(\cdot,x)$. A.21(2) gives $u(\cdot,b) - u(\cdot,a) = -\int_a^b \varphi_x(\partial_t) u(\cdot,x)\,dx$ in X_0, hence $u(t,b) - u(t,a) = -\int_a^b g(t,x)\,dx$ for almost every t; both sides are continuous in t, so the identity holds for every t, and since g is continuous in x the fundamental theorem of calculus yields the pointwise derivative $\partial_x u(t,x) = -g(t,x)$, jointly continuous.

The sign. At a past-dominating point every bracket $u(t,x) - u(t-xr,x)$, $r > 0$, is nonnegative, so the integral term of $-g$ is ≤ 0, and the drift term $-b_0 \partial_t u$ vanishes under either proviso; the dual statement replaces $\geq$ by $\leq$ throughout. The path clause is the chain rule — u is C^1 in each variable with jointly continuous partials, hence totally differentiable — with the ∂_t term killed by criticality, at whatever speed the path moves. □